NOKIA

## Safaricom Ethiopia

# High Level Design

## Wakanda Cloud Packet Core (CPC)

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# 1. About this document

## 1.1 Purpose

The purpose of this document is to provide a high-level view on the service architecture and logical design for Nokia's Cloud Packet Core, deployed in Safaricom Ethiopia's NFV cloud infrastructure as a fully functioning virtualized Core solution supporting 3GPP defined network features.

It outlines the technical requirements imposed into the CPC by the services/feature scope baselined in the Technical Solution Description(TSD) document [R1] and enumerated in Requirements Traceability Matrix(RTM) [R2].

This document captures the information available at the time of writing, as well as the design assumptions and conventions relevant for planning the implementation of CPC functionality in Safaricom Ethiopia network.

Following the delivery of the solution, this document should not be subject to regular change. Addendums to assumptions and scope maybe carried out be newer versions of this document.

## 1.2 Audience

The primary intended audience is Safaricom Ethiopia's project governance and engineering teams, as well as the Nokia engineering community assigned to this project.

It is assumed that the reader is familiar with 3GPP Mobile Architectures, related services and possesses a general knowledge on Telco terminology.

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## 1.3 Scope

The scope of this document is to detail the High-Level Design considerations that ensure a stable and scalable Cloud Packet Core implementation in Safaricom Ethiopia EPC Sites using the Nokia cloud-based Packet Core product portfolio.

This document frames the scope of functionality delivered by the Cloud Packet Core solution and the collection of functional interfaces that compose it. It also describes base design guidelines for networking, Network Function Virtualization Infrastructure (NFVI) and considerations for efficient integration with Operating Support Systems (OSS) and Business Support Systems (BSS) where applicable.

Details on Hardware components, product features/roadmap and system configuration are available either on the Low-Level Design (LLD) documentation set delivered under the scope of this project, or in customer product documentation. As such, these will not be covered under the scope of this document.

In sum the HLD provides the overall system design, covering the service requirements that govern the end to end architecture and the relation between various modules and functions of the system.

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# 1.4 Design Criteria

## 1.4.1 Product Release Matrix

The CPC solution addressed in this document encompasses a list of products under the following software release line-up:

| Function | Product Name | Software Release |
|---|---|---|
| SGSN/MME | Cloud Mobility Manager (CMM) | R21.8 |
| SGW/PGW | Cloud Mobility Gateway (CMG) | R21.8 |
| NFVM | CloudBand Application Manager (CBAM) | R21 |
| VIM | VMware – NFV3.2.1 | NFV3.2.1 |

## 1.4.1 Assumptions & Design Decisions

This section provides a table that lists all design decisions made and technical dependencies for the solution.

| ID | Section | Descriptions |
|---|---|---|
| AS-01 | | The proposed CPC architecture baselines CMG CUPS design on a vlan-aware, Hybrid (MG+/LB) architecture.<br>The proposed CPC architecture baselines CMM design on a vlan-unaware, CMM TA architecture. |
| AS-02 | | |
| AS-03 | | |
| | | |
| | | |

**Table 1: Assumptions**

## 1.4.2 Exclusions & Design Constraints

The following exclusions and constraints have been considered in the architecture process. These use-cases are supported. They have not been tested within the Cloud Packet Core solution due to time constraints.

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| ID | Section | Descriptions |
|---|---|---|
| EX-01 | | |
| EX-02 | | |
| EX-03 | | |
| EX-04 | | |
| | | |
| | | |
| | | |
| | | |

**Table 2: Exclusions**

### 1.4.3 Project Open Items

This section provides visibility on the design topics pending closure during the document's lifecycle.

| ID | Section | Descriptions |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

**Table 3: Open Points**

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# 1.5 References

## 1.5.1 Project documents

| Nº | Reference | Title |
|---|---|---|
| [R1] | VF_Ethiopia_vEPC_Technical-Solution Description_NOKIA_ed2 | Technical_Solution_Description |
| [R2] | Safaricom Ethiopia VNF CNF BoM - July 2021 v1.6 | vBOM |
| [R3] | | |
| [R4] | | |
| [R5] | | |
| [R6] | | |
| [R7] | | |
| [R8] | | |
| [R9] | | |
| [R10] | | |
| [R11] | | |
| [R12] | | |

**Table 4: Project Documents**

## 1.5.2 3GPP References

Specify 3GPP TS relevant for this implementation, identifying the release compliance. References bellow are given as an example of the syntax and information that should be included when filling this table.

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| Nº | Reference | Title |
|---|---|---|
| [R13] | TS 36.413 R11.2.0 | Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP) |
| [R14] | TS 23.401. R11.8.0 | GPRS Enhancements for E-UTRAN Access |
| [R15] | TS 24.301 R10.5.0 | Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS); Stage 3 |
| [R16] | TS 29.274 R11.9.0 | 3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunnelling Protocol for Control plane (GTPv2-C); Stage 3 |
| [R17] | TS 29.281 R10.3 | General Packet Radio System (GPRS) Tunnelling Protocol User Plane (GTPv1-U) |
| [R18] | TS 32.299 R11.10.0 | Telecommunication management; Charging management; Diameter charging applications |
| [R19] | TS 29.061 R10 | Interworking between the Public Land Mobile Network (PLMN) supporting packet based services and Packet Data Networks (PDN) |
| [R20] | TS 23.501 | System Architecture for the 5G System |
| [R21] | TS 23.799 | Study on Architecture for Next Generation System |
| [R22] | TS 28.500 | Telecommunication management; management concept, architecture and requirements for mobile networks that include virtualized network function. |
| [R23] | TS 28.801 | Study on management and orchestration of network slicing for next generation network |
| [R24] | TS 37.340 | E-UTRA and NR Multi-Connectivity |
| [R25] | TS 23.214 | Architecture enhancements for control user plane separation of EPC nodes |
| [R26] | TS 29.244 | Interface between the Control Plane and the User Plane of EPC Nodes |
| [R27] | TS 23.007 | Restoration Procedures |
| [R28] | TS 29.303 | DNS procedures for UP function selection |
| [R29] | TS 23.060 | General Packet Radio Service (GPRS); Service description; Stage 2 |
| [R30] | TS 23.236 | Intra-domain connection of Radio Access Network (RAN) nodes to multiple Core Network (CN) nodes |

**Table 5: 3GPP References**

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# 2. Solution overview

## 2.1 Target Solution, Elements and Sites

The target of this opportunity is to provide Safaricom with a Cloud Packet Core Network based on Nokia CMM and CMG, with intention of providing LTE Data, 5G NSA, VoLTE and VoWIFI. The CPC will be installed on a Safaricom managed VMWare Cloud Infra, the VMWare will be NFV3.2.1

| Network Element | Version | Functionality |
|---|---|---|
| Nokia CMM | 21.8 | MME/SGSN |
| Nokia CMG | 21.8 | S/P-GW, Gi Firewall, CGNAT |
| Nokia CMG | 21.8 | ePDG, TWAG |

Table 6: Target Solution and Elements

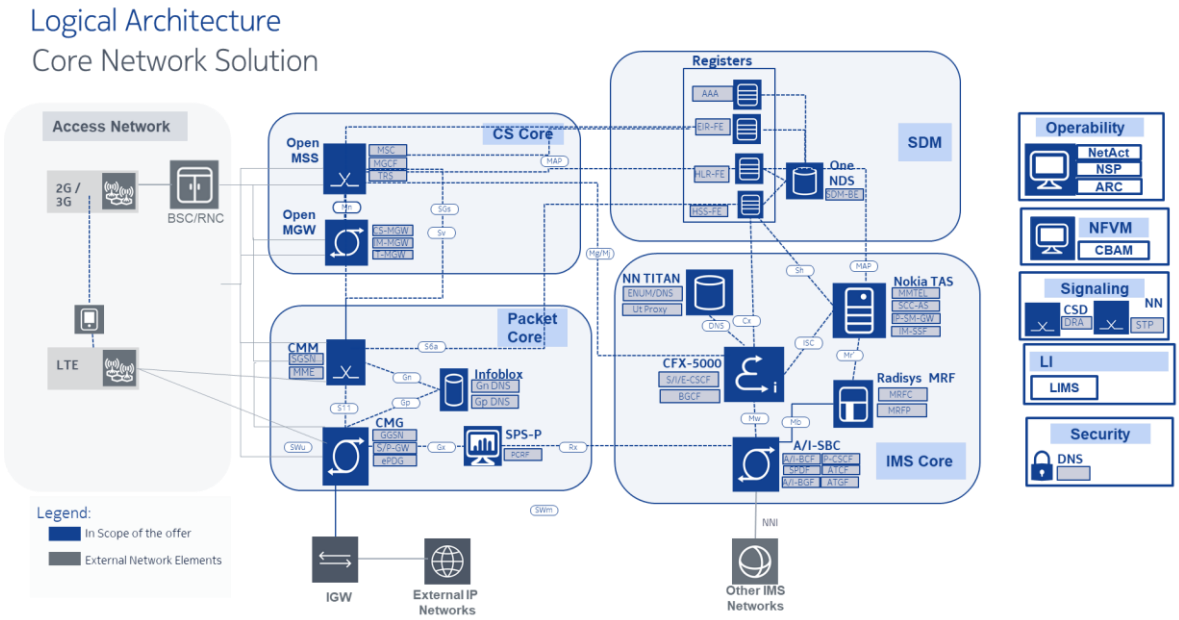



Figure 1: Core Network Solution

Officially, Nokia will deploy in the 1st 8 datacentres below

| Site Number | Site Name (Names to be corrected) | Rollout Priority | Rollout Year | Type |
|---|---|---|---|---|
| 1 | Addis Ababa | 1 | 0/1 | Central |
| 2 | South West Shewa | 1 | 0/1 | Central |
| 3 | West Gojjam | 2 | 2 | Central |
| 4 | East Gojjam | 2 | 2 | Central |

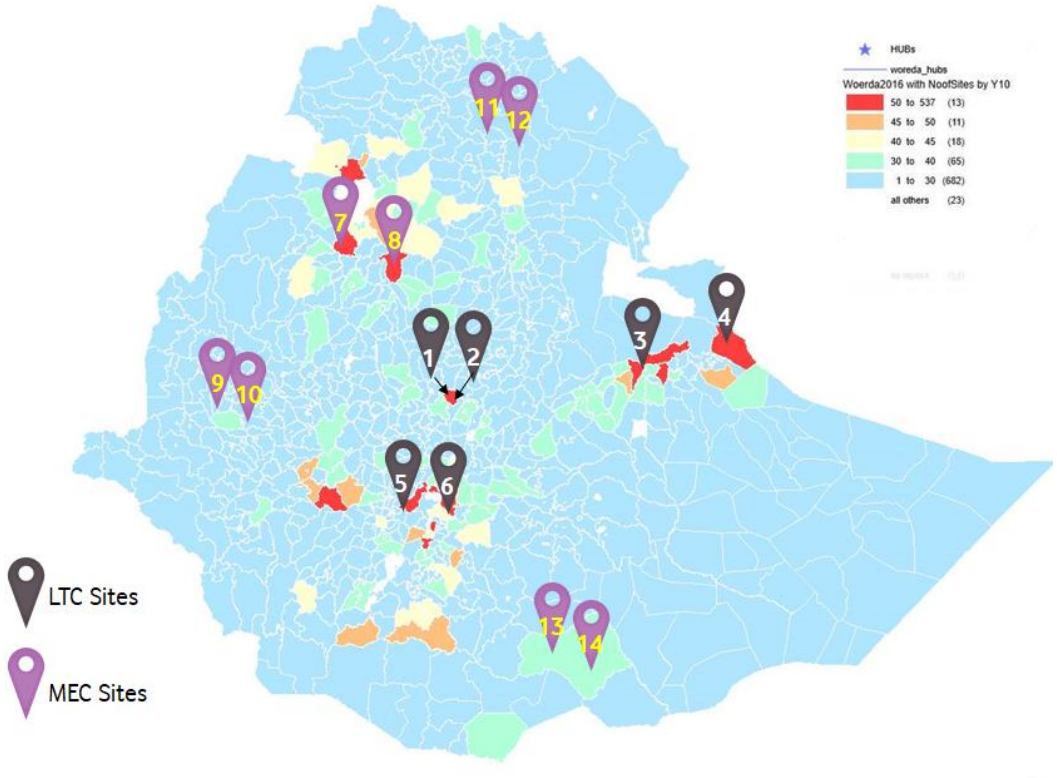
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| | | | | |
|---|---|---|---|---|
| 5 | Dire Dawa | 3 | 3 | Central |
| 6 | Jigjiga | 3 | 3 | Central |
| 7 | Borena | 4 | 4 | Regional |
| 8 | Afder | 4 | 4 | Regional |
| 9 | Kamashi | - | 5 | Regional |
| 10 | East Wellega | - | 5 | Regional |
| 11 | South Tigray | - | 6 | Regional |
| 12 | North Wello | - | 6 | Regional |
| 13 | Oromia | - | 7 | Regional |
| 14 | Zone 5 | - | 7 | Regional |

Table 7: Sites and the nodes

- All data centres (core sites) will be deployed in pairs with 1+1 redundancy. Each pair of core sites is dedicated to a specific geographical area.
- The control plane VNFs will be deployed at the central data centres and user plane VNFs deployment will be extended across the regional data centres.






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Figure 2: Core Data Centres

Below is the core network element and the VNF distribution

| Site Number | VNFs Deployed |
|---|---|
| 1'-3 | **SDM** (SDL, HSS, HLR, EIR, AAA), **VoLTE** (CSCF, TAS, A/I-SBC, PCRF, MRF, ENUM, Ut Proxy, DRA), **CS** (MSS, MGW, STP), **EPC** (CMM (MME/SGSN), CMG-C (GGSN/SPGW), CMG-U (GGSN/SPGW), ePDG) |
| 3'-6 | **VoLTE** (CSCF, TAS, A/I-SBC, PCRF, MRF, ENUM, Ut Proxy, DRA), **CS** (MSS, MGW, STP), **EPC** (CMM (MME/SGSN), CMG-C (GGSN/SPGW), CMG-U (GGSN/SPGW), ePDG) |
| 7'-8 | **VoLTE** (A/I-SBC), **CS** (MGW), **EPC** (CMG-U (GGSN/SPGW)) |

Table 8: Nodes per Site

Since all the sites above aren't ready, the project has a temporary approach, below is the plan that the project has in deploying the various nodes for a first call, in the table below Temp site1 is the priority.

| Functionality group | Network Element | Release | Functionality | Platform | Testbed | No. NE - Temp Site 1 | No. NE - Temp Site 2 | No. NE - Perm Site 1 | No. NE - Perm Site 2 |
|---|---|---|---|---|---|---|---|---|---|
| Packet Core | Nokia CMM | 21.8 | MME/SGSN | VNF | 2 | 1 | 1 | 6 | 6 |
| Packet Core | Nokia CMG | 21.8 | CMG-U | VNF | 2 | 1 | 1 | 2 | 2 |
| Packet Core | Nokia CMG | 21.8 | CMG-C | VNF | 2 | 1 | 1 | 2 | 2 |
| Packet Core | Nokia CMG | 21.8 | CMG-SSG | VNF | 2 | 1 | 1 | 2 | 2 |

## 2.2 3GPP Logical interfaces

Bellow follows the list of logical interfaces implemented under the scope of the current CPC deployment and their sate of compliancy with 3GPP standards:

| Interface | Protocol | Related NE | Standard compliancy | 3GPP Release |
|---|---|---|---|---|
| Gb | BSSGP | BSC-SGSN | 3GPP TS 48.018 | Rel 12.7.0 |
| Iu | RANAP | RNC-SGSN | 3GPP TS 25.411 | Rel 12.7.0 |
| S1-MME | S1-AP | eNodeB-MME | 3GPP TS 36.413, RFC 2960, 3GPP TS 24.301 | Rel 12.7.0 |
| S1-U | GTPv2/v1 | EnodeB – SGW | 3GPP TS 23.401, 23.007 | Rel 13.9.0 |
| S5 | GTPv2/v1 | SGW – PGW | 3GPP TS 29.060, 29.274, 3GPP TS 29.281 | Rel 13.8.0, Rel 12.7.0 |
| S6a | Diameter | MME-HSS | 3GPP TS 29.272 | Rel 12.7.0 |
| S8 | GTPv2/v1 | SGW – PGW | 3GPP TS 29.060, 29.274, 3GPP TS 29.281 | Rel 13.8.0, Rel 12.7.0 |
| S10 | GTPv2 | MME-MME | 3GPP TS 29.274 | Rel 13 |
| S11 | GTPv2 | MME-SGW | 3GPP TS 29.274 | Rel 13 |
| S13 | Diameter | MME-EIR | 3GPP TS 29.272 | Rel 12.7.0 |

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| Interface | Protocol | Related NE | Standard compliancy | 3GPP Release |
|---|---|---|---|---|
| Gn | GTPv1 | SGSN-GGSN | 3GPP TS 29.060 | 13.6.0 (2016-12) |
| Sxa | PFCP/GTP | SGW_UP-CP | TS29.244 | Rel 14 |
| Sxb | PFCP/GTP | PGW_UP-CP | TS29.244 | Rel 14 |
| SGi | IP | PGW | 3GPP TS 29.061 | n/a |
| Gx | Diameter | PGW-PCRF | 3GPP TS23.203/23.401/29.212/29.213 | Rel 13.10/13.9/13.8/13.8 |
| Ga | GTPv2 | PGW-CGF | 3GPP TS 32.295 | Rel 11.0 |
| Bp(SAEGW) | sFTP | SGW/PGW-OFCS | 3GPP TS 32.240/32.297 | Rel 13.2 |
| Bp(SGSN) | sFTP | SGW/PGW-OFCS | 3GPP TS 32.240/32.297 | Rel 8 |

Table 9: Logical interfaces and 3GPP compliancy

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# 3. Design and Dimensioning Considerations

## 3.1 CMG Design Assumptions

### 3.1.1 CMG – GGSN/SPGW-c/SPGW-u

- Nokia CMG offered as GGSN/S_PGW-C on 2(1+1) Data Centers with Geo Redundancy and GGSN/S_PGW-U on 2 (1+1) Data Centers with Geo Redundancy
- 100% DPI traffic is considered
- 70% of the traffic is considered for TCP Optimization
- 5% of the total throughput traffic is considered for Corporate APN
- Dimensioning considers 70% CPU load
- Dimensioning considers hyperthreading OFF, HT = OFF At VM Bios in-case HT = ON at host level
- Software licenses proposed based on the number of subscribers
- DPI and TCP-O functions integrated into the CMG VNF
- Packet size: 650 B
- CUPs (Control & User Plane Separation) deployment model proposed
- CMG will be managed by Nokia Network Services Platform (NSP)

### 3.1.2 CMG – SSG (CGNAT & GiFW) & ePDG/TWAG

- Nokia CMG offered as SSG (Service Steering Gateway) for Gi Firewall/CG NAT functionality on 2 Data Centers with Geo Redundancy
- CMG also offered as ePDG and TWAG on 2 Data Centers with Geo Redundancy. Excluded from Temp sites
- Dimensioning considers 70% CPU load
- Software licenses proposed based on the number of subscribers.
- SSG & CMG offered as CMG VNF.

## 3.2 CMM Design Assumptions

- Nokia CMM as SGSN/MME is offered on 2 (1+1) Data Centers with Geo Redundancy
- Nokia CMM is offered as SGSN/MME to support 2G, 3G, 4G, 5G NSA
- CMM dimensioning considers 70% CPU load
- Dimensioning considers 90% Direct Tunnelling traffic
- Dimensioning considers hyperthreading as ON
- Software licenses proposed based on the number of subscribers each year
- MM will be managed by Nokia Network Services Platform (NSP)

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# 3.3 Projected Traffic Profile

For Temp sites, dimensioning is redone to account for 1.5M subscribers.

- Subscriber Count:

| Number Subscribers | Temporary Site | Permanent Site 1 / 2 |
|---|---|---|
| Total subs | 5 000 000 | 10 000 000 |
| VoLTE - Active | 1 000 000 | 2 000 000 |
| ~~VoWiFi - Active~~ | ~~100 000~~ | ~~200 000~~ |

- Dimensioning (Including **20% extra buffer**)

| | Temporary Site | Permanent Site 1 / 2 |
|---|---|---|
| 2G/3G Attached Users - SAU | 4 260 000 | 8 520 000 |
| 4G/5G Active Users - SAAU | 1 740 000 | 3 480 000 |
| **Total Users** | **6 000 000** | **12 000 000** |
| Tput 2G/3G BH UL+DL[Gbps] | 14 | 27 |
| Tput 4G BH UL+DL[Gbps] | 32 | 63 |
| **Total Tput BH UL+DL[Gbps]** | **46** | **90** |
| 2G/3G PDP Context | 3 973 485 | 7 946 970 |
| 4G PDP Context | 3 485 157 | 6 970 315 |
| **Total PDP Context** | **7 458 643** | **14 917 285** |

| | 6 000 000 | 12 000 000 |
|---|---|---|
| Split (2G SAU) | 31% | 31% |
| 3G SAU | 40% | 40% |
| 4G SAU | 29% | 29% |

| | | |
|---|---|---|
| EPC 2G+3G SAU | 4 260 000 | 8 520 000 |
| 2G SAU | 1 860 000 | 3 720 000 |
| 3G SAU | 2 400 000 | 4 800 000 |
| EPC 4G SAU | 1 740 000 | 3 480 000 |
| Total SAU | 6 000 000 | 12 000 000 |
| | 30.00% | 70.00% |
| 2G+3G TH (Gbps) | 14 | 27 |
| 2G TH | 4.34 | 8.37 |
| 2G th/sub(kbs) | 2.33 | 2.25 |
| 3G TH | 9.66 | 18.63 |
| 3G TH/SUB(Kbps) | 4.03 | 3.88 |
| 4G TH | 32 | 63 |
| total TH | 46 | 90 |

| | | |
|---|---|---|
| 2G+3G PDP | 3 973 485 | 7 946 970 |
| 2G PDP | 1 231 780 | 2 463 561 |
| 3G PDP | 2 741 705 | 5 483 410 |
| 4G PDP | 3 485 157 | 6 970 315 |
| total PDP | 7 458 643 | 14 917 285 |

Table 10: Projected Traffic Profile

# 3.4 CMM and CMG Dimensioning

Temp1 site

| VNF/CNF Number | VNFC Number | VNF Component Description (E.g. GTP-C Session Management, Charging, Data Plane Forwarding, GTP-U Data Plane, Encryption, L7 Firewall, etc.) | Vendor Product Name | Affinity/Antiaffinity | VNF type (management, control, data planes) | I/O acceleration (SR-IOV, DPDK, None) | Nnumber of VMs (including HA) | VM size (vCPU) | CPU Pinning (Yes/No) | Total vCPU | VM Memory size (GB) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Totals | | | | 11,968 | |
| VNF 27 | VNFC 27.1 | MGVM | CMG -U | Antiaffinity | Data | SR-IOV | 16 | 8 | YES | 128 | 64 |
| VNF 27 | VNFC 27.2 | LBVM | CMG -U | Antiaffinity | Data | SR-IOV | 2 | 6 | YES | 12 | 24 |
| VNF 27 | VNFC 27.3 | OAM VM | CMG -U | Antiaffinity | Data | SR-IOV | 2 | 8 | YES | 16 | 24 |
| VNF 27 | VNFC 27.4 | DBVM | CMG -U | Antiaffinity | Data | None | 1 | 8 | YES | 8 | 64 |
| VNF 27 | VNFC 27.5 | VNF Template / Catalog | CMG-U | | | | 1 | | | | |
| VNF 28 | VNFC 28.1 | MGVM | CMG -C | Antiaffinity | Control | SR-IOV | 11 | 8 | YES | 88 | 64 |
| VNF 28 | VNFC 28.2 | LBVM | CMG -C | Antiaffinity | Control | SR-IOV | 2 | 6 | YES | 12 | 24 |
| VNF 28 | VNFC 28.3 | OAM VM | CMG -C | Antiaffinity | Control | SR-IOV | 2 | 8 | YES | 16 | 24 |
| VNF 28 | VNFC 28.4 | DBVM | CMG -C | Antiaffinity | Control | None | 1 | 8 | YES | 8 | 64 |
| VNF 28 | VNFC 28.5 | VNF Template / Catalog | CMG -C | | | | 1 | | | | |
| VNF 29 | VNFC 29.1 | MGVM | CMG-ePDG/TWAG | Antiaffinity | Data | SR-IOV | 6 | 8 | YES | 48 | 64 |
| VNF 29 | VNFC 29.2 | LBVM | CMG-ePDG/TWAG | Antiaffinity | Data | SR-IOV | 2 | 6 | YES | 12 | 24 |
| VNF 29 | VNFC 29.3 | OAMVM | CMG-ePDG/TWAG | Antiaffinity | Data | SR-IOV | 2 | 8 | YES | 16 | 24 |
| VNF 29 | VNFC 29.4 | VNF Template / Catalog | CMG-ePDG/TWAG | | | | 1 | | | | |
| VNF 30 | VNFC 30.1 | NECC | CMM | Antiaffinity | Control | None | 3 | 16 | NO | 48 | 48 |
| VNF 30 | VNFC 30.2 | IPDS | CMM | Antiaffinity | Control | None | 2 | 20 | NO | 40 | 52 |
| VNF 30 | VNFC 30.3 | DBS | CMM | Antiaffinity | Control | None | 3 | 8 | NO | 24 | 28 |
| VNF 30 | VNFC 30.4 | 4G-CPPS | CMM | Antiaffinity | Control | None | 5 | 10 | NO | 50 | 28 |
| VNF 30 | VNFC 30.5 | PAPS | CMM | Antiaffinity | Control | None | 19 | 10 | NO | 190 | 24 |
| VNF 30 | VNFC 30.6 | IPPS | CMM | Antiaffinity | Control | None | 2 | 10 | NO | 20 | 32 |
| VNF 30 | VNFC 30.7 | VNF Template / Catalog | CMM | | | | | | | | |

Table 11: CMM and CMG Dimensioning

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# 4. Telecom Use-Cases

The Cloud Packet Core as an EPC solution can be designed to support multiple types of access technologies and services. This release targets the following use-cases for validation as a solution.

## 4.1 Evolved Packet Core – LTE Data

The Nokia Cloud Packet Core (CPC) base capability complies with the 3GPP defined Evolved Packet Core (EPC) set of functions required for the support of LTE Radio Network Access. Under this use-case the set of network function provided by CPC under the scope of the current deployment are the following:

The Mobility Management Entity (MME) - TS 23.401 4.4.2, which deals with the control plane and manages User Equipment (UE) registration, authentication, and mobility. In the EPC architecture the MME is responsible for the tracking and the paging of UE in idle-mode and is the termination point of the Non-Access Stratum (NAS).

The Serving GW (S-GW) - TS 23.401 4.4.3.2, which terminates the user plane interface towards E-UTRAN. It connects to the MME via S11 interface and to the eNodeB via the S1-U interface. The SGW function routes and forwards user data packets. It acts as the mobility anchor for the user plane during Inter-eNodeB handovers. The SGW may also serve as the anchor for mobility between LTE and other 3GPP technologies.

The PDN GW (P-GW) - TS 23.401 4.4.3.3, which in turn, serves as the link between the mobile device and the services that reside in an external packet network, such as the Internet. It provides connectivity to the UE to external data networks by being the point of entry and exit of traffic for the UE. The PGW is also responsible for several IP functions, such as address allocation, policy enforcement, packet screening and routing, and lawful interception. It supports per-user based packet filtering such as deep packet inspection. The PGW supports Flow Based Charging under control of the PCRF. The PGW serves as the enforcement point for the policy decisions coming from the PCRF.

The Policy and Charging Rules Function (PCRF) – TS 23.203, functioning as the policy and charging control element it supports service data flow detection, policy enforcement and flow-based charging. It interfaces with the PGW through Gx to manage the service policy and sends QoS setting information for each user session and accounting rule information.

The Diameter Agent (DA) – which acts as both Diameter Edge Agent (DEA) and Diameter Routing Agent (DRA), functioning as a focal point for all diameter-based interfaces traversing the Evolved Packet Core. The DRA/DEA may take specific actions on Diameter flows such as: topology hiding for S6a interface in roaming scenarios (DEA) or session “stickiness” for Gx guaranteeing that different sessions relating to the same subscriber are managed by the same PCRF.

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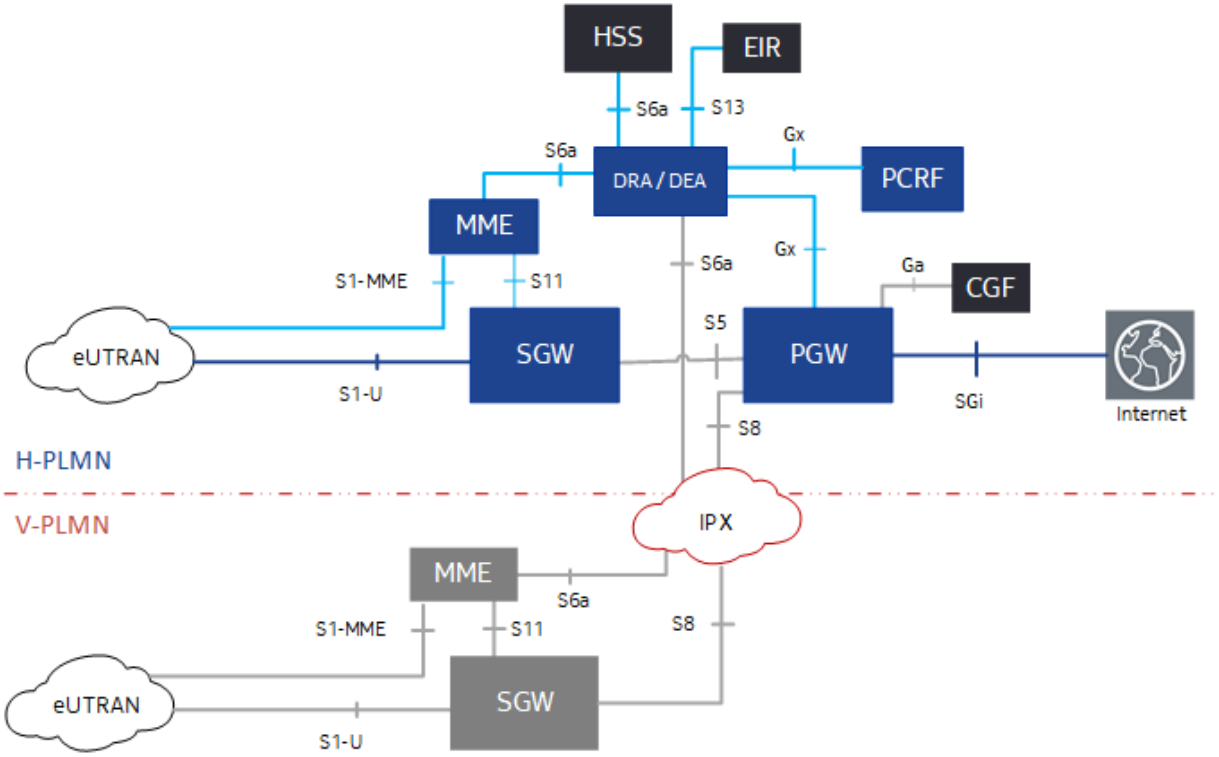



Figure 3: EPC functional elements in the support of greenfield LTE data

In the addition to the delivery of these EPC network elements and corresponding functional interfaces mentioned above, the enablement and integration of the following cloud external facing interfaces is also considered under this scope:

- S6a / S13 – for standard compliant HSS and EIR integration
- S1-MME / S1-U – Standard compliant eNodeB integration
- Ga / Bp – For basic CDR generation

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## 4.2 Support for 5G Non-Standalone (NSA) Option 3X

5G is the new generation of radio systems and a new network architecture, delivering extreme broadband and ultra-robust, low latency connectivity, and massive networking. In the initial phases, extreme Mobile Broadband (eMBB) is expected to be the main business use-case and driver for 5G introduction. Later, use-cases and applications for ultra-reliable low latency communications (URLCC) and massive IoT will follow.

3GPP is introducing several architecture options for 5G, which can be divided into Non-Standalone (NSA) and Standalone (SA) option families. In NSA options, 5G radio is complemented with LTE radio and served by 4G core (Option 3X) or 5G core (Option 7). In SA Option 2, 5G radio is served by 5G core. The figure below outlines the foreseen relevant 3GPP architecture options for Nokia E2E 5G Solution at generic level.

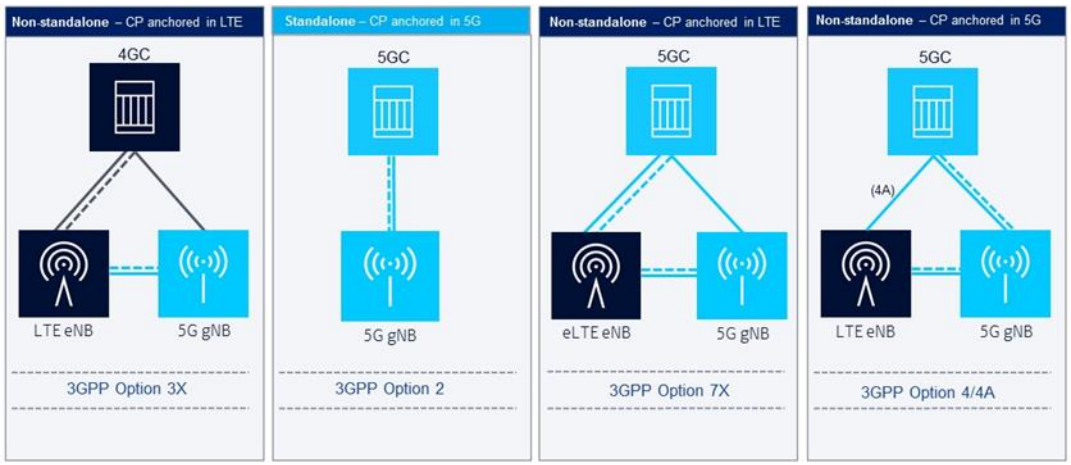



Figure 4: 3GPP 5G architecture options for Nokia E2E 5G Solution

The selected option from a single operator viewpoint naturally depends on the operator specific strategy for introducing 5G. The consensus shared between standardization bodies, is that Option 3X and Option 2 are expected to be implemented first in the operator networks, followed by Option 7 and Option 4 in later phases. Therefore, Nokia E2E 5G Solution is also introduced in phases, starting from Option 3X defined in 3GPP Rel-15.

The current Cloud Packet Core implementation will exercise 5G NSA Option 3X.

1. The major advantage of Option 3X architecture is its ability to support LTE-5G Dual Connectivity service over existing LTE service:
    - Continue to support existing LTE services (e.g. VoLTE and regulatory services)
    - Additional dual connectivity with 5G access allows higher bandwidth services
2. For initial deployment, Internet and enterprise services are candidates for utilization of LTE and 5G access
    - Internet services can include Internet Access to public domain (ISP Service) and hosted content access

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NSA Option 3X requires Dual Connectivity capabilities to be supported by RAN elements. Dual Connectivity enables data to be routed directly to the NR gNB to avoid excessive user plane load on the existing LTE eNB, which was designed for 4G LTE traffic load and not additional NR traffic load. As service continuity after the loss of 5G radio coverage is more graceful in this variant, it also minimizes excessive signaling traffic between RAN and core.

NSA Option 3X defines a "Master Cell Group (MCG) bearer" and "Secondary Cell Group (SCG) bearer" (see TS 36.300) concept. For E-RABs configured as "MCG bearers" the U-plane termination points are maintained, while E-RABs configured as "SCG bearers" can change S1-U termination point in the E-UTRAN by means of S1-MME signaling without changing the S1-MME termination point. 5G NSA Option 3X is realized through the "split bearer" alternative of the Dual Connectivity feature set. The "split bearer" in the E-UTRAN is transparent to the core network entities (e.g. MME, S-GW etc.).

For option 3X, the involved core network entity is the MME. S1-MME is terminated in MN and the MN and the SN are interconnected via X2-C for Control Plane purpose.

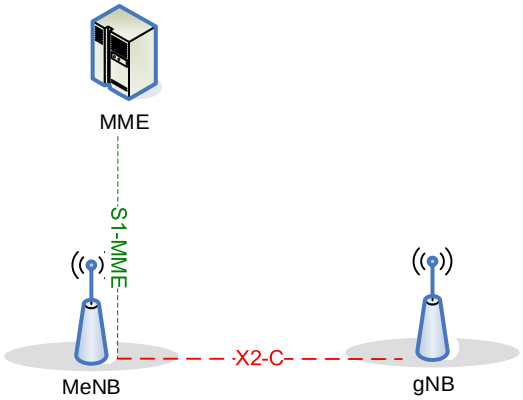



Figure 5: S1 CP: Option 3X deployment

The eNB, at which the S1-MME terminates, performs all necessary S1-MME related functions for both Master and Secondary Nodes such as mobility management, relaying of NAS signaling, E-RAB handling, etc. and manages the handling of user plane connection of S1-U.

There are different U-plane connectivity options of the MN and SN involved in MR-DC for a certain UE. The U-plane connectivity depends on the bearer option configured:

- The user plane connection to the CN entity is terminated in the MN.
- The user plane connection to the CN entity is terminated in the SN.
- The Transport of user plane data over the Uu either involves MCG or SCG radio resources or both.
  - For MCG bearers, only MCG radio resources are involved;
  - For SCG bearers, only SCG radio resources are involved;
  - For split bearers, both MCG and SCG radio resources are involved.
- For split bearers, MN terminated SCG bearers and SN terminated MCG bearers, PDCP data is transferred between the MN and the SN via the MN-SN user plane interface.

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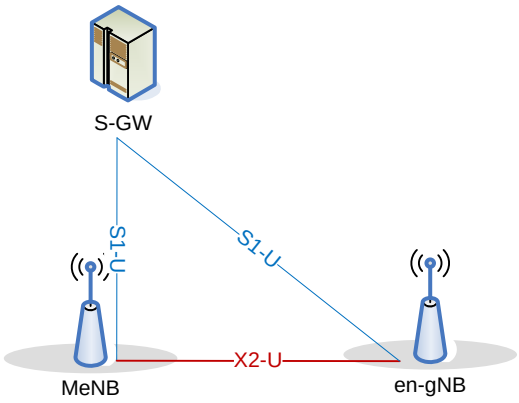



Figure 6: U-Plane connectivity for EN-DC

Under the scope of the CPC BP lab deployment, scenario with split bearer is considered as the most interested so to be validated.

5G NSA is not included in Temp sites.

## 4.3 Control and User Plane Separation (CUPS)

For the majority of mobile operators, user data traffic has been doubling on an annual basis in recent years. The reasons for this growth in traffic are the rapidly increasing use of smart devices, the proliferation of video and other applications that they support and the use of USB modem dongles & personal hotspots using 3GPP networks.

As the penetration of these terminals increases worldwide and the interest in content-rich multi-media services (e.g. OTT video streaming services, person to person video, content sharing) rises, this trend of rapidly increasing data traffic is expected to continue and accelerate. At the same time, there is a strong consumer demand for user experience improvements, with lower latency being one of the critical KPIs to be met on the way.

Separating User Plane and Control Plane functions in the Mobile Packet Core design addresses this problem by providing an architecture that allows:

- Reducing latency on application service, e.g. by selecting user plane nodes which are closer to the RAN or more appropriate for the intended UE usage type without increasing the number of control plane nodes.
- Locating and scaling the CP and UP resources of the EPC nodes independently.
- Independent evolution of the CP and UP functions.
- Enabling Software Defined Networking to deliver user plane data more efficiently.

Control and User Plane Separation (CUPS) provides the architecture enhancements for the separation of the functionality necessary to handle signaling vs user generated data in the Evolved Packet Core's SGW and PGW network elements. This grants additional flexibility in the way the network is deployed and operated, by facilitating distributed (as opposing to centralized) deployment models and the independent scaling between control plane and user plane functions. Separating core functions into their Control-Plan and User-Plane counterparts, shall be performed while the functionality of the existing nodes subject to this split is not affected.

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Control and User Plane Separation (CUPS) design principled shall be applied under the scope of this deployment for S-GW and PDN-GW network functions, where the following high-level principles shall be obeyed:

- A legacy SGW and PGW can be replaced by a split node without effecting connected legacy nodes.
- A CP function can interface multiple UP functions, and a UP function can be shared by multiple CP functions.
- A UE is served by a single SGW-CP but multiple SGW-UPs can be selected for different PDN connections. A user plane data packet may traverse multiple UP functions.
- The CP function controls the processing of the packets in the UP function by provisioning a set of rules in Sx sessions, i.e. Packet Detection Rules for packets inspection, Forwarding Action Rules for packets handling (e.g. forward, duplicate, buffer, drop), Qos Enforcement Rules to enforce QoS policing on the packets and Usage Reporting Rules for measuring the traffic usage.
- All the 3GPP features impacting the UP function (PCC, Charging, Lawful Interception, etc) are supported, while the UP function is designed as much as possible 3GPP agnostic. For example, the UPF is not aware of bearer concept.
- Charging and Usage Monitoring are supported by instructing the UP function to measure and report traffic usage, using Usage Reporting Rule(s). No impact is expected to OFCS, OCS and the PCRF.
- The CP or UP function is responsible for GTP-u F-TEID allocation.
- All legacy Control Plane protocols such as GTP-C, Diameter (Gx, Gy, Gz) shall be terminated in CP functions alone.

### 4.3.1 CUPS Sx interface overview

The architecture reference model in the case of separation between control plane and user plane for S-GW and PDN-GW functional elements is depicted in Figure **7**.

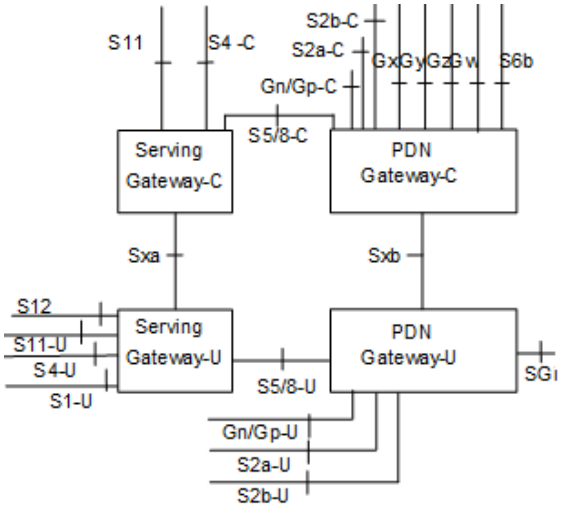



Figure 7: SGW / PGW CUPS architecture and logical interface

The architecture reference model above covers non-roaming as well as home routed and local breakout roaming scenarios. The logical interfaces added to the reference points defined in TS 23.401, TS 23.402 and TS 23.203 for the support of CUPS are the following:

- Sxa: Reference point between SGW-C and SGW-U.

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- Sxb: Reference point between PGW-C and PGW-U.

As depicted, the Sx interface is the interface between the control plane and the user plane. There is the Sxa between SGW-CP/UP and the Sxb PGW-CP/UP. The Sx interface can further be separated into its control plane and user plane functions, however usage of Sx for User Plane forwarding is not considered under the current scope of design.

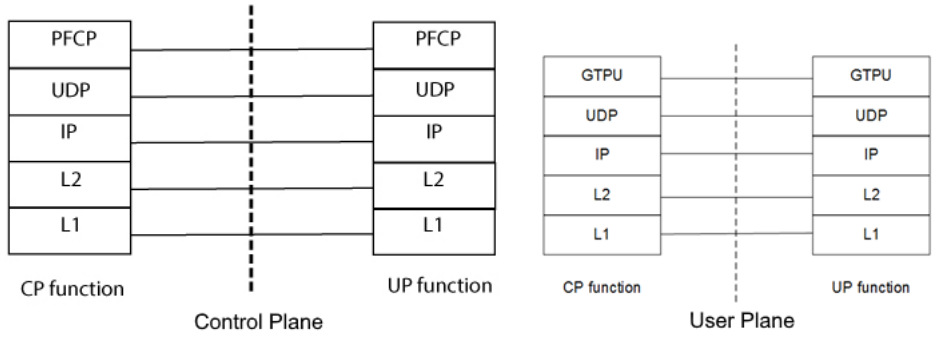



Figure 8: Sx Protocol Stack

For control purposes Sx uses PFCP protocol running on top of UDP/IP. The UDP destination port for a Request message uses port 8805. PFCP is a binary, TLV encoded protocol and is design similarly to GTP-C protocol (e.g. message encoding, session endpoint identification, delivery of messages, load & overload control).

The Sx may also make use of a user plane component running over GTP-U/UDP; it is used for LI and when Downlink Data.

For each subscriber's individual PDN connection there is a corresponding session over Sx interfaces. These sessions are defined around procedures and rules.

### 4.3.2 Sx procedures

The Sx node association procedures are used to setup, update, and release an Sx association between a CP function and a UP function.

The Sx procedures are the following:

- **Establishment** Procedure, to establish an Sx session.
  - CP function provides a new F-SEID (Fully qualified session endpoint identifier) identifying the Sx session in the CP function and the Rules to be created.
  - UP function returns its own F-SEID (identifier of the session in the UP function).

- **Modification** Procedure, to modify an Sx Session (e.g. add, modify, release Rules)
- **Deletion** Procedure, to delete an Sx session
- **Report** procedure, to report information related to a particular Sx session:
  - Downlink Data Report, to report arrival of a DL user plane packet
  - Usage Report, to report traffic usage or specific events (e.g., start of an application)
  - Error Indication Report, to report an Error Indication received from the GTP-U peer
  - User Plane Inactivity Report, to report an Sx session w/o user plane activity

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#### 4.3.2.1 Sx rules

Once CP selects the UP function anchoring the user data and the session is established, the CP Function controls the UP function by provisioning (adding, modifying, deleting) Rules:

The Sx rules are the following:

- **Packet Detection Rule (PDR)**, with Packet Detection Information (PDI) describing what packets should receive a certain treatment (e.g. forwarding and other types of enforcement).
- **Forwarding Action Rule (FAR)**, with information on whether (and how) to forward, drop, buffer (with or without notifying the CP function) or duplicate (interception) packets.
- **Usage Reporting Rule (URR)**, with information defining a certain measurement to perform in the UP function and how it shall be reported.
- **QoS Enforcement Rule (QER)**, with information related to QoS enforcement of traffic.
- **Buffering Action Rule (BAR)**, with information on how to buffer packets in the UP function.

One Sx session shall have at least one **PDR** and at least one **FAR**
One Sx session may have URRs and **QERs** and may have 1 **BAR**.

On receipt of a UP packet, the UP function first looks up for the Sx session to which the packet corresponds, and then find the first **PDR** matching the incoming packet among all PDRs provisioned for the Sx session, in decreasing order of PDR's precedence.

The UP function then applies the processing defined by the FAR, URR(s), QER(s), BAR associated to this PDR.

## 4.4 IPv6 support on User Plane

The increasing popularity of mobile devices such as smartphones and tablets enables a great number of users to enjoy cellular data services. This new wave of growth together with improvements in connectivity provided by 4G and 5G technologies have been observed to increase Internet usage via mobile devices, along with demand for Internet Protocol (IP) addresses. This need has been even more pressing when we consider emergence of Internet of Things (IOT) and Machine 2 Machine (M2M) use-cases, which pose connectivity environments that need to cater to millions of devices.

As such, mobile network operators need to make strategic decisions to sustain business growth by maximizing their IPv4 holdings, and as a long-term solution; plan the transition to IPv6 enabled production services to cope with the increasing demand for IP addresses.

IPv6 has many features that make it significantly more powerful than its predecessor, a summary of which is provided in Table **12**. These features include extended address space, autoconfiguration, header structure, extension headers, IPsec, mobility, quality of service, route aggregation, and efficient transmission.

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| Property | IPv4 | IPv6 |
|---|---|---|
| **Address size and network size** | 32 bits, network size 8-30 bits | 128 bits, network size 64 bits |
| **Packet header size** | 20-60 bytes | 40 bytes |
| **Header-level extension** | limited number of small IP options | unlimited number of IPv6 extension headers |
| **Fragmentation** | sender or any intermediate router allowed to fragment | only sender may fragment |
| **Control protocols** | mixture of non-IP (ARP), ICMP, and other protocols | all control protocols based on ICMPv6 |
| **Minimum allowed MTU** | 576 bytes | 1280 bytes |
| **Path MTU discovery** | optional, not widely used | strongly recommended |
| **Address assignment** | usually one address per host | usually multiple addresses per interface |
| **Address types** | use of unicast, multicast, and broadcast address types | broadcast addressing no longer used, use of unicast, multicast and anycast address types |

Table 12: Summary of Differences between IPv4 and IPv6

IPv6 provides a long-term, scalable solution with fewer operational and maintenance issues than IPv4 networks deployed in NAT (Network Address Translator) environments.

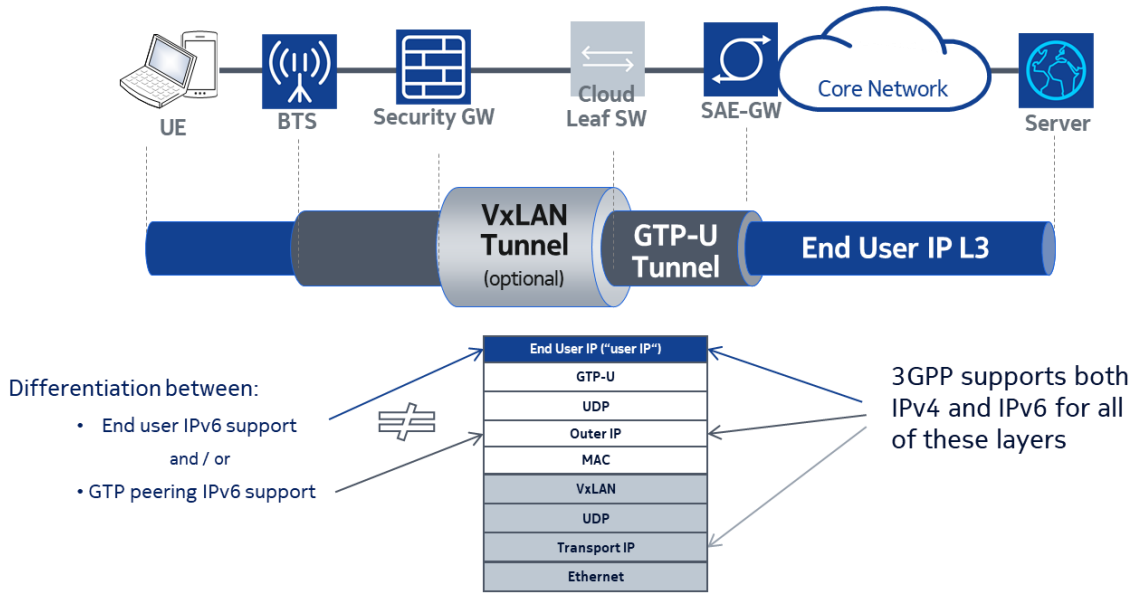



Figure 9: IPv6 adoption on the different IP layers of CPC

In what concerns IPv6 adoption in the Cloud Packet Core ecosystem, three distinct scopes of IPv6 support may be pursued:

- User Plane - IPv6 is supported at UE IP-CAN bearer. PGW is responsible to assign an IPv6 address to Subscriber UE thus enabling access to IPv6 content on the SGI facing PDN.
- Transport – Logical interfaces support IPv6, peering between 3GPP NEs is made using IPv6 subnets and related protocol variants

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- Underlay – Where the NFVI stack itself uses IPv6 for networking between its different components

**The scope of the current deployment only targets IPv6 support at User Plane level,** the focus is thus providing the subscriber devices and applications access to IPv6 capabilities.

### 4.4.1 IPv4v6 Native Dual Stack

Under this scope the transition to IPv6 will be introduced by providing both IPv4 and IPv6 stacks within the same PDP Context/Bearer in parallel. The protocol used to establish a connection is determined at UE or at application level, depending what type of service is required and which type of network capabilities are available. This technology is known as native Dual stack or Dual IP layer; it does not require any change to the packet header throughout the end-to-end connectivity path and avoids making use of encapsulation between IPv4 and IPv6 like in other alternatives.

The following considerations make the Dual-Stack approach perceived industry wide as one of the smoothest IPv6 transition technologies:

- When adding IPv6 to a system, it does not delete IPv4
  - This multi-protocol approach is familiar and well-understood (e.g., for AppleTalk, IPX, etc.)
  - no additional SW is required for tunneling interfaces. In most handsets IPv6 is be bundled and available as Dual stack with all the OS release.

- The choice to use either the IPv4 or IPv6 stack is made by the Application (or libraries)
  - When initiating, based on DNS response (if dest has AAAA record) use IPv6, else use IPv4)
  - when responding, based on version of initiating packet

This allows indefinite co-existence of IPv4 and IPv6, and gradual app-by-app upgrades to IPv6 usage. Multiple IPv6 implementations may coexist within the service provider network. This design stream shall cover only native Dual stack implementation in CMG and the network requirements it imposes to cloud overlay networks and peering with the Service Provider PE routers. The current design is not delving into further considerations related with the following IPv6 adoption options:

- IPv6 + CGN
- IPv6-only
- IPv6-only with NAT64 (with or without DNS64 )
- 464XLAT
- Other transition technologies

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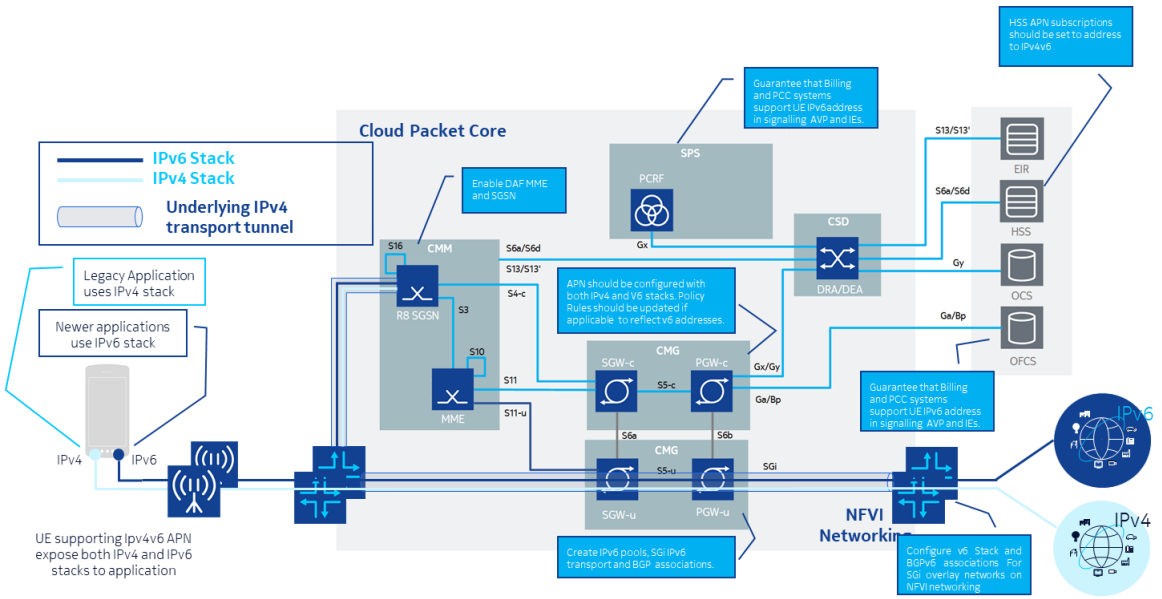



Figure 10: PDP/PDN IPv4v6 Dual Stack support

In alignment with the considerations provided above, CPC shall be deployed to support PDP contexts/Bearers of type IPv4v6, as such the following requirements apply to its underlaying functions:

- **Cloud Network:** SGi VRF and all subnets and ports that connect to it within the NFV Cloud must be configured with both IPv4 and IPv6 stacks. All BGP associations already created in this VRF must be extended to support IPv6 routes.

- **PGW:** APNs “IPv6 enabled” must be explicitly configured to support PDP/PDN Type IPv4v6. DNS PCO field configurations should be extended to include primary/secondary DNSv6 addresses and IPv6 subnet ranges for handsets should be set under the APN pool.

- **SGSN/MME:** Needs to be set to support 3GPP-R9 and set with Dual Address Flag to enabled.

- **PCRF:** PCC use-cases should be validated against the use of IPv4v6 PDN type and IPv6 addressing related AVPs in Gx interactions.

External elements surrounding the CPC should also be configured to facilitate the use of IPv4v6 PDP/PDN Type, where successful parsing of IPv6 IEs should be validated on Billing systems (both OCS and OFCS) and HSS subscriptions must be updated to support PDP/PDN Type IPv4v6 in the APNs targeted to support IPv6. Integration requirements on NEs outside of the CPC are not within the scope of this design.

### 4.4.2 3GPP Dual Address Flag

IP assignment criteria is first and foremost defined by the handset APN configuration, that sets what PDP/PDN Type is requested towards the network. When a subscriber's UE succeeds in the attach stage, the IP allocation process takes place to retrieve IP addresses. In general, a PDP/PDN Type IPv4v6 request implicitly allows the network side to make several IP assignment options, including IPv4-only, IPv6-only, IPv4 and IPv6 in single PDP/PDN bearer, and IPv4 and IPv6 in separated PDP/PDN bearers.

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The IP assignment decision depends on the configurations in place in the PGW regarding its support for IPv4v6 and on the MME/SGSN configuration for Dual Address Bearer Flag. The general rule is that if the requested PDN Type matches one of the subscribed (HLR/HSS Provisioned) PDP/PDN types, then the final assignment will align to that. When the UE requests PDP/PDN Type IPv4v6, the PGW shall perform an additional validation and take the value of the Dual Address Bearer Flag to determine if IPv4v6 should be initiated.

The MME/SGSN shall set the Dual Address Bearer Flag when the PDN type is set to IPv4v6 and all SGSNs which the UE may be handed over to are Release 8 or above supporting dual addressing, which is determined based on node pre-configuration by the operator. Table **13** summarizes all PDN Type Selection and fullback rules, **where only the 1st and the very last one of which are considered as valid and tested under the scope of this deployment.**

| UE Req | Subscribed | DAF | NW Req | PGW config | Result |
|---|---|---|---|---|---|
| **IPv4** | **IPv4, IPv6, IPv4v6** | **-** | **IPv4** | **IPv4, IPv4v6** | **IPv4** |
| IPv4 | IPv4v6 | - | IPv4 | IPv4, IPv4v6 | IPv4 |
| IPv6 | IPv4,IPv6,IPv4v6 | - | IPv6 | IPv6, IPv4v6 | IPv6 |
| IPv6 | IPv4v6 | - | IPv6 | IPv6, IPv4v6 | IPv6 |
| IPv4v6 | IPv4 | - | IPv4 | IPv4, IPv4v6 | IPv4 |
| IPv4v6 | IPv4, IPv4v6 | - | IPv4(pre-r9) | IPv4, IPv4v6 | IPv4 |
| IPv4v6 | IPv6 | - | IPv6 | IPv6, IPv4v6 | IPv6 |
| IPv4v6 | IPv6 | - | IPv4(Pre-9) | IPv6, IPv4v6 | Rejected |
| IPv4v6 | IPv4v6 | - | IPv4v6 | IPv4 | IPv4 (ESM casuse #50) |
| IPv4v6 | IPv4v6 | - | IPv4v6 | IPv6 | IPv6 (ESM casuse #51) |
| IPv4v6 | IPv4v6 | unset | IPv4v6 | IPv4, IPv6 | IPv4 or IPv6 (ESM casuse #52) |
| **IPv4v6** | **IPv4v6** | **set** | **IPv4v6** | **IPv4, IPv4v6** | **IPv4v6** |

Table 13: PDN Type Selection and fallback rules

## 4.5 SGi VRF separation – Corporate APNs

Corporate APNs are often provided as a service that extends private corporate IP network (almost as a VPN) over the cellular network, allowing UEs seamless access to corporate applications and services. As such, in these cases, L3 separation must be enforced in SGi guaranteeing that traffic for these APNs is completely isolated from other networks and services. Different breakout models may be used to enforce this separation:

1. Tunnels: PPP/L2TP regeneration, IPoGRE and IPSec. The L2TP tunnel was either established:
   - to a MNO PE router (LNS) (resulting in unscalable /32 and /64 host route advertisements by those PE-LNS!) or
   - directly to the corporate CPE (LNS) (through pinholes in the corporate FW or by placing the entire CPE in the DMZ)
2. VLAN tags leading to tagged SAPs (Service Access Points) on local Provider Edge (PE) routers (in which the VPN services would be configured)

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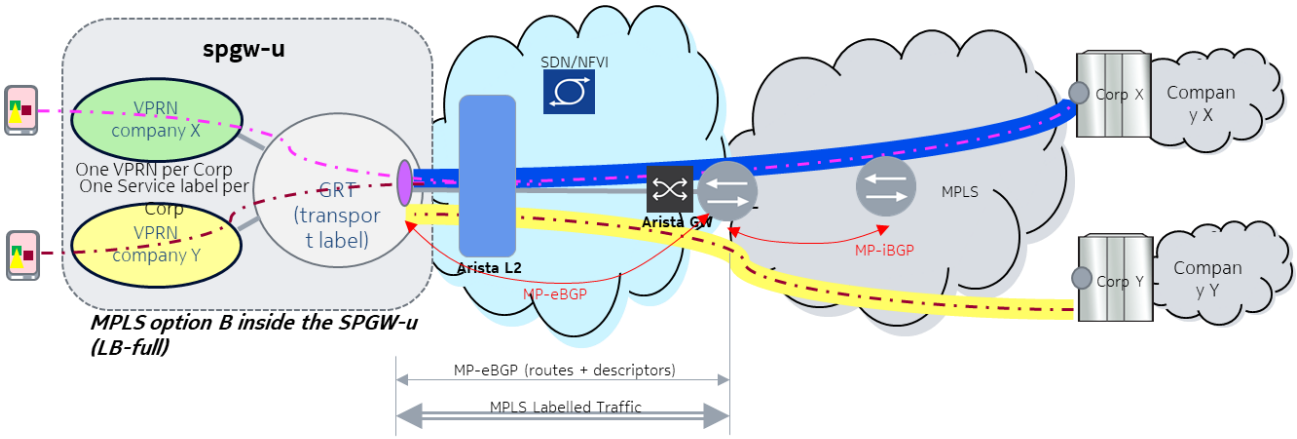

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3. MPLS-VPN Model B peering (for PGW capable of MP-BGP)

This implementation shall make use of the third approach mentioned above - MPLS-VPN Model B peering – to enforce SGi separation to up to 500 APNs.

Model B removes the need to create sub-interfaces for each shared VPRN between the PGW and its next hop L3 Gateway, which on MPLS terms is referred as the PE router.

In the MPLS-VPN Model B peering a single Layer 3 interface is required between both PGW-PE(s) to peer all VPRNs (even though x2 are to be used in this design for redundancy purposes). In order to distinguish the packets from each VPN, packets must be labeled in the data plane. The architecture essentially creates a kind of tunnel for each VPN on the interface between the PGW-DCGW, by using MPLS labels. The Multi-protocol exterior border gateway protocol (MP-eBGP) is used to pass the VPN routes between the ASBR-PE(s), and it assigns the labels to be used for each VPN prefix.




Each LB-VM has two eBGP neighbors, one on each subnet (/30)
One Service MPLS per VPRN Corp is created in the PGW (number of subnet: 2 x LB).
The PGW sits in a separate AS (autonomous system) than the Arista GW and acts as a CE. The PGW MP-eBGP peers with the Arista GW.

i. Routes for all VPNs supported on the Arista GW are exchanged with the PGW.
ii. Routes belonging to different VPNs are logically separated, via separate VRFs.
iii. Routes for each VPN are advertised as VPN-IPv4 routes where route distinguishers are prepended to regular v4 routes to allow them to be unique within the routing table. Route targets added to the BGP extended-community-attributes identify different VPN address spaces.
iv. The particular upstream BGP peer routing domain (VPN) from which a route is to be imported by the downstream peer into appropriate VRF is identified an extended community in the advertised NLRI.

Figure 11: MPLS-VPN Model B peering between VMG and DCGW

Furthermore, MP-BGP on CMG can only be supported through LB VMs, as such the design requires the deployment of CMG in Hybrid mode.

## 4.6 Roaming

In wireless telecommunications, roaming helps to make sure that a travelling wireless device, typically a mobile phone, is kept connected to a network, without breaking the connection. For example, if you travel beyond your mobile phone company's transmitter range, your mobile will automatically "hop on to" another phone company's service, where it is available.

Benefits of roaming

- Sustain seamless user experience in both home and visit network;
- Use data to access internet.
- Make and receive voice calls;

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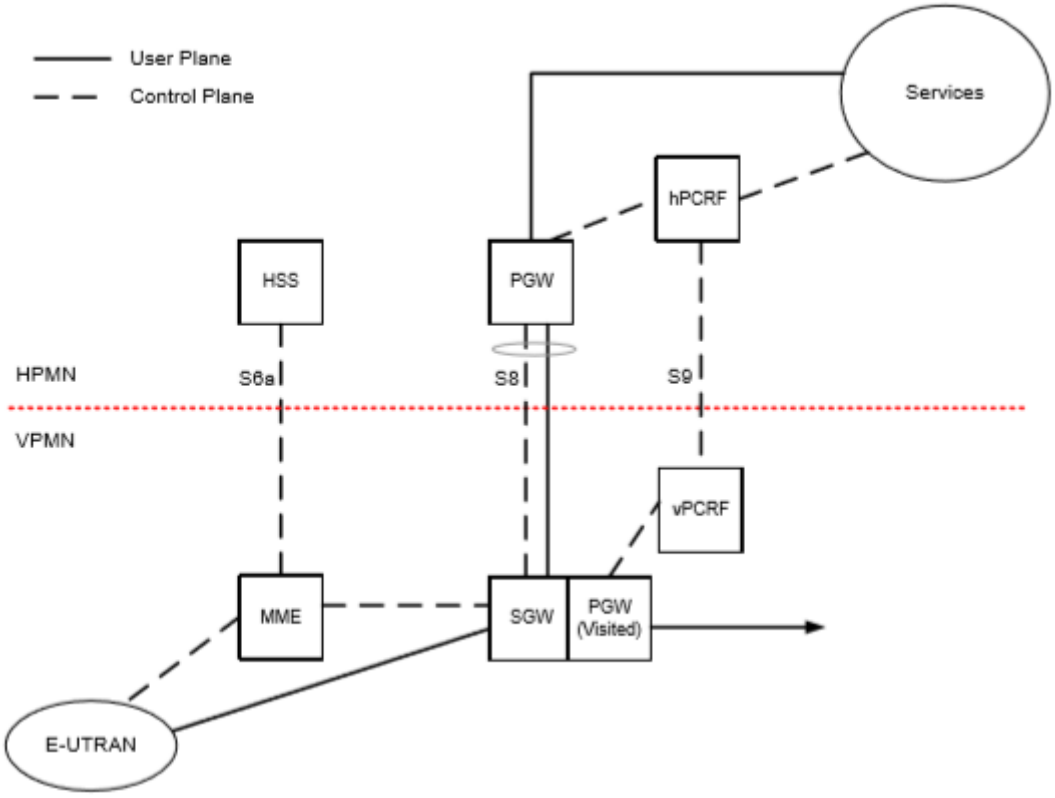

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- Send and receive messaging; and
- Access other services (e.g. home data services) while travelling outside the geographical coverage area of the home network.

### 4.6.1 Architecture

The following diagrams are produced based on the network diagrams from 3GPP TS 23.401 [1] and 3GPP TS 23.402 Section 4.2 [2].

We consider Home Routed user traffic for this architecture.






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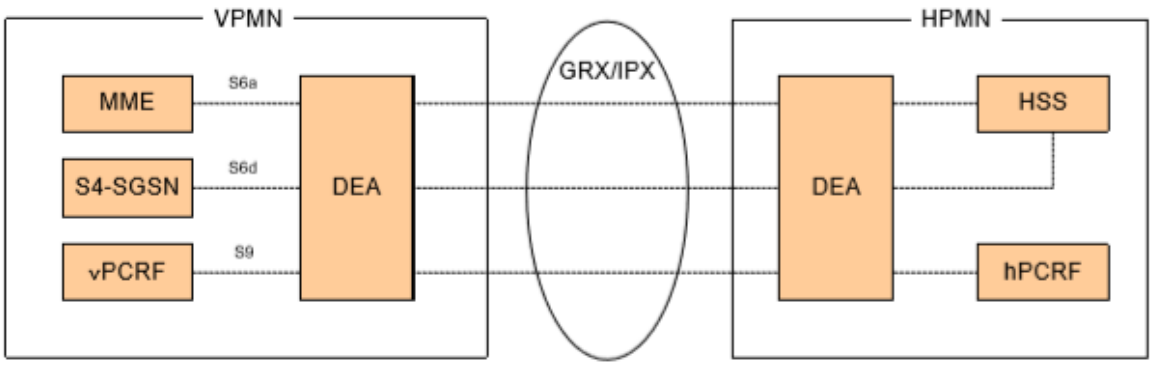
### 4.6.2 Interfaces

| Interface | Nodes | Protocols |
|---|---|---|
| *S6a* | MME – HSS | Diameter RFC 3868<br>3GPP TS 29.272 |
| *S6d* | S4-SGSN – HSS | Diameter RFC 3868<br>3GPP TS 29.272 |
| *S8* | SGW – PGW | GTP-C TS 29.274<br>GTP-U TS 29.281 |
| *S9* | hPCRF – vPCRF | Diameter RFC 3868<br>3GPP TS 29.215 |
| | DRA/DEA supported for DIAMETER traffic flows | |
| | DNS | |

### 4.6.3 Diameter Architecture

The logical architecture illustrates, at the Diameter application level, the position of DEA in the PMN.

Border gateways are not presented as they are not involved with Diameter procedures. DEA's must be secured by Border Gateways as any other equipment exposed to GRX/IPX, unless they are outsourced to IPX Providers.




### 4.6.4 DNS

Diameter routing on international networks shall be performed based on the destination-realm AVP.

Therefore it is mandatory to use the standard realm as detailed in 3GPP TS23.003 section 19.2.

"**epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org**"

#### 4.6.4.1 SGW Selection

SGW selection is performed by the MME/SGSN at initial attach or PDN connection establishment procedure. This occurs in the VPMN or the HPMN (non-roaming scenarios).

SGW selection is performed by using the S-NAPTR procedure with:

- "Service Parameters" = {desired reference point, desired protocol}
- "Application-Unique String" = the TAI FQDN (per 3GPP TS 23.003 [7])

For example, in a roaming scenario with Home routed traffic (S8) and GTP protocols, the MME/SGSN performs SGW selection using the S-NAPTR procedure with:

- "Service Parameters" = {"x-3gpp-sgw:x-s8-gtp"
- "Application-Unique String" = tac-lb<TAC-low-byte>.tac-hb<TAC-high-byte>.tac.epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org

#### 4.6.4.2 PGW Selection

PGW selection is performed by the MME/SGSN at initial attach or PDN connection establishment.

PGW selection is performed by using the S-NAPTR procedure with:

- "Service Parameters" = {desired reference point, desired protocol}
- "Application-Unique String" = the APN FQDN (per 3GPP TS 23.003 [7])

For example, in a roaming scenario with Home routed traffic (S8) and GTP protocols, the MME/SGSN performs PGW selection using the S-NAPTR procedure with:

- "Service Parameters" = {"x-3gpp-pgw:x-s8-gtp"}
- "Application-Unique String" = <APN-NI>.apn.epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org

#### 4.6.4.3 DCNR-capable SGW and PGW selection

An MME or S4-SGSN may select specific SGWs and PGWs for UEs supporting dual connectivity with NR and not restricted to use NR (by user subscription or local operator policy).

In addition, to regular SGW/PGW selection procedure, DNS should return additional “Service Parameters” in SNAPTR DNS response:

- "x-3gpp-pgw:x-s5-gtp+nc-nr"
- "x-3gpp-pgw:x-s8-gtp+nc-nr"

By default, when NR-specific “Service Parameters” included, CMM will always choose the "nc-nr" capable nodes first over the collocated.

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# 5. Cloud Mobility Management (CMM)

The Nokia Cloud Mobility Manager (CMM) is a virtualized network function (VNF) that under the current scope of deployment implements the LTE Mobility Management Entity (MME) and Service GPRS Support Node (SGSN) functionality in a cloud environment running on top of standard, multi-purpose IT hardware.

The figure illustrates the position of the CMM within common radio access networks, referencing all logical interfaces supported by the CMM product software.

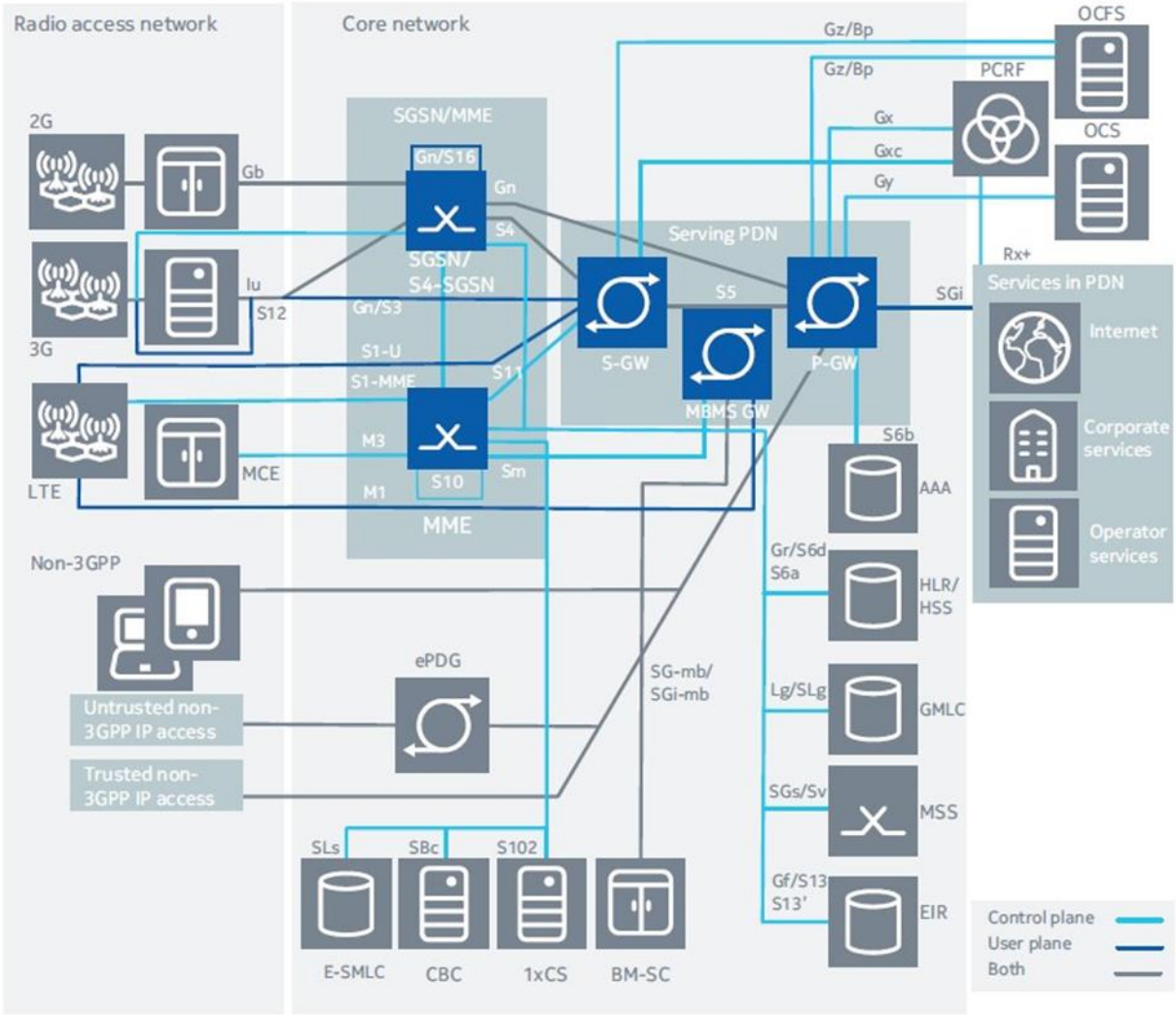



Figure 12: CMM in 3GPP LTE and non-3GPP networks

## 5.1 Functional requirements

The cMM as a VNF has a modular design with several sub-components, that fit at different levels of a multi-tier design paradigm, and together realize a fully compliant 3GPP network element.

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### 5.1.1 MME/SGSN Pooling

As introduced in previous chapters CMM is deployed within the CPC solution to fulfil the roles of Mobility Management Entity (MME) and release 8 compliant Service GPRS Support Node (SGSN) in the Service Provider central Datacentres.

CMM instances geographically distributed between Service Providers core sites will be deployed as MME/SGSN pools to support the Following RAN types:

- GERAN
- UTRAN
- EUTRAN
- NB-IOT
- 5G NR (NSA option 3X)

When core network operates in a pool, radio access nodes are connected to several CMMs. When pooling is enabled, the radio nodes try to maintain the initially chosen core network elements for the UE as long as the UE stays within the pool area. As a result, when a UE moves between different access types (2G/3G and 4G) within the same geographical area covered by the same core network pool, radio access chooses the same CMM. For example, radio access chooses the collocated SGSN or MME which previously served the UE.

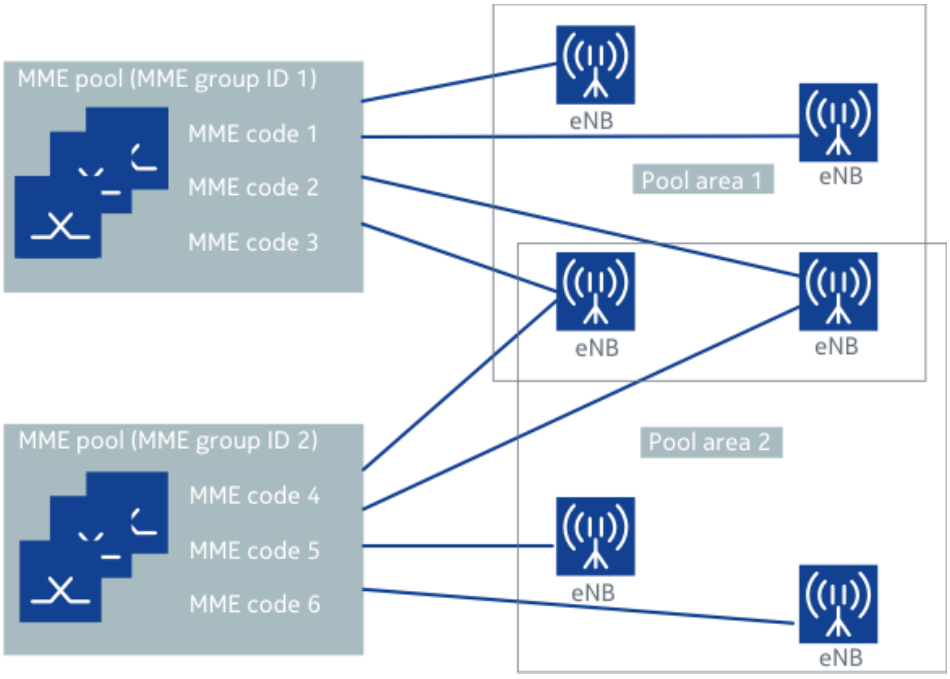



Figure 13: MME pooling principle

The current deployment makes use of S1-flex and other CMM functionalities to implement pooling mechanisms on both MME and SGSN network functions and support the following Handover procedures:

- GERAN -> UTRAN -> EUTRAN -> 5G NSA option 3X
- 5G NSA option 3X-> EUTRAN -> UTRAN -> GERAN

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### 5.1.1 Functional interfaces

Bellow follows the list of CMM logical interfaces used under the current scope of Nokia Cloud Packet Core deployment. For many of these interfaces different transport protocols alternatives may be considered, the table specifies what transport options are implemented in the deployment.

| Interface | Protocol | Transport | Remote Endpoint | Vendor / Solution | Description |
|---|---|---|---|---|---|
| Gb | BSSGP | UDP | BSC | Nokia SRAN | |
| Iu | RANAP | SCTP | RNC | Nokia CMG | |
| S1-MME | S1-AP | SCTP | EnodeB | Nokia SRAN | |
| S6a / S6d | diameter | SCTP | DRA - HSS | Nokia CSD / Nokia EIR | |
| Gn | GTP | UDP | GGSN | Nokia CMG | |
| Gp | GTPv2 | UDP | GGSN | ANY | |
| S10 | GTPv2 | UDP | MME | Nokia CMM | |
| S11-C/S11-U | GTPv2 | UDP | SGW | Nokia CMG | |
| S13 / S13' | Diameter | SCTP | DRA - EIR | Nokia CSD / Nokia HSS | |
| Bp(SGSN) | sFTP | TCP | OFCS | Proprietary | |

Table 14: Logical interfaces and respective transport protocol options

### 5.1.2 Multipoint Gb/ Multipoint-Iu

The Multipoint Gb and Multipoint Iu feature enables radio access sharing in a 2G and 3G environment respectively. With this feature, RNCs and BSCs can be connected to more than one SGSN as described in 3GPP TS 23.236.

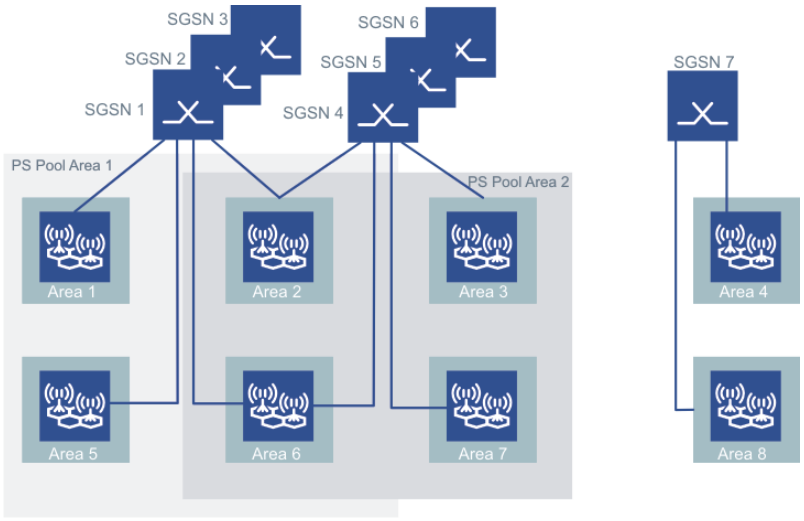



Figure 14: Multipoint Iu concept overview

These features enable the operator to add several SGSNs to serve the same routing area. It increases service availability, as other SGSNs may provide services in case of an SGSN in the pool area fails. It also enlarges the served area compared to the service area of one SGSN. This results in reduced network load, that is, reduced inter-SGSN location updates and handovers.

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### 5.1.1 S11-U

In the current NB-IOT use-case, S11-U interface is used for the exchange of short data between the MME and the S-GW. Indirect data forwarding tunnel creation and deletion procedures are not supported for S11-U and are instead handled by its control interface (S11-C).

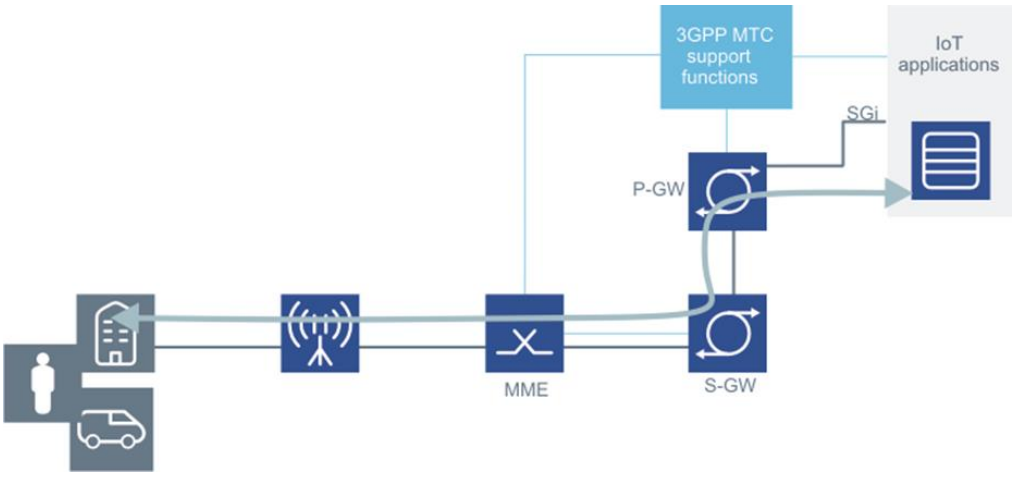



Figure 15: S11-U interface for small user data

NB-IOT not included in scope

### 5.1.1 Direct Tunnel

Usually, SGSN handles the user plane processing by receiving data from one user plane tunnel and transferring the data to another tunnel. This kind of user plane handling requires some processing power in the SGSN and ties up part of the SGSN hardware capacity.

The direct tunnel feature provides a direct user plane tunnel between the RAN and the GGSN/S-GW when transporting user plane information. This means that the 3G access user plane tunnel bypasses SGSN. The user plane tunnel is created directly between the RAN and the GGSN/S-GW through the IP backbone.

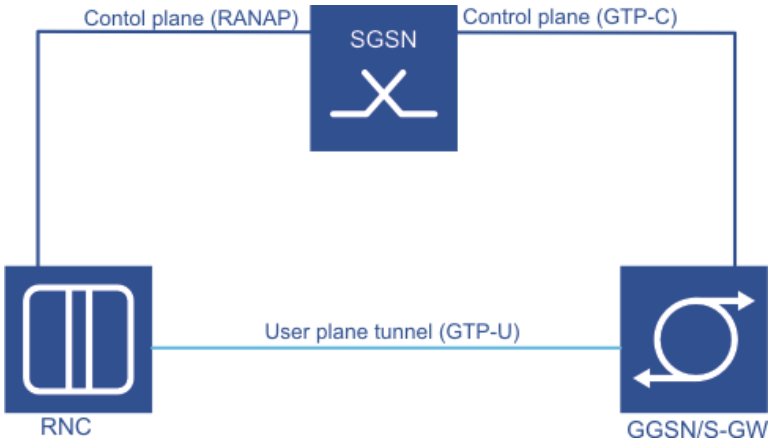



Figure 16: Direct Tunnel for 3G traffic

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Direct Tunnel shall be enabled for all 3G traffic in the Service Provider network. After configuring the direct tunnel parameter in RNC Parameter Set, it is also required to enable the DT license and identify what APNs are eligible for Direct Tunnel.

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## 5.2 VNF Components

CMM VNFCs are organized according to a multi-tier design strategy in which, the load balancing, transaction handling, and database functions are strategically placed in a hierarchy that can scale independently.

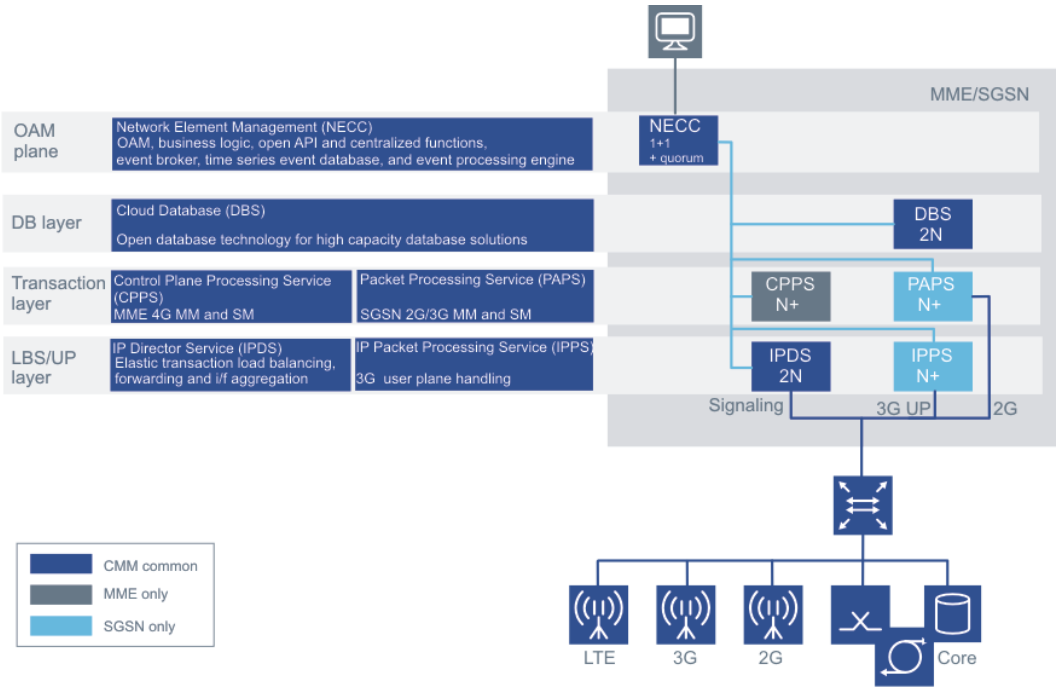



Figure 17: CMM functional units

**IPDS load balancer (IPDS)**
IPDS terminates all post-R8 3GPP interfaces logical interfaces from the external network's viewpoint. It terminates application-level protocols and performs internal and external load balancing. IPDS implements link management. IPDS uses in-house high performance SCTP stack that recovers 100k S1 link failover in less than one second. IPDS performs stateful assignment and reassignment of UEs to the CPPSs for efficient call processing. IPDS implements advanced signalling storm and network load/overload controls.

**Control Plane Processing Service (CPPS)**
CPPS Manages UE registration, authentication and mobility. CPPS manages UE bearer setup and management for data and Voice over LTE services. CPPS provides hybrid state efficient call processing with local cache that increases signalling capacity, reduces database latency, and optimizes compute resources for both DBS and CPPS. It provides extensive service optimization and overload control mechanism.

**IP Packet Processing Service (IPPS)**

To achieve high bandwidth and low latency for 3G user plane packet forwarding comparable to native HW implementations, 3G user plane packet processing, buffering, and data forwarding functions are decoupled into a dedicated VNFc, IPPS. In this sense, IPPS deals exclusively with user plane data and terminates externally both the Gn/Gp (for 3G) and IU-PS-u user-plane interfaces.

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PDPs user plane processing is assigned to a given IPPS by PAPS on a per-subscriber/per-PDP basis based on internal load-balacing criteria. PAPS and IPPS support full mesh connectivity, which means that any PAPS may select any IPPS to serve any primary PDP for any subscriber.

**The Packet Processing Service (PAPS)**

PAPS role is to manage authentication and mobility for GPRS subscribers as well as 2G/3G bearer management. Link management messages are routed via the IPDS to the leader PAPS, which is responsible for coordinating link management among the PAPS instances. To maximize Gb throughput capacity, the established Gb links bypass the IPDS and are terminated directly on PAPS instances.

IPPS provides 3G Iu and MAP signaling is initially handled in the M3UA load balancer in the IPDS, which terminates the SCTP interface to the RNCs and HLR/EIRs. The M3UA and application layer SS7 protocols are terminated in the PAPS.

**Database Service (DBS)**
DBS implements the persistent UE Context database designed to meet the high performance and low latency requirements of the MME application. DBS supports advanced data import and export capabilities to enable flexible integration towards other solutions. DBS Offers programmable views for data mining, for example, for statistics enhancements.

**Network Element Cluster Controller (NECC)**
Provides a centralized operation and management unit that provides interfaces between the CMM and the NMS (the NSP, NetAct, or third-party NMS). NECC supports CMM distributed VNF aspects such as CPPS redundancy mechanism, VM liveness, and so on. NECC supports the interworking of the CMM with the NSP (EMS), which in turn supports the interaction with the CBAM (Nokia VNFM) to support life cycle management functionality and VNF assurance. NECC VNFCs are always deployed in a group of three, where one of the NECCs in the group is referenced as NECQ for planning and dimensioning purposes.

In alignment with the functional definitions above, termination point and message handling for MME/SGSN logical interfaces are split across the different CMM VNFCs as depicted in Figure **18**.

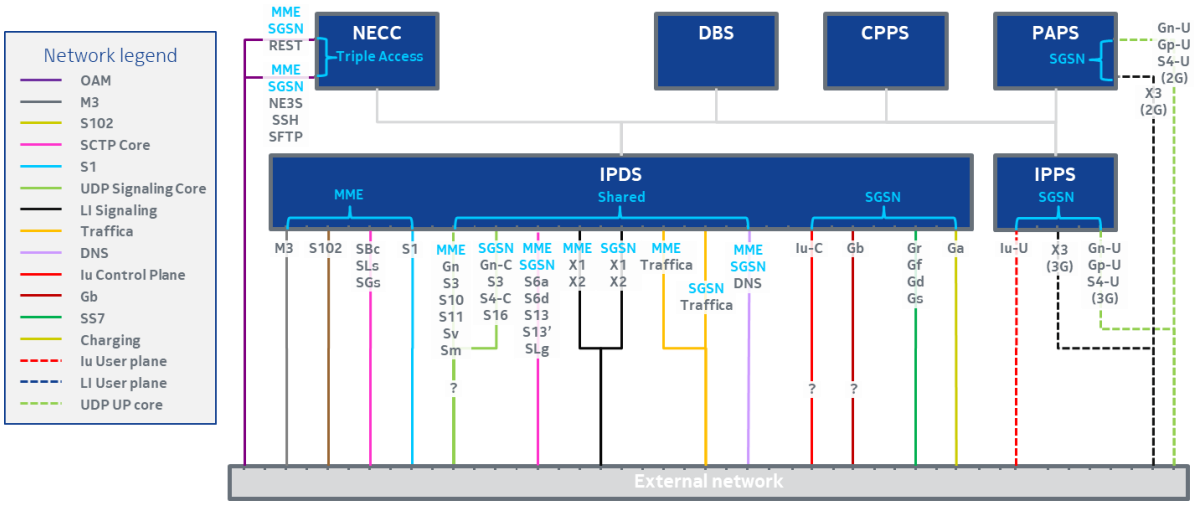



Figure 18: Cloud Mobility Manager VNFC architecture an interface mapping

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# 6. Cloud Mobile Gateway (CMG)

The Cloud Mobile Gateway (CMG) supports mobile gateway functionality that can be deployed on a generic computing infrastructure in a cloud environment. CMG supports multiple GW functions including ePDG, UPF, SGW and PGW.

Only SGW/PGW functions and related VNFCs are considered under the current scope of CPC deployment, as such the focus of the following chapters will be restricted to those.

## 6.1 VNF Components

A CMG instance consists of multiple virtual machines (VMs) running on a generic computing infrastructure. Each VM is dedicated to a specific set of functions that can be replicated across many similar VMs. A group of VMs is represented as a single instance of an application as they operate in-sync with other similar VMs in the group to support a network function.

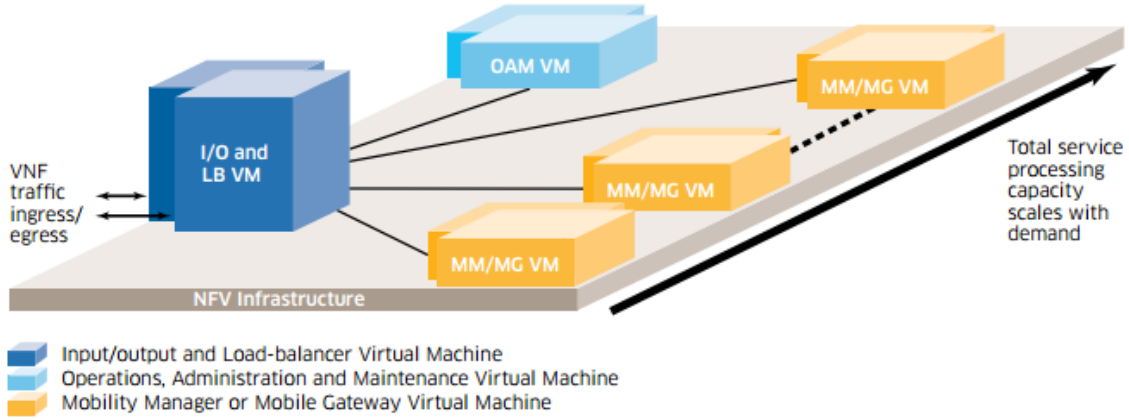



Figure 19: CMG functional units

The CMG as a VNF is composed of three different virtual machine (VM) types aka VNFCs. These are:

**OAM VM**
It performs control plane functions that include VNF management, routing protocols, management interface for SNMP/TELNET/SSH/CLI for the configuration. Each OAM-VM has Cinder volume attached representing CF1/CF2 for KPI/KCI, CDR, and log storage. CF3 is part of the image (qcow2 file). The OAM-VMs map directly to CPM-A/B cards on former Hardware based SR7750 MG platform and are as such reference in many of CMG configuration views.

**Load Balancer (LB) VM**
This VNFC takes the same role as interface IOMs cards in MG Hardware based solution and provides external network connectivity (input/output) to mobile gateway function and load distribution across Mobile Gateway VMs. It forwards GTP-/GTP-U and user equipment (UE)-addressed packets to the Mobile Gateway VM.

Virtual IPs that deal with Control Plane 3GPP functional Interfaces, sit behind the LB-VMs. Packets destined to 3GPP Loopback IPs reach VIPs via the LB-VMs, which work in Active/Active mode.

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**Mobile Gateway (MG) VM**

Fulfil the function of the MG-ISM card, holding the subscriber sessions in control and user plane. MG-VMs performs the following functions:

- 3GPP call processing (Control and Data Plane), Policy and Charging Enforcement Function (PCEF), and Application Assurance (PCEF enhanced with ADC for application detection and control, and L7 service classification for policy charging control).
- 3GPP gateway functions such as SGW, PGW/GGSN, SAE-GW/GGSN, ePDG, TWAG, and so on.
- Network connectivity for UP traffic for the CMG configured in LB-less mode.
- Supported service functions depending on the configurable personality of the MG-VM. As an SGW, the MG-VM provides bearer management services and the mobility anchor point. In an MG-VM that is configured as a PGW, the VM provides bearer management, IP anchor point support, lawful interception functionality, charging functionality, QoS enforcement, and advanced DPI functionality. When configured as an SAE-GW, all the SGW and PGW functions are supported on a single MG-VM.
- External IP interfaces are configured on the MG-VM for a CMG instance deployed in LB-less mode

Mobile Gateway VMs are typically grouped in active-backup pairs(when 1:1 redundancy scheme is in place) which can perform all previous mentioned functions or be dedicated just to a specific function.

**DataBase (DB) VM**

This is an optional VNFC that is only used when CMG N:K redundancy model is used. DB-VM is the common session storage to support N:K redundancy. DB-VM supports proxy function and hosts DB instances. During normal operation, each MG-VM stores its session state in the DB-VM, when the session is created and on any event that triggers change in the session's state.
Only one DB-VM per CMG instance is supported in the current release.

### 6.1.1 MG+ Architecture (LB-less)

CMG supports GTP-U traffic (User Plane) to be processed directly on MG-VM, without the need of it having to go first through LB-VMs. CMG without GTP-U load balancer uses GTP-U redirection method to distribute sessions between MG-VMs Groups. Each MG-VM Group, composed of an active-standby pair of MG-VMs, is configured with one or more GTP-U interfaces and SGi IP pools. GTP-C is used to anchor GTP-U traffic on an individual MG-VM allowing user plane traffic to be sent and received directly.

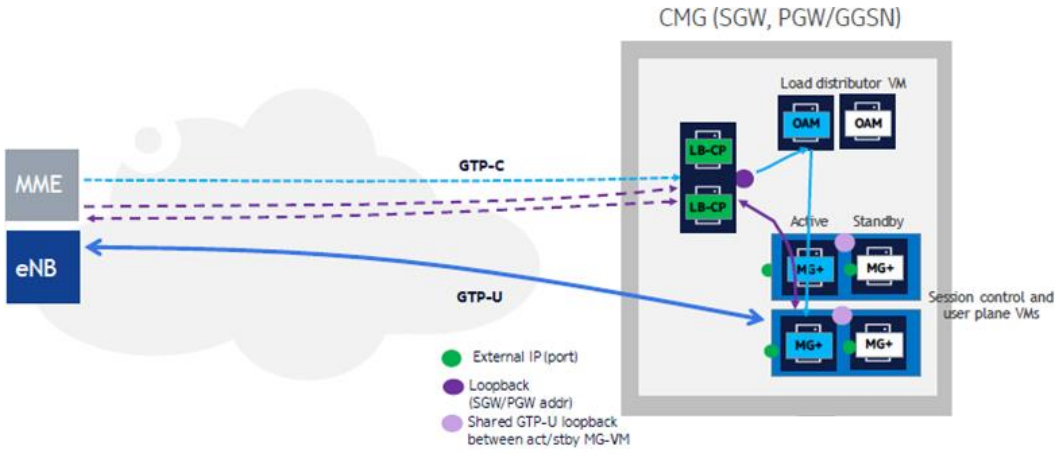



Figure 20: MG+(LB-less) CMG VNF Architecture



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The LB-VMs will receive all GTP-C traffic, e.g. Create Session Request, forward it to the active OAM VM, where OAM performs the load-distribution function and selects anchor MG+ group. The active MG VM of the anchor MG+ group will respond to the initial Create Session Request and set GTP-U in the Create Session Response to a new IP address, the loopback of MG+. Each MG group will use a unique GTP-U address, for example with 3 MG group, there will be 3 S1-U addresses on the CMG. All the data plane traffic will be sent directly to MG+, therefore no bandwidth on the data switch fabric will be used for data plane traffic.

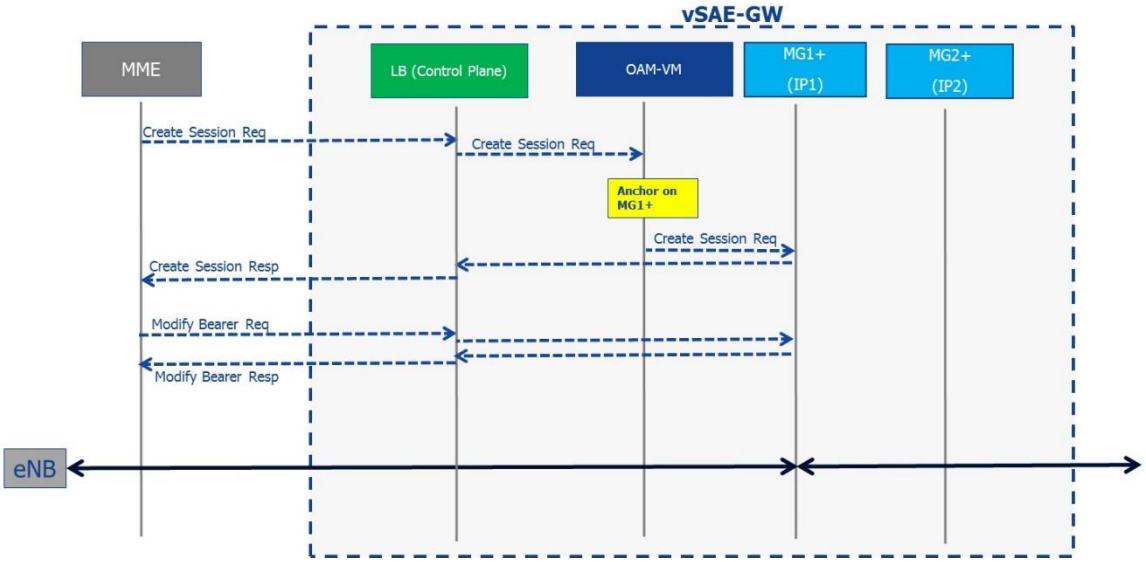



Figure 21: MG+(LB-less) CMG VNF Architecture

This improves the CMG throughput performance in the user plane. By removing the LB-VM from the path of the user plane traffic, the gateway instance capacity scales better in infrastructures with limited I/O capacity on the LB-VM.

### 6.1.2 CMG in LB-mode

In CMG in LB-mode deployment option, all Control- and User-Plane traffic is anchored on LB-VM and no GTP redirection mechanisms are used.

Multiple LB-VMs can be configured per CMG instance to allow for a scalable system. The peering router is configured with an ECMP path to the common IP address so that traffic is distributed across multiple LB-VMs. GTP-U UP traffic is received on the LB-VM and forwarded to the MG-VMs based on the identified session. The CP traffic processing is identical on both LB and LB-less deployments of the CMG.

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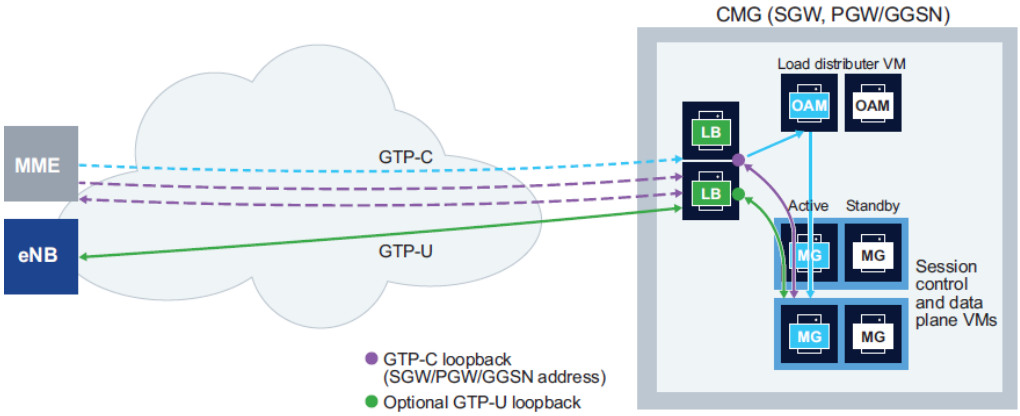



Figure 22: CMG LB-mode VNF Architecture

CMG in LB-mode significantly simplifies IP-related design as no need for MG-VM specific GTP-U IPs and IP Pools. The most significant drawback is limited CMG scalability (UP traffic growth requires extensions of both LB-VMs and MG-VMs) and costly East-West traffic between LB-VMs and MG-VMs (especially important when software-based switching on computes is in use). However, use-cases (e.g. Enterprise GW) that require tunnelling on SGi (GRE, L2TP, MP-BGP with MPLS) can be only realized based on LB-VM.

The current design makes use of LB-mode, to accommodate the use of MPLS-VPN Model B peering for SGI separation on Corporate APN scenarios introduced in section 4.5.

### 6.1.3 CMG in Hybrid Mode

CMG supports a variant of the MG+ architecture, branded Hybrid Mode, where besides enabling direct traffic flow between the MGs and external GW on SGi, user-plane traffic coursing through LB is also supported (mix of LB-mode and LB-less). In this mode traffic on the SGi interface is supported either on the LB or the LB-less option per APN basis. For example, the Enterprise APN is configured to use the LB-VM option while the Internet APN is configured to avoid LB (that is, using the LB-less mode). Its advantage over LB-mode even in case all APNs are using LB-mode connectivity, UP data connectivity towards RAN (S1-U) can leverage from MG+ connectivity (half of forwarding based on MG+).

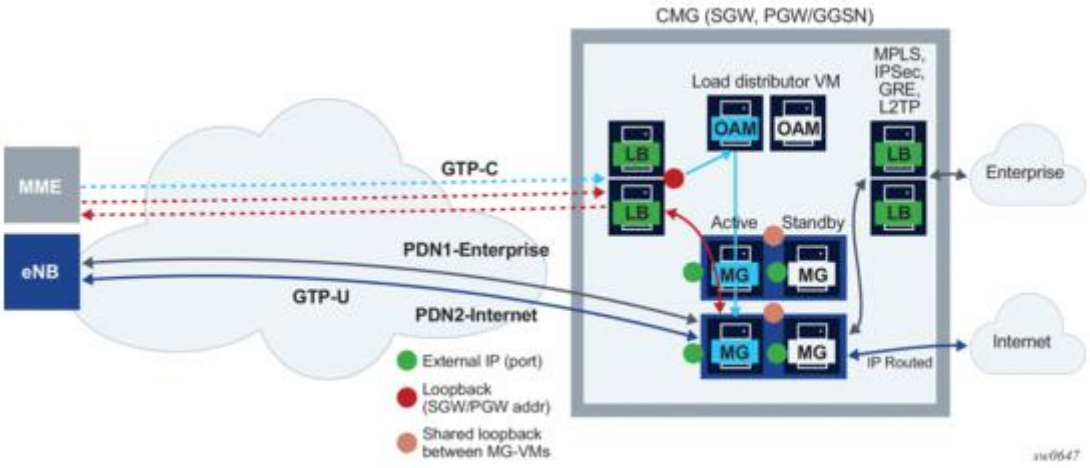



Figure 23: CMG in Hybrid Mode

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Using the LB-VM on the SGi interface allows tunneling options like MPLS, IPSec, GRE, and L2TP (only available in LB-VMs of non-CUPS architecture) to be used in conjunction with Enterprise PDN session.

### 6.1.4 CMG N:K Redundancy deployment model

To cater for a more efficient CMG VM footprint, the CPC BP deployment makes use of the new redundancy mode introduced in CMG 11.0 (used for CMG instances acting as User Plane gateways and running in LB-less mode; for LB-mode N:K will be considered in future CPC BP releases). Besides the current 1:1 option, based on active/standby pairs of MG-VM, it is now possible to create a pool of MG-VM resources that will be dynamically assigned to the configured redundancy groups. When the MG-VM allocated to a redundancy group fails, it will be replaced with one of the standby MG-VMs available in the resource pool.

This deployment option, thanks to its flexibility will be considered to be included as a part of CPC BP design in future releases, especially when N:K option will be supported.

In the N:K redundancy mode, N is the number of active (working) MG-VM and K the number of standby MG-VM in the resource pool. In the current release, the upper limit for N is 15 and K is maximum 3.

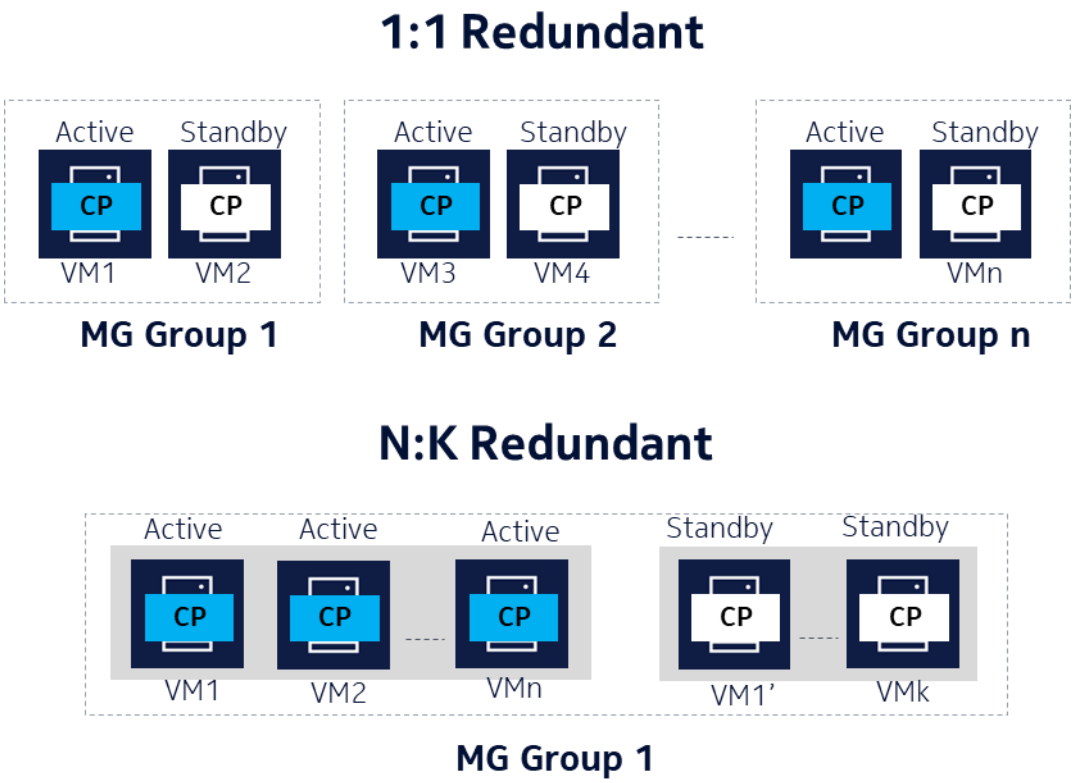



Figure 24: N:K redundancy

A DB-VM is introduced to store a copy of the session data from each active MG-VM. The DB-VM runs several instances of an in-memory database to allow efficient access to the data. A proxy server transparently distributes and indexes the content over the available database instances. The proxy receives and handles the external queries and update requests from the MG-VM. Only one instance of the DB-VM is allowed in the current release.

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When a new session is created, the OAM allocates a MG-VM for it. A context with the session information is stored in the MG-VM and a copy is uploaded to the DB-VM.

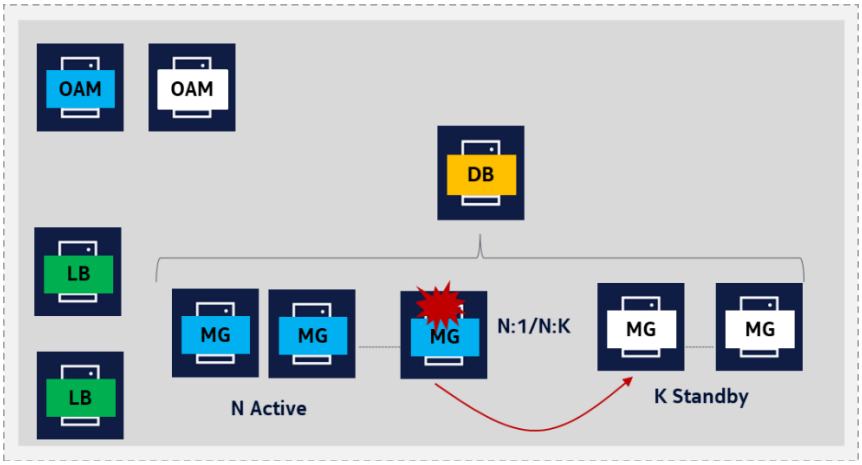



Figure 25: MG failover with session recovery from DB

If the session is modified, the DB copy must be updated as well as the local context. If an active MG-VM fails, the CPM detects it and assigns a new MG-VM to the redundancy group, chosen among those in standby. The new VM is initialized with the redundancy group configuration and retrieves the sessions from the DB-VM before starting to forward traffic.

Recovery time for 500-600k sessions is expected to be under 15 seconds. Also, prioritization schemes shall be put in place to help speed up the restoration process: active sessions should be recovered before idle ones and real-time APN before low priority ones.

### 6.1.5 MG+ and network requirements

N:K redundancy imposes new connectivity requirements between VMs, namely that all the MG-VM be connected to the same broadcast domain(s). The reason is that a standby MG-VM does not "own" configuration, it inherits the configuration of the VM it is replacing.

A new option in the sap, labelled "dyn", is introduced in the configuration. Instead of statically defining the interface sap in the VPRN, e.g. `sap 3/3/1 create`, the VM/card id part of the identifier is going to be dynamically populated by the system, based on the actual resource (VM id) allocated to the redundancy group by the system. e.g. `sap dyn/3/1 create`, where 'dyn' takes the VM id value. With this option, any MG-VM in the pool can be allocated to a redundancy group. This is illustrated in the diagram below in .

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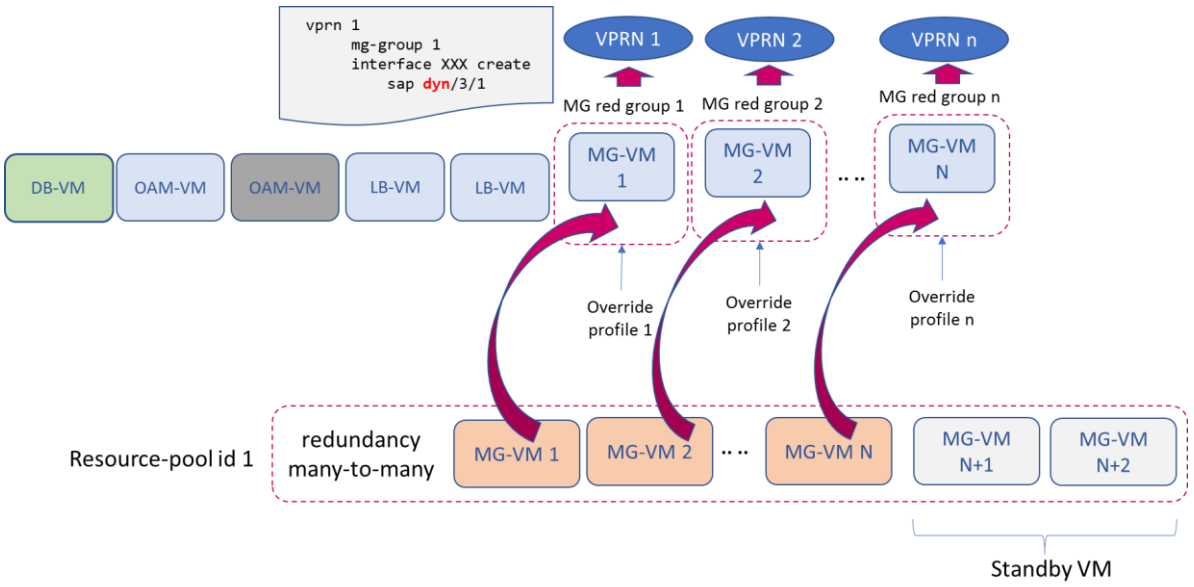



Figure 26: Dynamic resource allocation

The DB-VM hosts three interfaces: two signaling (VMXNET3) interfaces and an OAM (OVS) port, as displayed in the next figure.

The signaling traffic on the VMXNET3 interface will be routed through either an existing or dedicated VRF. Nokia recommendation is to use a dedicated VRF to ensure correct isolation and prioritization of this critical traffic. The DB OAM interface can be added in the OOB management subnet defined for the CMG.

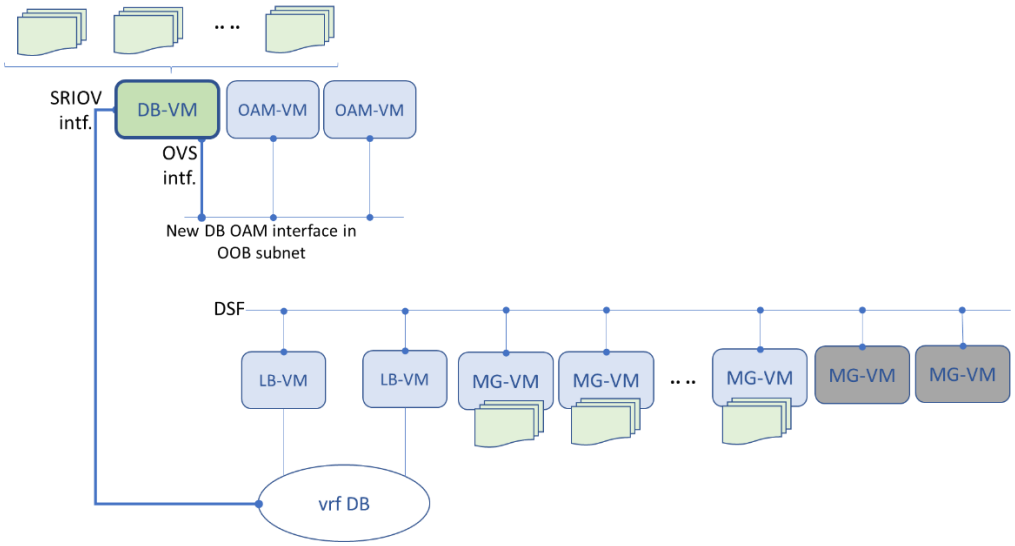



Figure 27: DB-VM connections

Link redundancy in VMware deployments, as shown in Figure 32, is achieved by configuring active-standby uplinks under the Teaming and failover settings of the VMware vSphere Distributed Switch.

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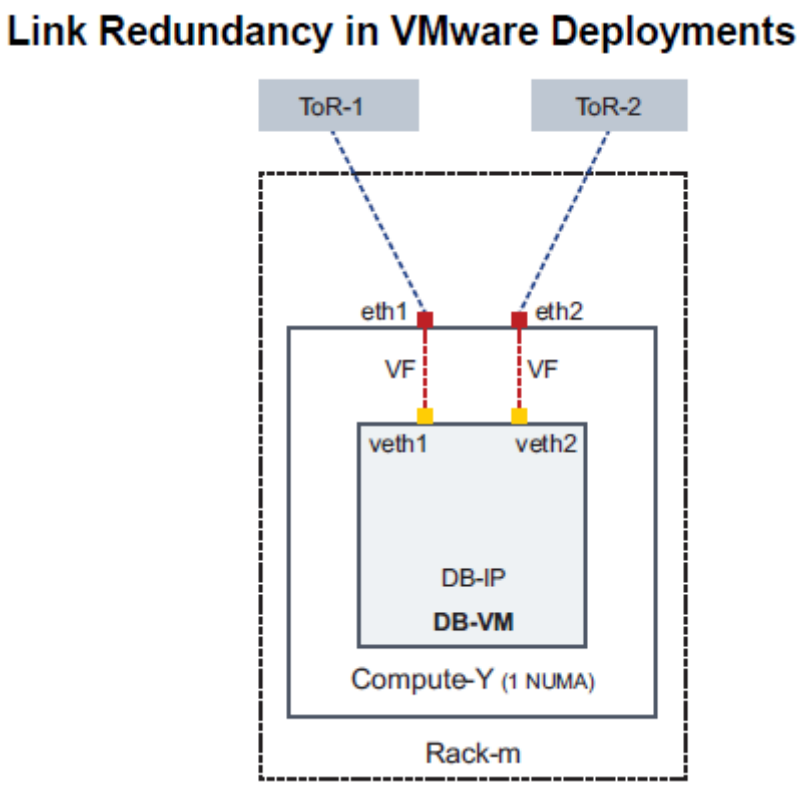



Figure 28: DB-VM Link Redundancy

The veth1 and veth2 ports on the DB-VM can be connected to the same or different physical NICs, that is, they can be part of the same or different port groups. If the ports are added to the same port group, redundancy is achieved at the vNIC (veth1 and veth2) level. If the ports are added to different port groups, redundancy is achieved using the active-standby uplink configuration under the Distributed Port Group settings of the Teaming and failover tab as shown in Figure 33. The selection of either of these port group options does not have any impact on the DBVM behaviour.

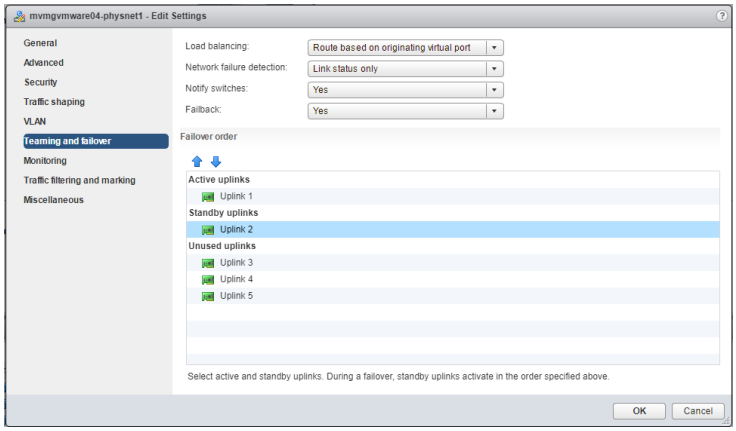



Figure 29: Sample Port Group Settings on Distributed Switch

Note: **This active-standby uplink configuration must be done only for the DB-VM link redundancy.**

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## 6.2 Functional requirements

The Control and User Plane Separation (CUPS) design principle shall be applied under the scope of this deployment for S-GW and PDN-GW network functions. This requirement in conjunction with the distributed data centre use-case, pushes into CMG design, the definition of **3 types of CMG instances**, topologically organized into Central and Regional/Aggregate-Edge Datacentres.

In Central Datacentres, three types of CMG instances are considered:

- **CMG-CP**: Delivering on SGW-C and PGW-C network functions. This instance should use CMG LB mode and run 1:1 redundancy
- **CMG-UP:** Delivering on SGW-U and PGW-U network functions serving Corporate/Enterprise APNs. To support SGI tunnelling functionality these instances should be deployed in LB-mode (see section 1.1), running 1:1 redundancy (in next releases Hybrid-mode architecture and N:K redundancy will be considered as well)
- **CMG-UP:** Delivering on SGW-U and PGW-U network functions for gross market Mobile Broad Band (MBB) type of APNs. To cater to high throughput demands of MBB, these instances use LB-less (MG+) deployment mode and run on the 1:1 or N:K redundancy model

While Regional Datacentres is considered to host only one type of CMG (no technical constraints for the others as well):

- **CMG-UP:** Delivering on SGW-U and PGW-U network functions for gross market Mobile Broad Band (MBB) type of APNs. To cater to high throughput demands of MBB, these instances use LB-less (MG+) deployment mode and run on the 1:1 or N:K redundancy model

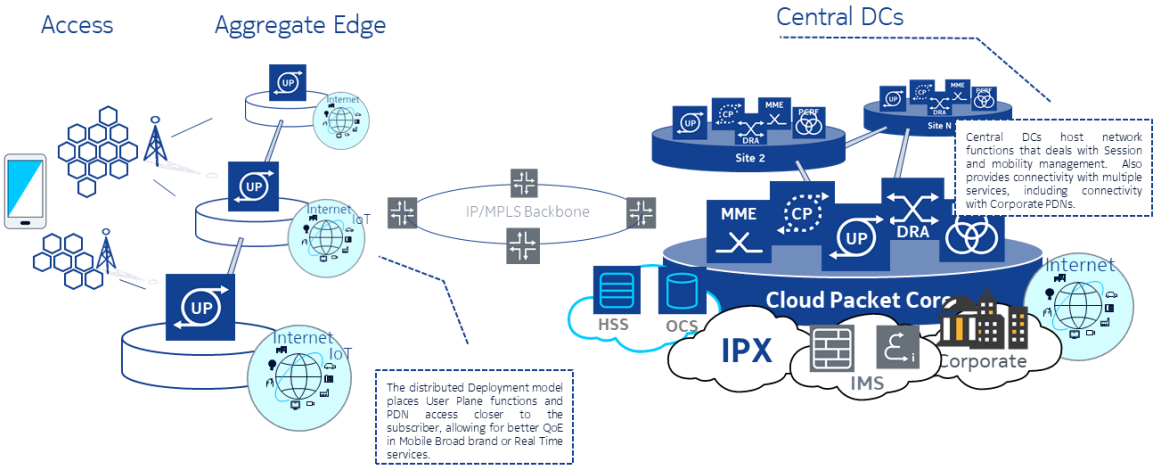



Figure 30: CMG-CP/UP distribution across data centers

The distribution of CMG instances across data centers is depicted in Figure **30**. Under the intended network scenario Control Plane functions are exclusively hosted in central sites, upon receiving a new session request SGW/PGW-Cs residing in central sites, shall select base on APN where user plane data should be anchored, either in the Central DC or Regional/aggregate Edge DC. This behavior is depicted in Figure **31**. By default, all APNs (including Enterprise) are directed to User Plane functions on the Central Data Center which have local connectivity points with multiple PDNs, gross market MBB internet however is primarily directed to SGW/PGW User Plane functions on Aggregate-Edge

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Data Centers (when available). SGW/PGW-U nodes on Regional/Aggregate Edge Data centers are selected based on the APN identified in the incoming request and on a Geographically closeness criteria grounded on the contents the User Location Info IE.

UUU

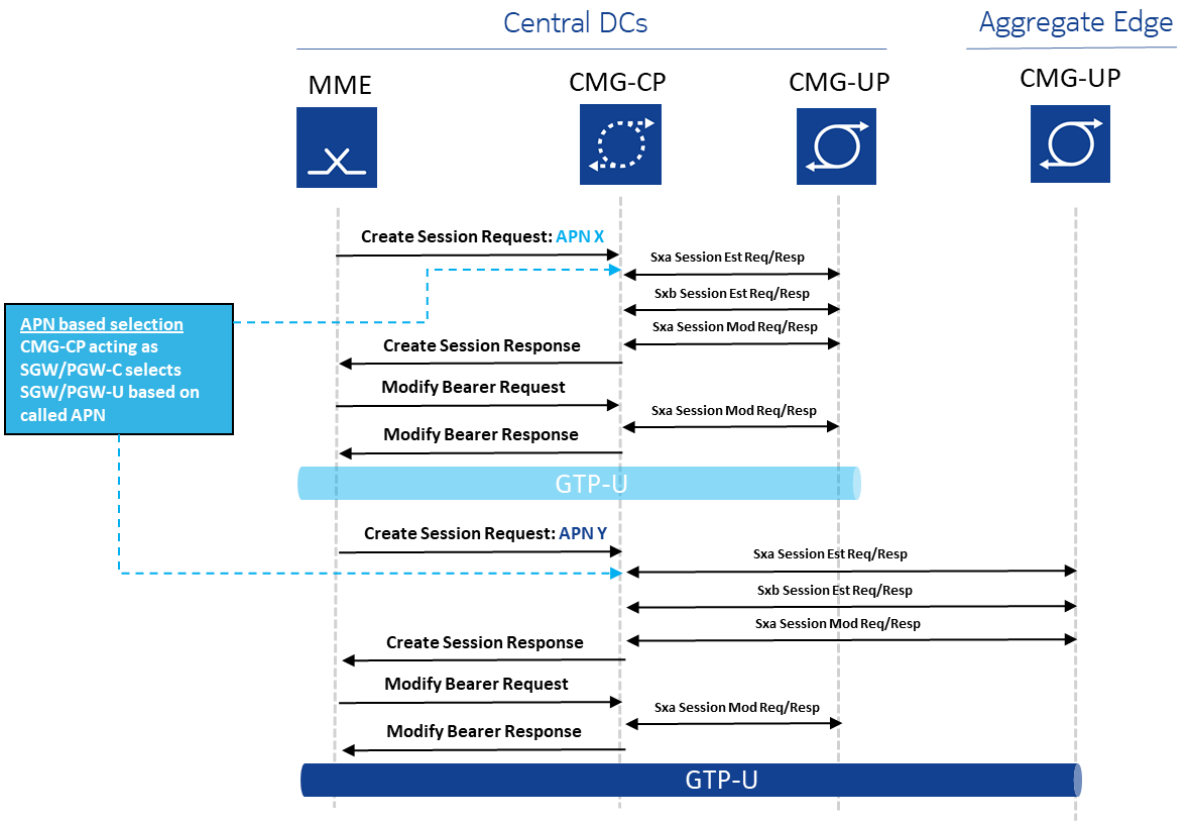



Figure 31: SGW/PGW-C selects (S/P)GW-U based on called APN

### 6.2.1 Functional interfaces

Bellow follows the list of CMG logical interfaces used under the current scope of Nokia Cloud Packet Core deployment. For many of these interfaces different transport protocols alternatives may be considered, the table specifies what transport options are implemented in the deployment.

| Interface | Protocol | Transport | Remote Endpoint | Vendor / Solution | Description |
|---|---|---|---|---|---|
| S1-U | GTPv2/v1 | UDP | EnodeB | Nokia SRAN | |
| S5 | GTPv2/v1 | UDP | SGW – PGW | Nokia CMG | |
| S4 | GTPv2/v1 | UDP | SGSN-SGW | Nokia CMG | |
| S8 | GTPv2/v1 | UDP | SGW – PGW | Nokia CMG | |
| S12 | GTPv1 | UDP | RNC | Nokia | |
| Gp | GTPv1 | UDP | SGSN-GGSN | ANY | |
| Sxa | PFCP/GTP | UDP | SGW_UP-CP | Nokia CMG | |
| Sxb | PFCP/GTP | UDP | PGW_UP-CP | Nokia CMG | |
| SGi | IP | - | Packet Data Network | | |

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| Interface | Protocol | Transport | Remote Endpoint | Vendor / Solution | Description |
|---|---|---|---|---|---|
| Gx | Diameter | **SCTP** | DRA / PCRF | Nokia CSD / Nokia SPS | |
| Ga | GTPv2 | UDP | CGF | Nokia | |
| Bp | sFTP | TCP | OFCS | Nokia | |

Table 15: Logical interfaces and respective transport protocol options

#### 6.2.1.1 E2E alignment of GTP time-out timers

It's common for different Network Elements on the EPC to be deployed using the same default GTP re-transmit timeout. This approach can hinder end-to-end timer optimization effort for different call flow scenarios, where signaling procedures can be carried out in core elements when the initial signaling RAN message that triggered them as already timed-out.

To optimize GTP signaling within the Packet Core network it is then good practice to engineer the message re-transmit timeout in the GTP layer, so that it is handled decrementally from eNodeB->MME->SGW to PGW.

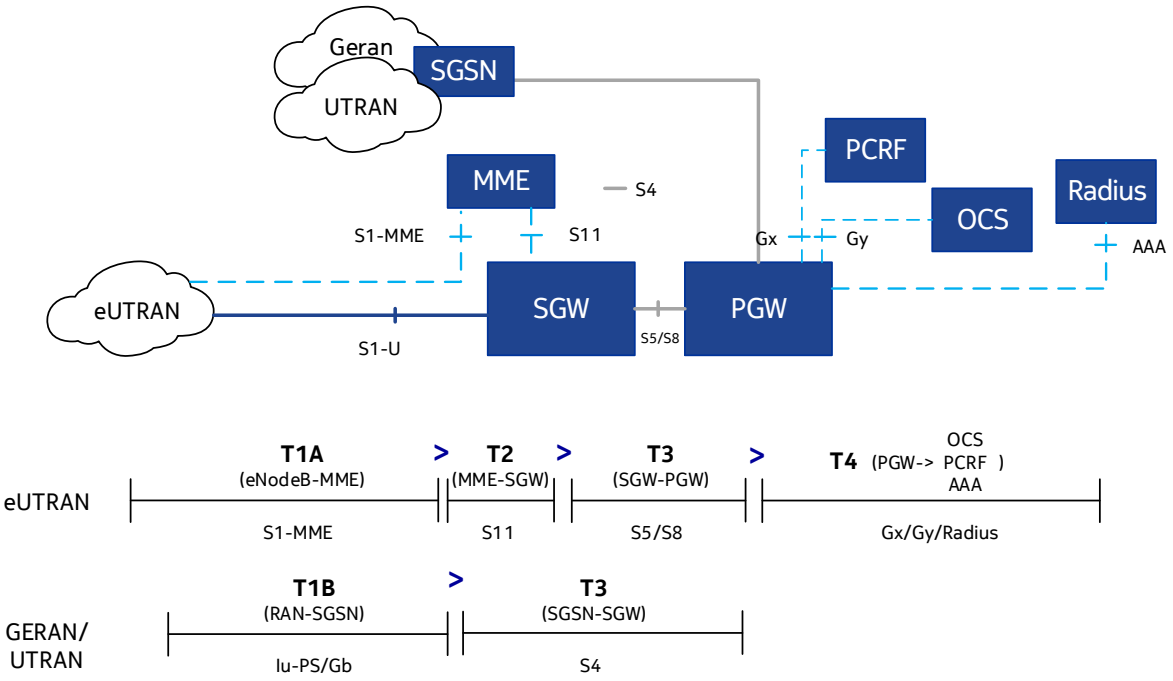



Figure 32: End 2 End Network Engineering of message timeout for signaling optimization

Figure 32 depicts the relationship that is required to be in place so that upstream elements timeout more quickly. Guaranteeing that for example, the SGW S5 reference point, times-out before the MME's S11 reference points. Making the MME able to differentiate between a case where the SGW is simply not responding and the case where the SGW has not yet timed out on its respective S5 interface.

So that timer manipulation is not restricted to "seconds" granularity, the GTP Profile configuration element allows "msg-retransmit" time-out to be configured in steps of 100 ms.

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## 6.3 APN

APN is used as an identifier to the PDN a user can connect. APNs need to be configured in PGW to identify the service provided by the external IP networks. Operators assign APNs to UE during user's subscription. During a PDN connection setup (i.e. when the user attaches to the network), the user provides the APN to indicate to the network the PDN that it wants to connect to. The APN has two main components:

- APN NI: name of the external PDN. It should follow the rule <APN_name> e.g. "internet"
- APN OI: encoded as "mnc<MNC>.mcc<MCC>.gprs"

The scope of this deployment is to provide a Cloud Packet Core to cater to both mass market mobile internet service and APNs used in either VAS or Enterprise applications, the extensive list of APNs considered under the scope of the final implementation, site presence and high-level APN scope is provided bellow in table format.

| APN Name | Network Location | SGI peering option | Comments |
|---|---|---|---|
| **Internet.local** | Regional/Aggregate-Edge | LB-less | |
| **Internet.global** | Central Data Center | LB-less | |
| **Enterprise.1001** | Central Data Center | LB based, MPLS option B | |
| **Enterprise.1002** | Central Data Center | LB based, MPLS option B | |
| … | Central Data Center | LB based, MPLS option B | |
| **Enterprise.1400** | Central Data Center | LB based, MPLS option B | |

**Table 16 :** APN survey

Below are provided the functional requirements for APN internet, the majority of which are shared with other APNs. A full list of functional requirements is to be included in the solution LLD.

| VM type | Network | Interface Type |
|---|---|---|
| APN-NI | Internet.global | |
| APN-OI | mnc00X.mccYYY.gprs | |
| QCI | 9 | Only QCI 9 in scope |
| Bearer/PDP type | IPv4v6 | |
| Address allocation | Local pool or HSS assigned | |
| Session-setup Timer | 10 sec | |
| Session time-out | 48h | |
| Idle-timeout | 10 min | Counted on up-link |
| Max. sessions | 5000000 | |
| tcp-mss-adjustment | 1358 | |
| Roaming-out | allowed | |
| Charging | Offline (Ga/Bp) | |
| PCO options | Primary & secondary DNS<br>link-mtu 1358 | |
| Router (VPRN) | SGi | |
| PCC Rules | PCRF assigned statically defined | |

**Table 17 :** APN service profile

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## 6.4 Policy and Charging

This section summarizes the functional requirements involved in the implementation of the Policy and Charging Enforcement Function (PCEF) within CMG, that together with the OCS and PCRF, comprise the main elements of the 3GPP PCC Architecture Figure **33**33.

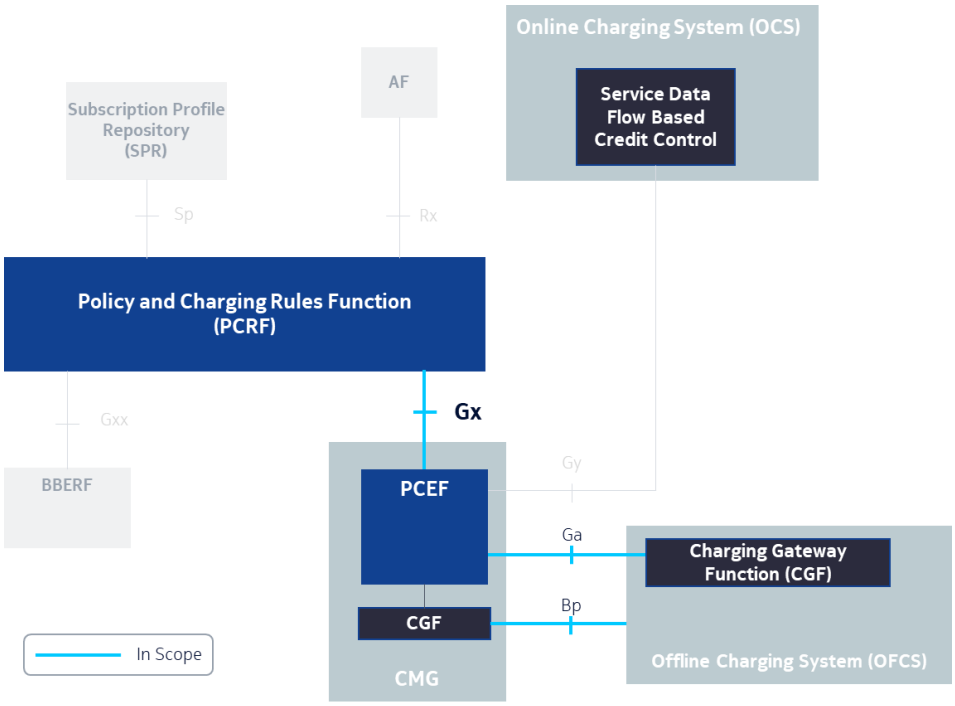



Figure 33: Current deployment scope of PCC Architecture

The Gx functional interface shall be configured in the SGW/PGW to support interactions between CMG PCEF engine and the Policy and Charging Rules Function (PCRF). PGW enforces QoS policies and charging rules (policy and Charging Control rules) forwarded from PCRF through the Gx interface.
Only Static Pre-Defined PCC Rules are considered under the current scope, these shall be locally configured on the SPGW and are activated or de-activated for a PDN session by the PCRF by referencing the corresponding rule name or rule base name. This is applicable for default bearers only.
Additionally, the following functionality is considered in scope of Gx integration with PCRF:

- Enhancement for Gx peering through **DRA**
- Enforcement of QoS changes through AMBR AVP at bearer level.
- Installation and removal of Static Pre-Defined PCC Rules through Gx
    - With a corresponding gating action and using 5 tuple filters as match criteria
    - Classify traffic flows under specific RG's (for offline charging purposes only)

LLD Documents [R5][R7] provide additional details surrounding PCC Rules and Gx specific flows.

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## 6.4.1 Online Charging

Online Charging – a 3GPP's term for Pre-paid charging, is a mechanism where charging information can affect, in real-time, the service rendered.

The integrated PCEF within PGW/GGSN performs flow-based online charging as specified in 3GPP TS32.251, and using the Diameter Credit Control Application defined in 3GPP TS 32.299 and RFC 4006.

IP-CAN bearer and Service Data Flow online charging are supported. In both cases, charging data pertain to sessions (IP-CAN bearers) and are associated to session based online charging (SCUR) with centralized rating and centralized unit.

The PGW/GGSN as the Charging Trigger Function (CTF) monitors the signalling, to identify the user's consumption of network resources and/or services. The PGW assembles the accounting metrics into charging events and forwards them to the OCS/Charging Gateway. The authorization for the network resource usage must be obtained from the OCS prior to using the service.

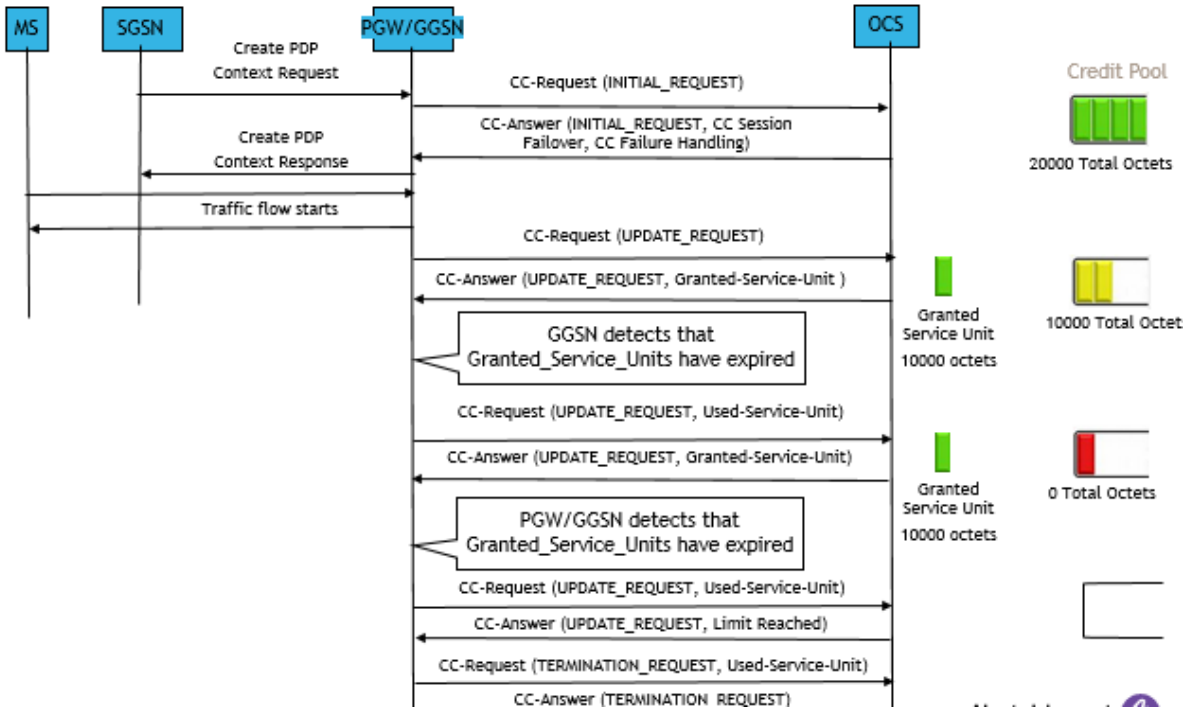



**Figure 34 – Data session Online Charging**

The OCS is defined as a Gy Diameter Peer and the online charging can be configured dynamically via a Charging rule AVP from the PCRF or statically via Charging rule unit config via static config.

The baseline for all charging interfaces in CMG is 3GPP Release 13 December 2016.

The following specifications are applicable:

- 3GPP TS 32.240 v13.2.0 (2016-06) — Charging Architecture and Principle
- 3GPP TS 32.251 v13.6.1 (2016-10) — Packet Switched (PS) domain charging
- 3GPP TS 32.299 v13.7.0 (2016-12) — Diameter Charging applications

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#### 6.4.1.1 Gy Diameter Profile

The following parameters shall be defined for the Diameter profile applied to Gy interface.

| Parameter | Value |
|---|---|
| connection-timer | 30 secs |
| transaction-timer | 8 secs |
| watchdog-timer | 30 secs |

**Table 18 :** Diameter profile parameters for Gy

Gy diameter peering using transport port 3868 is performed over TCP with x redundant OCS IP endpoints, where Gy sessions are to be load shared evenly across the following OCS server

#### 6.4.1.2 Failure handling

In case of OCS failure (Tx expiration) or abnormal response no failover procedure to alternative OCS is to be performed and subscriber session shall be terminated immediately without any retransmissions of CCR message and with no holding time interval.
New sessions will be shared across OCS servers who are responsive. If all OCS servers are unresponsive / not-available, no new subscriber sessions are allowed to be activated and current sessions shall be terminated with no holding timer enforced.

#### 6.4.1.3 Advice of Charge

For non-Roaming subscribers only, upon receiving DIAMETER_CREDIT_LIMIT_REACHED (4012) from OCS either at command-level or mscc-level, system shall redirect the user session to a predefined Advice-of-Charge Portal specified through IP address.

### 6.4.2 Offline Charging (Post-Paid)

Offline Charging is where the Charging Data Records (CDRs) are transferred from the PGW (Charging Trigger Function CTF) to the Billing Domain which comprises of the post-processing systems that does billing and accounting.

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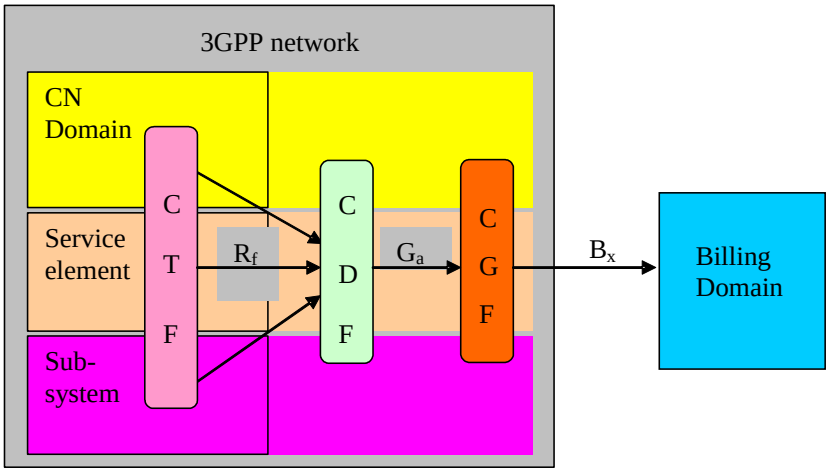



**Figure 35 : Offline Charging Architecture**

Charging Trigger Function (CTF): Collects information pertaining to chargeable events within the network element, and sends these charging events towards the Charging Data Function

Charging Data Function (CDF): Receives charging events from the Charging Trigger Function via the Rf reference point. It then uses the information contained in the charging events to construct CDRs (charging data records).

Charging Gateway Function (CGF): CDRs produced by the CDF are transferred immediately to the Charging Gateway Function (CGF) via the Ga reference point. The CGF acts as a gateway between the 3GPP network and the Billing Domain.

Session Based Ga Charging will be used on the SPGW supporting UDP only as the transport protocol. In the proposed architecture CGF role shall be performed by Nokia Data Refinery, responsible for converting the Ga UDP/GTP stream into CDRs that can then be retrieved by Safaricom Ethiopia mediation systems.

The following requirements shall be enforced in Ga/Bp to commit with the operator current network guidelines:

- CDRs shall contain IEs reporting on PGW assigned IP address and PLMN-ID of serving network
- CDRs shall be generated in compliance with TS 32.298 Release 9
- PGW generates an intermediate CDR when:
  - volume reaches 2MB
  - duration reaches 120 min
  - number of service containers in a CDR reaches 20

Values TBC with DR teams.

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#### 6.4.2.1 Failure handling

CDR files are transferred over the Ga interface and the Bp interface is used as backup if the Ga interface fails.

### 6.4.3 Charging Profiles

Charging Profiles can be used to customize the charging options for a PDN Connection/APN. It may also be useful to apply different charging profiles based on whether the Mobile is home, roaming or visiting. The EPC Node comprising of the MCC and MNC will be configured on the SPGW and is used for the decision criteria.

| UE Connection Type | Decision Criteria |
|---|---|
| HOME | IMSI MCC MNC = Serving Network MCC MNC = EPC-NODE MCC MNC (or in MCC MNC in home-plmn-list) |
| ROAMING | IMSI MCC MNC = EPC-NODE MCC MNC (or in MCC MNC in home-plmn-list) != Serving Network MCC MNC |
| VISITING | IMSI MCC MNC != EPC-NODE MCC MNC (or in MCC MNC in home-plmn-list) |

**Table 19 : Charging Profiles**

Home PLMNs can be configured as Lists on the SPGW.

Home, Roaming and Visiting UE's can have different default charging profiles.

### 6.4.4 Policy and Charging Control (PCC)

Dynamic PCC Rules are dynamically provisioned by PCRF to the SPGW (PCEF) over the Gx interface.

Pre-Defined PCC rules can be statically pre-configured in the SPGW and can be activated or deactivated by PCRF.

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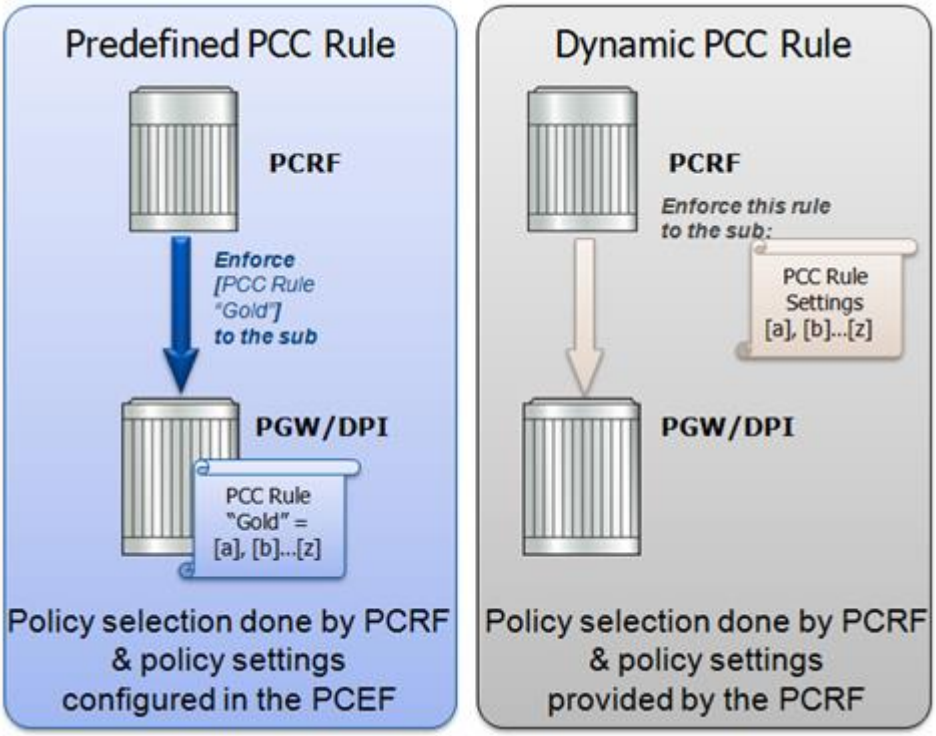



**Figure 36 :** *Pre-Defined PCC and Dynamic PCC*

When an event-trigger event occurs, the GGSN/SPGW sends a Credit Control Request (CCR-I in this case, because it's "Initial_Request") message that carries an event-trigger parameter to the PCRF, requesting to deliver PCC rules

In response to a request for PCC rules being made by the GGSN, the PCRF shall provision PCC rules in the CCA (Credit Control Answer)

#### 6.4.4.1 Static PCC Configurations

PCEF entity within PGW enforces QoS and charging rule (policy and Charging Control rules) forwarded from PCRF or static configured in PGW. PCC rule will either be pushed from PCRF to PGW (dynamic PCC rule) or statically configured on the PGW (static PCC rules) and configured for each APN.

The static PCC is called Policy Rule Base (PRB). Each PRB have multiple Policy Rules (PRs). Each PR includes the filters that identify an IP flow and assign QoS (qci and arp values) or gate control action defined in the PR. The gate control will decide to drop or discard the identified IP flow in uplink and/or in down-link direction.

The PCEF selects a Policy Charging Control (PCC) rule for each received downlink IP packet within a subscriber session by evaluating the packet against downlink service data flow filters in the order of the precedence (lower precedence value has higher priority). When a packet matches a service data flow filter, the PCC rule for that filter shall be applied. DL packets that do not match any PCC rule of the subscriber session are silently discarded.

Similarly, the UL packets are evaluated against uplink service data flow filters associated with the bearer/PDP context, in the order of the precedence of the PCC rules and once the packet matches

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a service data flow filter, the corresponding PCC rule is applied. UL packets that do not match any PCC rule of the IP-CAN session are silently discarded.

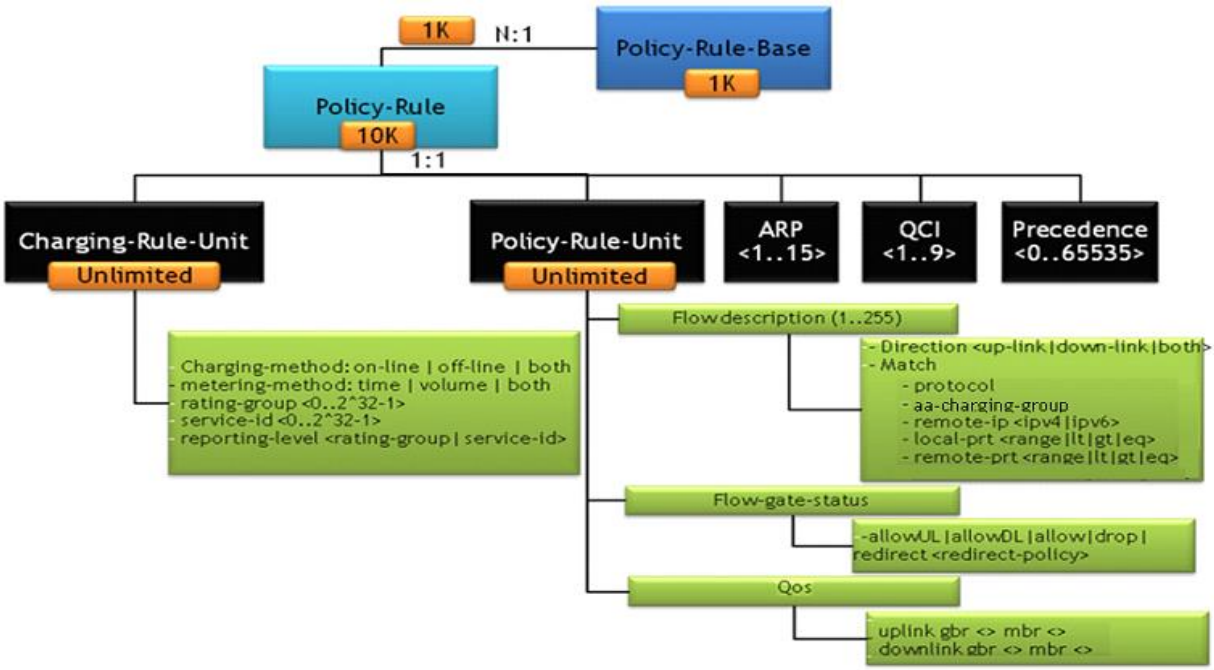



**Figure 37 :** Policy Rule/ Policy Rule Base Definitions

The configuration implementation for application identification policy rules is based on a two-level model:

1) At the first level, one or more Policy-Rule-Bases are configured, each with a set of policy-rules to:

   a) Identify all flows which can be identified just using 5-tuple classification ('shallow packet-inspection'). Unless there is a specific requirement to still perform signature-based inspection on these flows, identification of these flows completes at the policy-rule level and no processing by the L7 DPI processing engine is required.

   b) Filter traffic which requires L7 protocol analysis on the DPI inspection engine. The traffic-match criteria in the policy-rule will in this case refer to the name of a charging-group defined at the next hierarchy-level in an application policy partition on the AA engine.

**Example Policy Rule without DPI (> L4):**

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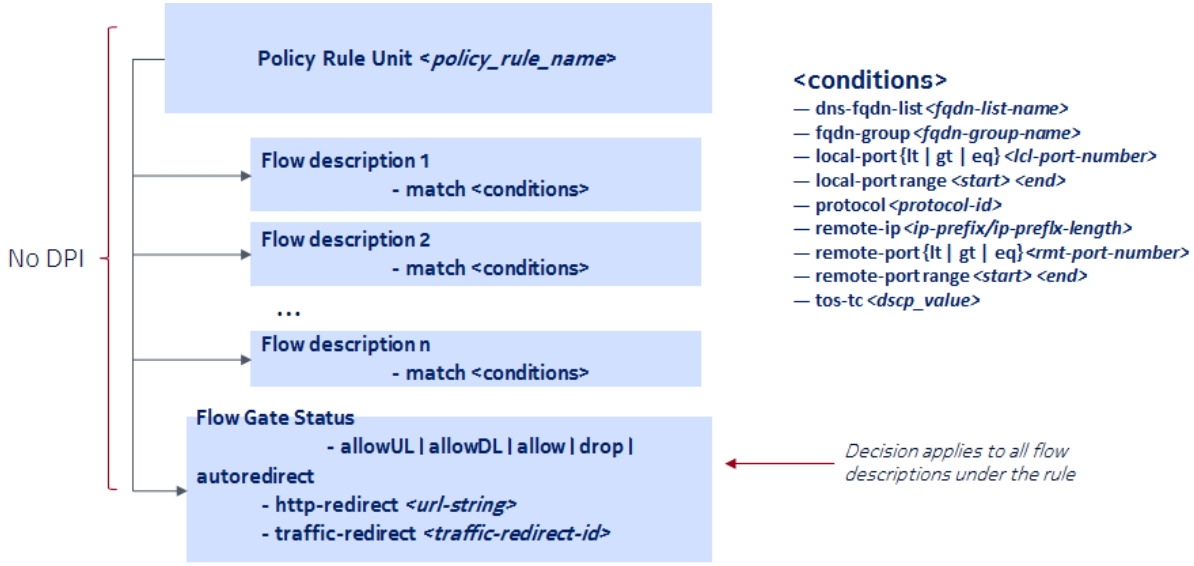



**Figure 38 :** Policy Rules without DPI

2) The second level is configured on the flow detection engine, to define L4-L7 flow match criteria for traffic which requires L7 protocol analysis including both well-known application protocols (like SIP) and heuristic analysis for encrypted protocol.

**Example Policy Rule with DPI:**

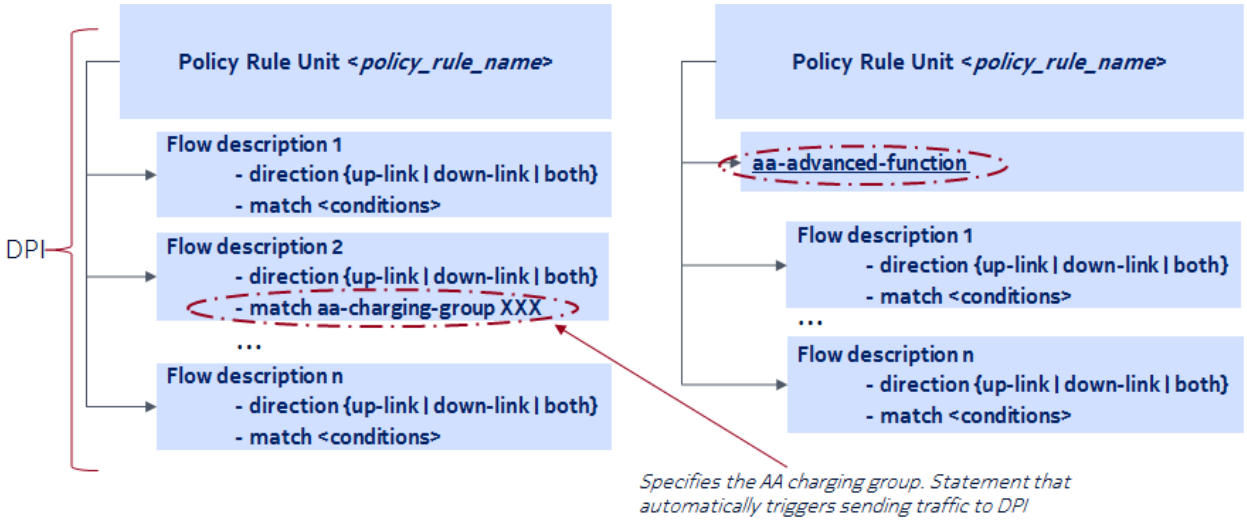
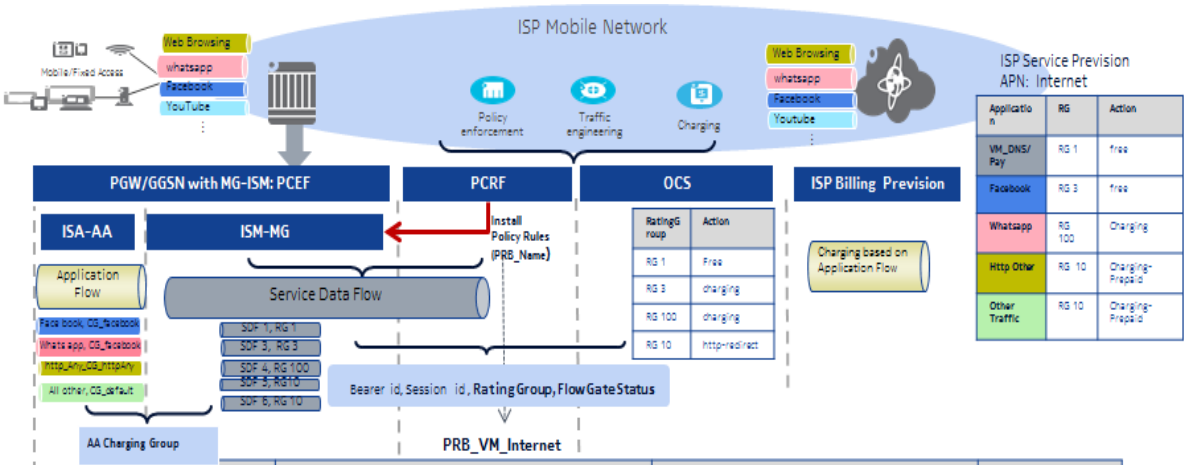



**Figure 39 :** Policy Rules without DPI



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**E2E View of Policy and Charging Rule:**




| PR /SDF: Name | PRU Name: SDF filters : flow- description, ip filter match criteria | | CRU Name: Map to Rating Group, charging method | | | Qos(Qci, ARP, Prece) |
|---|---|---|---|---|---|---|
| PR_VM_Pay | PRU_VM_PAY | src p/destip/port/protocal .. | CRU_VM_Pay | On/offline | RG 1 | 7,6, 98 |
| PR_FaceBook | PRU_Facebook | Match aa-charging-group Facebook | CRU_Facebook | On/offline | RG 3 | 7,6 ,320 |
| PR_whatsapp | PRU_whatsapp | Match aa-charging-group Whatsapp | CRU_Whatsapp | On/offline | RG 100 | 7,6, 350 |
| PR_http_any | PRU_http_any | match aa-charging-group httpany | CRU_http_any | On/offline | RG 10 | 7,6 ,65534 |
| All_Other | PRU_Other | match -any( catch all) | CRU_other | On/offline | RG 10 | 7,6,65535 |

**Figure 40 :** E2E view of Policy Rules

The picture above summarizes the concepts applied for Service flow identification and respective policies application

1) PCRF installs a Policy Rule Base (called Internet in this case)
2) Different Policy Rules are defined inside this Policy Rule Base
3) Each Policy Rule is then composed by a Policy Rule Unit and a Charging Rule Unit.
4) Each Policy Rule is composed also by a QoS field with QCI, ARP and Precedence

### 6.4.4.2 Application Level Control Policies

AQPs are a numbered list of configurable entries with match criteria and action configured at the group/partition level. They are complementing PCC rules.

AQP rules consist of match and action criteria:

- AQP Match Criteria

AQP can include the following match criteria: the source/destination IP address and port, flow direction Application name, Application Group or Charging Group name, one or more Application Service Option characteristics and value pairs.

- AQP Actions

It defines actions to be applied to the identified traffic. Several options are available:

  - Dual or single-bucket bandwidth rate limit policer;
  - Flow setup rate limit policer;
  - Total active flow count limit policer;
  - Remark QoS (one or a combination of discard priority, forwarding class name, DSCP). When applied, ingress marked FC and discard priority is overwritten by AA ISA

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and the new values are used during egress processing (for example, egress queuing or egress policy DSCP remarking).
- Drop (discard);
- None (monitor and report only);
- HTTP Redirect 404 redirect;
- HTTP Redirect 302 redirect: allows HTTP redirect for selective traffic steering of HTTP traffic while not affecting other traffic;
- HTTP header enrichment.
- Apply time-of-day criteria: where the policy action is only enforced when day or time-window criteria is met.

Application Profiles typically contain one or more characteristics defined as Application Service Options (ASOs). The operator can associate each APN profile (real or virtual) to a default application profile to customize services by matching a specific subset of application QoS policies.

The PCRF will customize characteristics values per subscriber on Gx interface.

#### 6.4.4.3 Application Assurance (AA) Protocols and Application Support

Nokia Application Assurance solution is a stateful packet and flow processing engine using state-of-the-art combinations of identification techniques to classify un-encrypted, encrypted, and obfuscated applications for data mining, charging, and control in Broadband, WiFi, Mobile, and Business VPN networks. Traffic identification accuracy with up-to-date traffic detection is achieved by constantly maintaining our database of supported application signatures, monitoring new functionalities and new software versions, and by adding new signatures based on customer feedback or market/region-specific requirements.

Application Assurance identification techniques:

- Layer 3 to 7 header and payload pattern-based signatures
- Inspection across packets and flows
- Control protocol monitoring to data flow association (for example, SIP/RTP, RTSP/RTP)
- Behavioural and numerical analysis

**APPLICATION SUPPORT LIST**: It provides a list of the latest supported protocols and applications per traffic categories: Business, Mail, Multimedia Streaming, Social Networking.

This grouping is fully flexible and can be configured to match specific service provider requirements in term for reporting and control.

AA Policy Configuration and expression-based applications (AA Policy Update include):

- Signature examples: **Facebook**, Twitter, **Youtube**, **Yourfreedom**, Whatsapp, **Pokemon Go**
- Capabilities: Various configurable match criteria besides protocol signatures are available for traffic refined from the following protocols: HTTP/HTTPS/HTTP2C, SSL/TLS, QUIC, DTLS, RTSP, RTMP, SIP, H323, and Citrix

Please refer to CMG 21.5 - AA Protocols and Application Support documentation for a detailed list of all protocols supported.

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## 6.5 TCP Optimization (TCP-O)

TCP is the dominant transport layer protocol of the current Internet, carrying around 90% of the total traffic. Furthermore, 50% of the TCP traffic is encrypted and unlike any L7 optimization techniques – such as video optimization- any optimization that functions at the TCP layer also benefits encrypted traffic.

Traditionally TCP connections have a number of factors that affects the throughput / latency:

- TCP window size
- Available bandwidth
- Round Trip delay (RTT)
- TCP reaction to error in both the “slow-start” and “congestion Avoidance”

The last two of which are especially problematic in mobile networks where RTT times and high packed drop are common factor when dealing with radio congestion, thus hindering twofold the throughput performance of TCP-based applications in mobile networks.

CMG leverages the functionality of its Application Assurance module to perform TCP optimization (TCP-O) on all subscriber TCP flows, allowing operators to leverage their existing access network deployments to provide higher throughput and improved user's experience. AA TCP optimization overcomes inherited inefficiencies of TCP when faced with larger round-trip delays, high bandwidth and non-congestion related packet loss which are common in radio access networks.

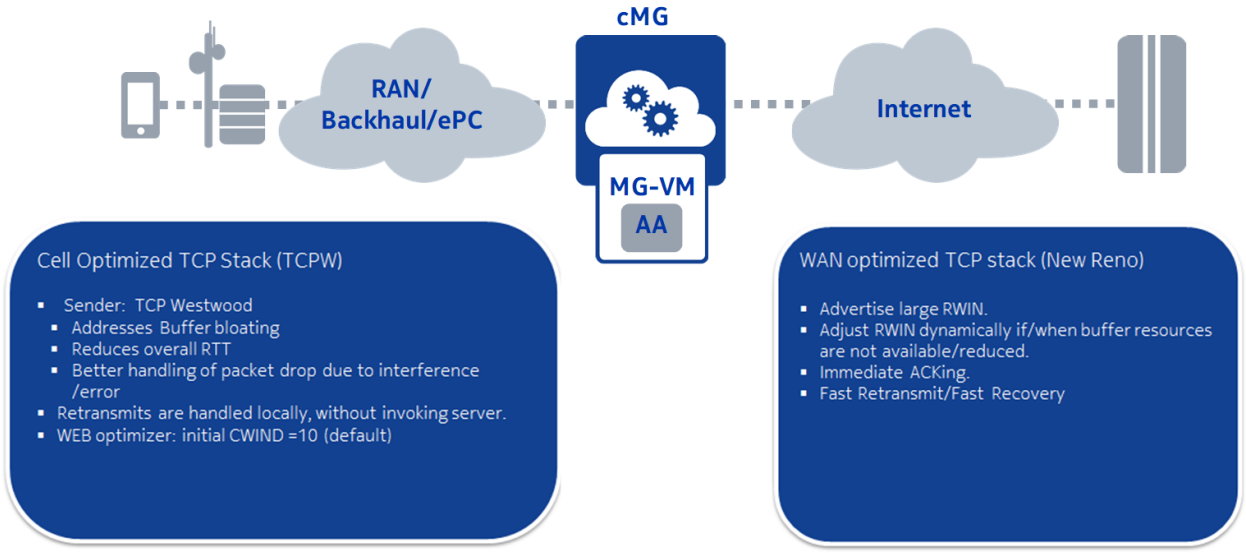



**Figure 41: - TCP-O function collocated with SGW/PGW**

As depicted in the figure above, TCP-O functions shall be collocated with SGW/PGW function under the same CMG instance. As its base functionality, AA TCP-O functions implement a TCP new-reno stack toward the core network and a TCP Westwood, TCP BBR, or TCP Illinois stack (configurable) towards the access network.

For interactions between AA TCPO and the rest of the AA feature set (such as policers), one can think as if AA-TCPO is logically implemented upstream from AA towards the core network, in essence it’s another service that gets chained to the IP flow after AA DPI.

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Once the functionality is enabled on CMG after diversion to Application Assurance module TCP traffic shall be subject to the following mechanism for optimization:

### Congestion Window Size

The initial congestion window size (cwind) is a key component of TCP slow start, also known as the exponential growth phase. Using standard TCP, a period of 3 RTTs is required for an average HTTP web load during slow start phase, which can happen after a packet loss.

Most web transactions are very short lived, and completed before slow start gets a chance to ramp up. To make full use of the significant bandwidth offered in today's mobile networks, AA TCPO sets the initial size of the cwind by default high enough to allow small web transaction to take place within one RTT period.

### Buffer Bloating

Deploy delay-based or bandwidth-based TCP sender stacks to overcome buffer-bloating issues, since these TCP congestion mechanisms use RTT or available bandwidth based on RTT measurements, as is the case with TCP Westwood, to adjust cwind size.

AA TCPO supports TCP Westwood, TCP BBR and TCP Illinois toward the mobile network. Running TCP Illinois, TCP BBR, or TCP Westwood stack does not only address the buffer bloating phenomenon, but also the inherent nature of TCP inefficiencies in mobile networks that are due to non-congestion related packet loss.

### Restrictions to TCP optimization in CMG

TCP-o does not perform any optimization for flows that have fragmented packets, IPv6 address extension headers, or unsupported/unknown TCP options (such as, TCPEncrypt). If such packets are received in the middle of the session, CMG shall cease ceases optimization for the rest of the session duration.

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## 6.6 AA Firewall

The Application Assurance firewall feature extends AA-ISA application level analysis to provide an in-line integrated stateful service that protects subscribers from malicious security attacks, AA stateful packet filtering feature, combined with AA Layer 7 classifications and control, empowers operators with an advanced, next-generation firewall functionality that is integrated within the PGW/GGSN/SSG.

### 6.6.1 Stateful Packet Filtering and Inspection

Stateless packet filtering does not take note of session initiator and hence, it discards or allows packets independent of the any previous packets. Stateless packet filtering can be performed in the system using IOM ACLs or PCC rules.

In stateful inspection, AA firewall not only inspects packets at Layers 3 to 7, but also monitors and keeps track of the connection's state. If the operator configures a "deny" action within a session filter, then the matching packets (matching both the AQP and associated session filter match conditions) are dropped and no flow session state/context is created.

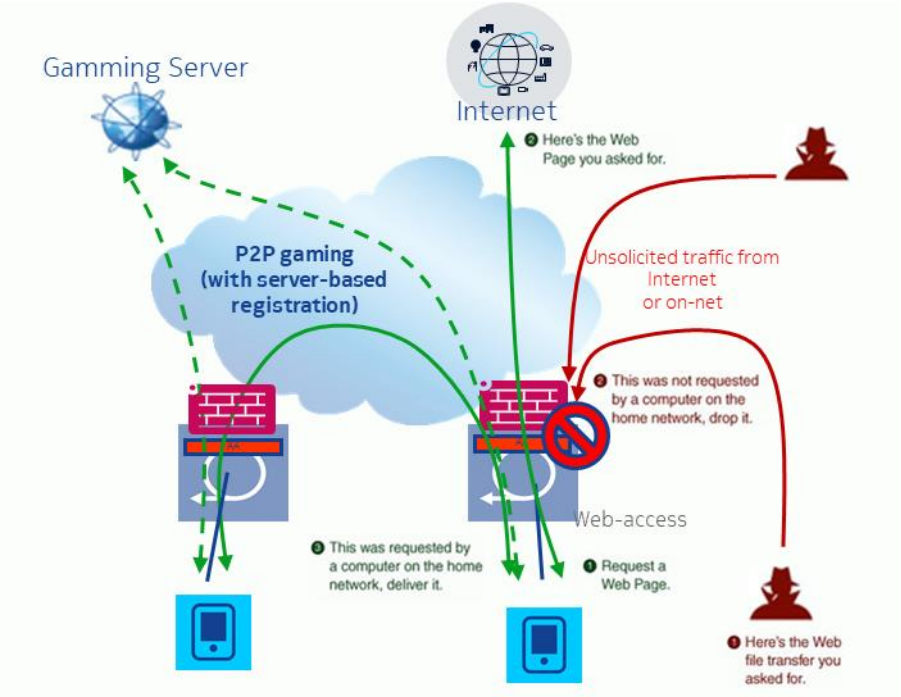



**Figure 42: – AA Stateful Packet filtering and inspection**

Stateful/stateless packet filtering and inspection with Application-level Gateway (ALG) support utilizes IP Layers 3/4 header information to build a state of the flow within AA ISA. Layer 7 inspection is used to provide ALG support. Stateful flow/session processing takes note of the originator of the session and hence, can allow traffic to be initiated from the subscriber, while denying, if configured, traffic originating from the network.

AA firewall inspection of packets at Layer 7 shall offer also Application Layer Gateway functionality for application make use of control channels/flows that spun other flows, inspecting the payload of these control flows so that it can open a pinhole for the associated required flows.

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| Application inspection | | |
|---|---|---|
| RTSP | Sybase | QVOD |
| SIP | Microsoft Exchange | "R" commands |
| H323 (IPv4 only) | Skinny | SC2 |
| Google Talk Voice | Ares | SOCKS |
| FTP | BitTorrent | VUDU |
| TFTP | DNS | WinMX |
| PPTP | IRC | Xunlei |
| Citrix | Mail.ru | |

### 6.6.2 Denial of Service (DoS) Protection

DoS attacks work by consuming network and system resources, making them unavailable for legitimate network applications. Network flooding attacks, malformed packets and port scans are examples of such DoS attacks. AA FW shall provide the following DoS mechanisms to protect subscribers and prevent any abuse of network resources:

- **Ports SCAN attacks:** Using AA FW stateful session filters, operators can protect their subscribers from any port scan scheme, by configuring the session filters to disallow any traffic that is initiated from the network
- **Flood Attacks Protection:** AA ISA provides configurable flow policers. These policers, once configured prevent all sort of flooding attacks (e.g. ICMP PING flooding, UDP flooding, SYN Flood Attack, ..etc).
  - These policers provide protection at multiple levels; per system per application/application groups and per subscriber per applications/applications groups.
  - AA ISA flow policers has two flavors:
    - flow setup rate policers to limit the number of new flows
    - flow count policers to limit the total number of active flows
- **Malformed Packets attacks:** In order to protect Hosts and network resources, AA FW performs validation on IP packets - at IP layer and TCP/UDP layer- to ensure that the packets are valid. Invalid packets are discarded (a configurable option). This provides protection against well know attacks such as Ping of Death attack, LAND attack (at IOM layer)…etc.

At the time of writing specific Firewall policies and DoS implementation specific requirements for Safaricom Ethiopia 's network have yet to be identified, this information shall be provided and thoroughly documented within CMG Low Level Design document.

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## 6.7 Carrier-Grade NAT (CGNAT)

With the shortage of IPv4 addresses, Service Providers (SPs) are many times obliged to provide a data access service where a public IPv4 address is shared by many subscribers. Each subscriber is assigned a private address, and a Network Address Translator (NAT) - RFC2663 - situated in the SP's network translates the traffic between private and public addresses.
Network Address Translation devices modify the IP headers of packets between a host and server, changing some or all of the following attributes: the source address, destination address, source port (TCP/UDP), destination port (TCP/UDP), or ICMP query ID (for ping).
It is common to refer to NAT solutions deployed in the Service Provider Core networks as Large-Scale Nat (LSN) or Carrier Grade Nat (CGN). In effect, the CGN/LSN terminology will be used interchangeably throughout this document and refers to a collection of Network Address Translation (NAT) techniques used in service provider network implemented on a highly scalable, high performance virtual function that facilitates various intra and inter-node redundancy mechanisms. The purpose of CGN/LSN semantics is to make delineation between high scale and high-performance NAT functions found in service provider networks, and enterprise NAT that is usually serving much smaller customer base at smaller speeds.
The target Nokia's Cloud Packet Core architecture uses the NAT features provided by CMG Application Assurance module to implement CGN/LSN capabilities collocated in the PGW.

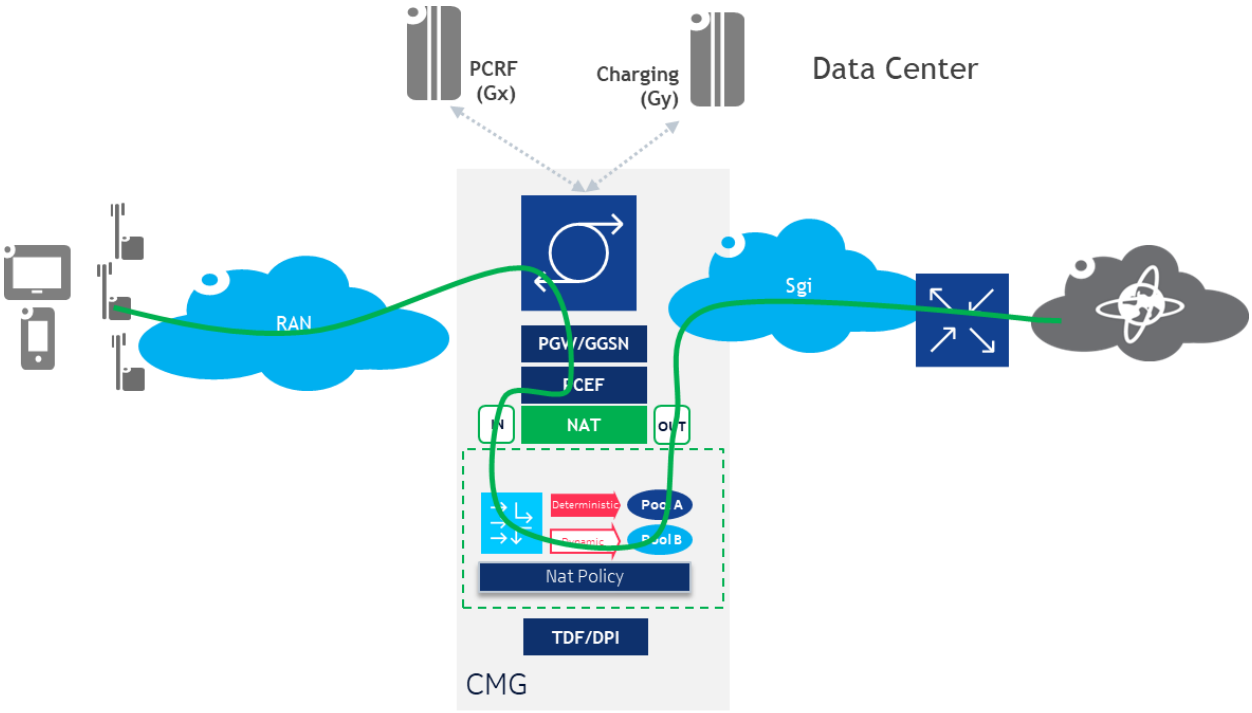



**Figure 43: NAT-ISA used in CMG for CGN/LSN**

CMG performs Source Network Address and Port Translation (S-NAPT). TCP/UDP connections use ports for multiplexing, with 65536 ports available for every IP address. Whenever many hosts are trying to share a single public IP address there is a chance of port collision where two different hosts may use the same source port for a connection. The resultant collision is avoided in S-NAPT devices

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by translating the source port and tracking this in a stateful manner. All S-NAPT devices are stateful in nature and must monitor connection establishment and traffic to maintain translation mappings

### 6.7.1 CGN/LSN requirements for application compatibility

Because subscribers do not receive unique IPv4 addresses, Carrier-Grade NATs introduce substantial limitations in communications between subscribers and the rest of the Internet.
To minimize these impacts CMG CGN/LSN implementation aligns itself with IETF's RFC 4787 -NAT behavioral requirements (and some of the updates introduced in RFC 7857), namely:

- The use of Paired IP address polling, where CMG uses the same external IP address mapping for all sessions associated with the same internal IP address, be they TCP, UDP, ICMP, or a mix of different protocols.
- Endpoint-Independent Filtering, in which sending packets from the internal side of the NAT to any external IP address is sufficient to allow any packets back to the internal endpoint.
- Endpoint-Independent Mappin, where CMG NAT heuristic reuses the port mapping for subsequent packets sent from the same internal IP address and port (X:x) to any external IP address and port.
- No port preservation, in the sense that aside from within the well-known ports range 0-1023 CMG does not attempt to make the external port numbers match the internal port numbers

Additionally, to minimize impacts on applications that operate behind NAT as servers, Port Forwarding is supported on CMG to allow servers which operate on well-known ports <1024 (such as HTTP and SMTP) to request the appropriate outside port for permanent allocation.
STUN support can also be enabled allowing STUN-capable applications to detect the NAT and allow inbound P2P connections for that specific service (i.e. SIP, IM , etc..) Both Port Forwarding and Stun support on CMG follow best-practice RFC for TCP (RFC 5382, NAT Behavioral Requirements for TCP) and UDP (RFC 4787, Network Address Translation (NAT) Behavioral Requirements for Unicast UDP).

### 6.7.2 Compression Ratio

The compression ratio is the ratio of

| <# nr of simultaneous users in BH> to <# of addresses available in the outside pool>. |
|---|

For example, 32 subscribers sharing 4 public addresses is a compression ratio of 32:4 or 8:1

Special care should be had when defining the compression ratio for NAT, given that larger compression ratios can impact the subscriber's Internet experience. While the traffic profile for mobile subscribers is typically less "session intensive" than for wireline, there is general consensus that Service Provider's should avoid compression ratios above 64:1.

The number of concurrent sessions per application (number of UDP and TCP flows an application triggers) and the number of applications used simultaneously by a single UE, are also important factors when defining the compression ratio of CGN, because there is trade-off between the efficiency of IPv4 address saving and the availability of those applications.
In this regard high session-consumption applications seen it the first years of CGN adoption no longer proliferate, applications such as Facebook went from abusing HTTP sessions too use several multiplexed requests over a single TCP connection, bringing the overall number of simultaneous sessions for a subscriber considerably lower.

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Factoring out impact on some peer-2-peer and gamming applications, for most traffic profiles, it should be safe to assume limitation of 1000 sessions per user would not affect most users while preventing the severe abuse from heavy users.
In any case compression ratio should always be determined by the Service Provider, weighting in three main factors:

- The number of simultaneous subscribers targeted for NAT in the BusyHour (BH)
- The number of Public IP addresses available in the Nat outside pool
- Traffic Profile of the target subscriber base, understanding the average number of sessions (TCP and UDP flows) of an active subscriber, and the gaussian distribution that goes with that average

At the time of writing, given the high restriction on the public addressing space available to Safaricom Ethiopia, the NAT compression ratio planned for deployment is roughly 250:1.

### 6.7.3 Deterministic vs. Non-Deterministic NAT

NAT in CMG can be implemented following a Deterministic or Dynamic LSN (Non-deterministic) outside IP-port mapping.

- **Deterministic NAT** deterministically maps an inside address to a specific port-block in the outside pool, ensuring that the inside address always gets the same port-block. This mapping is fixed by configuration and cannot change dynamically.

- **Non-deterministic NAT** allocates port-blocks to inside addresses on the fly, so the port-block allocated to an inside address may change over time. Given that ports blocks are assigned on demand, this allows for a more efficient scheduling of port blocks per subscriber when in comparison with deterministic NAT, where there is a static binding of “outside” port range per “inside” IP address.

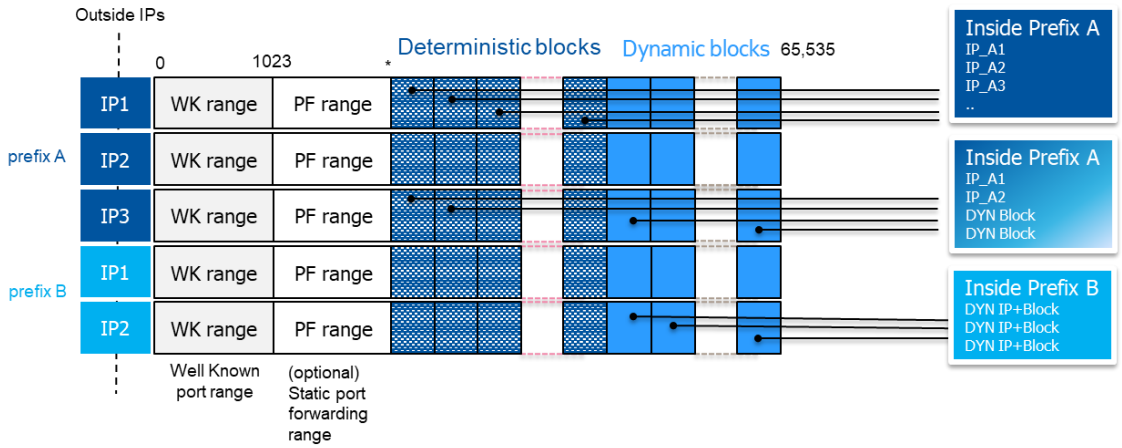



**Figure 44: Deterministic and/or Dynamic port-block allocation**

One downside of non-deterministic NAT is that the Service Provider typically needs to make a significant investment in logging infrastructure to log port-block allocations and de-allocations.
To process an abuse complaint triggered by law enforcement, the Service provider must determine the inside address corresponding to an outside address and port. With non-deterministic NAT, determining this mapping requires doing a search through logs (either manually or via scripts).
With deterministic NAT, determining this mapping requires running a program that algorithmically computes the mapping. The latter will usually be much simpler, more robust and not require a heavy investment in storage infrastructure.

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In general, Non-deterministic NAT is more appropriate in cases where the provider is really tight on free public IP address space and typically using high CGNAT compression ratios.
Furthermore, savings on logging infrastructure allowed by deterministic NAT may become negligible if the Service Provider has the requirements to record destination IP address and port information to unambiguously identify customers.

Given the 250:1 compression ratio planned per site and to align with the service Provider long-term requirement for destination IP logging, Non-deterministic NAT will be implemented.

### 6.7.4 Port-block allocation

There are two general models for a port-block allocation scheme:

- **Single Large Port-Block (SLPB):** Allocate a single port-block to each subscriber. Subscribers are no permitted to add additional ports.
- **Multiple Small Port-Blocks (MSPB):** Initially allocate a smaller port-block to a subscriber but allow subscribers to allocate additional port-blocks if required.

**SLPB** approach is simpler to describe, configure, and troubleshoot. Because there is never more than one port-block allocated to a subscriber, log entries related with this type of port-block allocation scheme are also fewer.

The **MSPB** model is most appropriate for Internet traffic at high compression ratios (32:1 or greater), as smaller chunk port-blocks are assigned on demand (up to a given maximum), this allows for a more efficient distribution of resources and most subscriber port requests can be honoured even at high compression ratios.

CMG CGN configuration shall enforce the **MSPB** model.

### 6.7.5 Connectivity State Optimization

Creating a NAT mapping is only one half of the problem – removing a NAT mapping at the appropriate time maximizes the shared port resource. Having ports mapped when an application is no longer active reduces solution scale and may impact the customer experience should they exhaust their port range block.

One such optimization on the timeout of NAT bindings, is reducing the TCP time-wait state. User's behavior normally correlates with the system performance. It's rather common that users often change video channels. It's investigated that 60% of videos are watched for less than 20% of their duration. The user's access patterns may leave a number of TIME-WAIT states. Therefore, acceleration of TIME-WAIT state transitions could increase the efficiency of port utilization.

CMG NAT application provides timeout configurations for TCP, UDP and ICMP which shall be set according to the best practice recommendations laid out in RFCs 4787, 5382, 5508 and 7857.

### 6.7.6 Watermarks

Once public IP Pools, port-block assignments and map binding timeouts have been set it is of critical importance to the Service Provider to be able to monitor actual usage of sessions and/or ports. A watermark is a construct that monitors the use of a particular resource and alerts the operator when

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consumption is getting high. CMG currently supports 4 different watermarks related to NAT resources.

- **NAT-group:** Watermarks can be placed to monitor the total number of sessions on an MDA.
- **Pool:** Watermarks can be placed to monitor the total number of blocks in use in a pool.
- **Policy:** In the policy it is possible to define watermarks on session and port usage. In both cases, it is the usage per host (for large-scale NAT).

For each watermark, a high and a low value shall be set following common network guidelines and in alignment with Service Provider monitoring requirements. The values for which shall be detailed in the Low Level Design.

## 6.7.7 Logging NAT binding records

CGN logging, which is referred in this context as the ability to generate records containing information related with NAT binding of "inside IP" to "outside IP + port block", is extremely important to the Service Providers (SP) who are typically required by government agencies to track source of suspicious Internet activities back to the users that are hidden behind the NAT service.

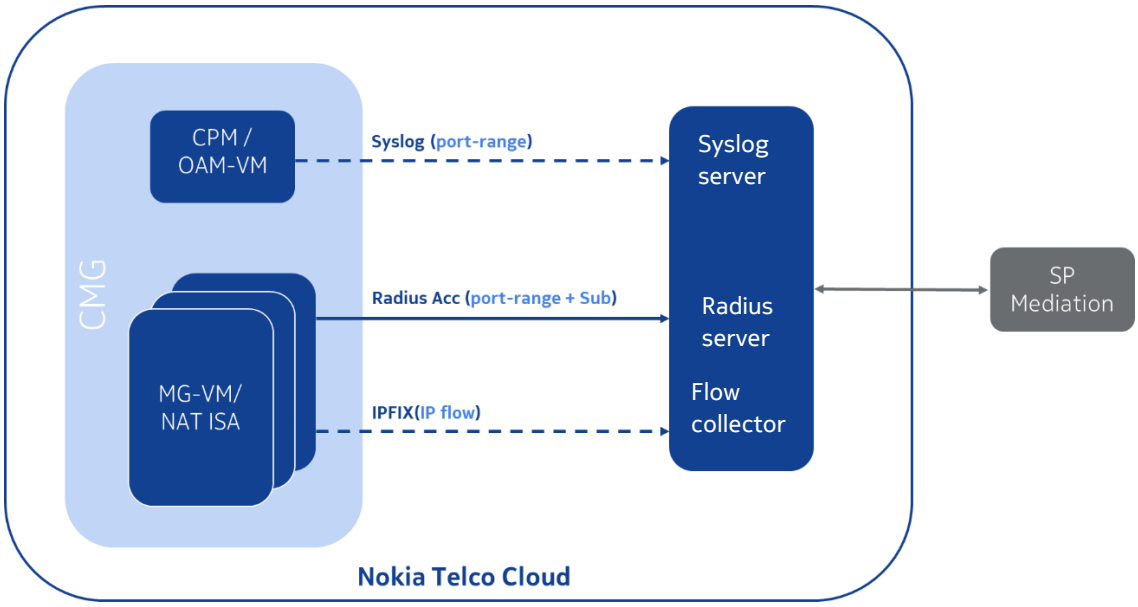



**Figure 45: NAT diversion methods**

CMG supports several modes of logging for NAT applications, which may be grouped into two main methods, Radius Nat Accounting or logging-facility. They are mutually exclusive, so only one method maybe pursued at any given time.

### 6.7.7.1 NAT binding records via logger

The simplest form of NAT logging is via the logging facility in CMG, commonly called logger. Each port-block allocation/de-allocation event will be recorded and sent to the system logging facility (logger). These events can then be:

- Recorded in the system memory as part of regular logs.
- Written to a local file.

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- Sent to an external server via syslog facility.
- Sent to a SNMT trap destination.

Choosing the right logging model will depend on the required scale, simplicity of deployment and granularity of the logged data. For most purposes logging of allocation/de-allocation of outside port-blocks and outside IP address along with the corresponding CGN/LSN subscriber and inside serviced will suffice.

### 6.7.7.2 RADIUS NAT accounting

CMG allows a Radius accounting policy to be tied to a non-deterministic NAT policy, so that NAT binding records can be propagated in real-time to a remote Radius server (over UDP but with message acknowledgment) for every port-block allocation and de-allocation.
Parameters such as IMSI, APN name, and PGW/GGSN IP address can be added to the RADIUS NAT binding record trough configuration, making this the only method under CMG release 10.R7 to provide built-in correlation of subscriber identity and NAT biding information.

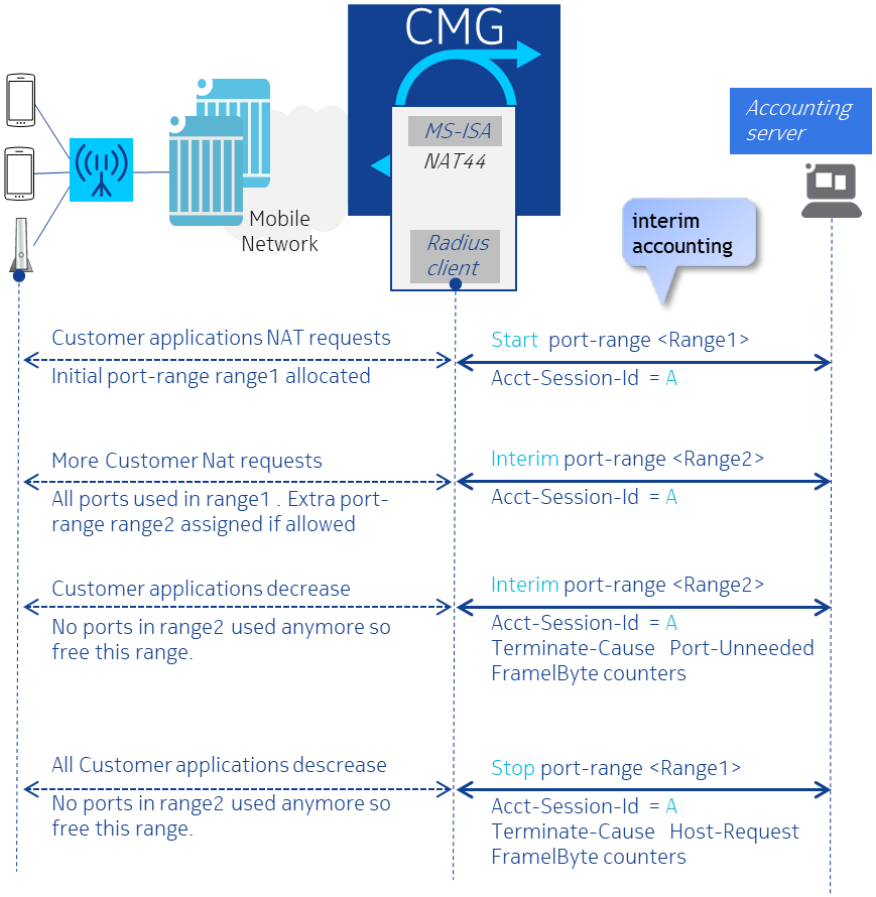



**Figure 46: NAT diversion methods**

The Service Provider can thus be exempt of the extra computational load related with the additional offline correlation of data. The only exception to this, are Bulk operations which are not captured by Radius NAT accounting methods.

A bulk operation, for example, represents the removal of the NAT-policy from the configuration, shutting down the NAT pool, or removing an IP address range from the pool, which may impact NAT bindings for millions of subscribers simultaneously. To prevent excessive logging during bulk operations, these events are instead captured in summarization logs. The summarization log for

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bulk operation while RADIUS logging is in effect will be generated only in the CPM (syslog). This means that for bulk operations with RADIUS logging, the operator will have to rely on RADIUS logging as well as on the CPM logging.

The alternative to the offline processing of summarization logs, would be for the Service Provider to put in place operational procedures that avoid bulk operations that impact NAT altogether. For example, offloading subscribers to another PGW before removing a NAT policy.

#### 6.7.7.3 Traceability of NAT subscribers

In many jurisdictions, service providers are legally obliged to provide the identity of a subscriber upon request to the appropriate authorities. However, where one public IPv4 address is shared between several subscribers, providing information about the external Address to a Service Provider is no longer sufficient to unambiguously identify customers.
There are two solutions to this problem, that are formalized in RFC 6269 as:

- Solution one, Internet content providers should additionally log the source port of incoming connections and include it in the legal request together with an exact timestamp for the period in which the abuse occurred.
- The second solution considers it unrealistic to expect all servers to log the source port number of incoming connections. To deal with this, service providers using IPv4 address sharing may need to log IP destination addresses.

To comply with this second requirement CMG supports LSN NAT flow logging, providing NAT MG pairs with the capability to export the creation and deletion of NAT flows to an external server. A NAT flow or a Fully Qualified Flow is formatted according to IETF IPFIX RFC 5101 and consists of the following parameters: Inside IP, inside port, outside IP, outside port, foreign IP, foreign port, protocol (UDP, TCP, ICMP).
Note that, destination logging is imperfect if multiple subscribers are accessing the same server at nearly the same time. If the destination IP address belongs to a popular internet website, it can still be impossible for the Service Provider to trace a particular activity to a specific subscriber. Thus, for traceability purposes, logging the destination address on the NAT is inferior to logging the source port at the server.

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# 7. Cloud Considerations and Requirements

## 7.1 VMware NFVi Solution

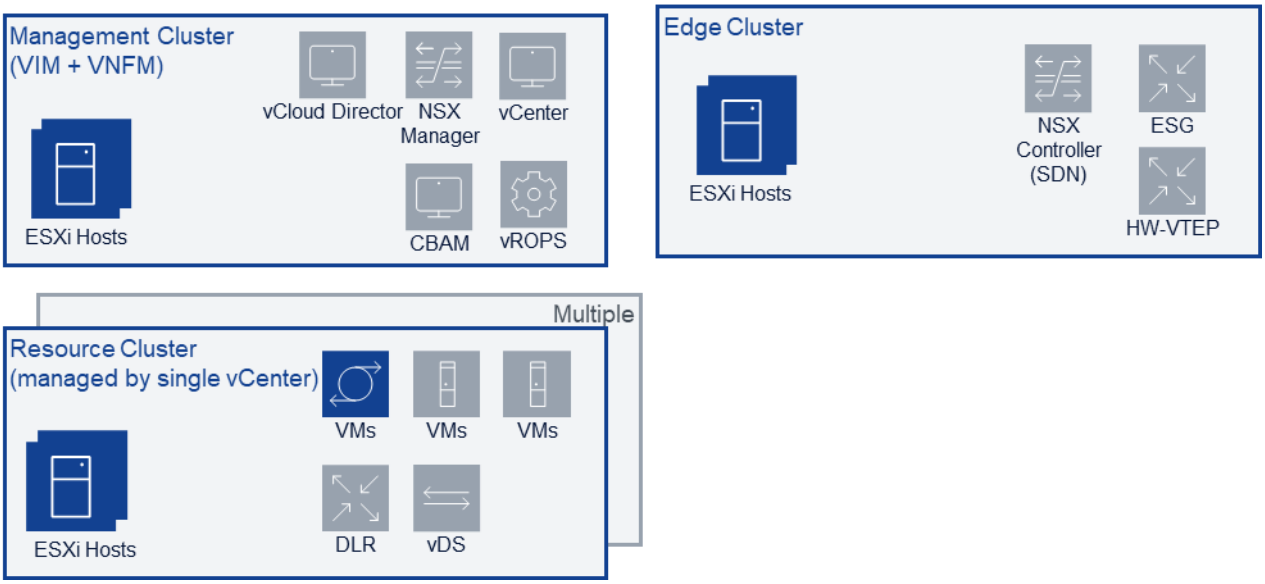



Figure 47: VMware NFVi Solution

**ETSI MANO Components**
VMware cloud software infrastructure version: NFV3.2.1
VMware NSX-V networking infrastructure: NSX-V with standard N-VDS and enhanced data path mode N-VDS
Interfaces: VMXNET3
vCloud Director version: 9.7
VMware ESXi version: 6.7
Virtual Hardware version: 14
VNFM: (VNF Catalog and TLM): Nokia CBAM

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## 7.2 VM Connectivity Options

Below are the CMG and CMM Traffic Segregation methods (VMXNET3) proposed

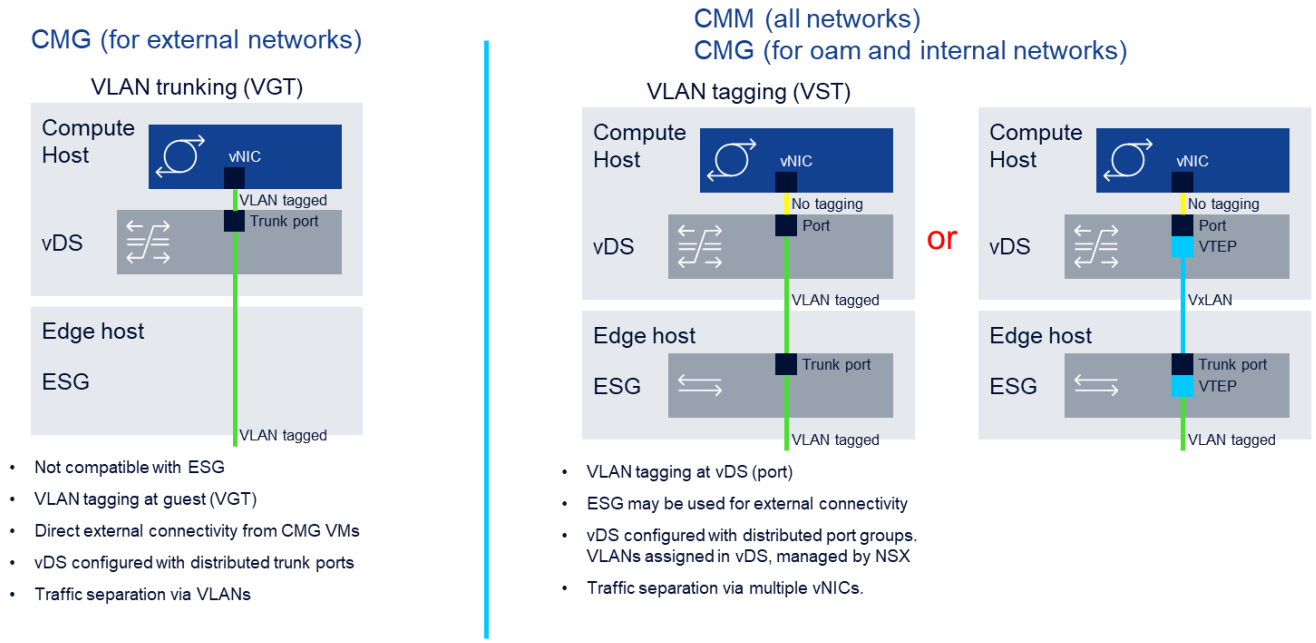



Figure 48: VM Connectivity Options

## 7.3 vSphere Cluster Level Considerations

| Cluster Parameters | Recommended Setting | Comments |
|---|---|---|
| vSphere HA | Disabled | Not Supported by CMG |
| vSphere DRS | Manual or Partially Automated | To ensure that vMotion is not triggered arbitrarily on any MG-VM carrying user traffic. |
| vSphere DRS Advanced Parameters: das.respectVmVmAntiAffinityRules = true | True | Respect vm-vm anti-affinity rules when restarting virtual machines after a failure. |
| vSphere DRS Advanced Parameters: LimitVMsPerESXHost = 2 | 2 | Maximum number of VMs that can be deployed per host |

Table 20: vSphere Cluster Level Considerations

Note : CMG supports the migration of the virtual machines with VMXNET3 connectivity using the vSphere vMotion for maintenance or resource optimization purposes.
The supported solution requires a separate network to be available for the vMotion traffic, as described in the VMware NFV2.0 and NFV3.2 reference architecture.

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## 7.4 vCD Organization vDC Design

Nokia recommends having a dedicated Organization vDC for each CPC VNF in order to better allocate and monitor the virtual resources per each VNF. That way each Nokia VNF composed out of multiple VNF Components (VNFCs) is better seen as one entity for networking and operability purposes. Also, the CPU/RAM/Disk resources can be guaranteed for the whole VNFs, not to individual VNFC/Virtual Machines only. In case the virtual resources were allocated on VM level, and not on the Organization vDC level, then some critical VMs for the whole VNF performance might not get virtual resources

| CPC VNF Organization vDC Design Specifications | |
|---|---|
| Allocation Model | Reservation Pool |
| CPU resources guaranteed | 100% |
| Memory resources guaranteed | 100% |
| Thin Provisioning and Fast provisioning | Disabled |

Table 21: vCD Organization vDC Design

## 7.5 VMware Host Performance Tuning

Align the host setting as per the VMWare white paper i.e. *Best Practices for Performance Tuning of Telco and NFV Workloads in vSphere* - TECHNICAL WHITE PAPER

| Tuning Method | Configuration | Technical Justification |
|---|---|---|
| BIOS - Power Management | Static High-Performance Mode | |
| BIOS - Hyperthreading | Enabled | |
| BIOS – Turbo Boost | Enabled | |
| BIOS – NUMA Node Interleaving | Disabled | NUMA Mode is Enabled |
| pNIC - MTU | MTU = 9000 | |
| pNIC - Rx/Tx ring buffer sizes | Settings depend on NIC vendor<br>Latest PMD driver must be installed all hosts | |
| vDS - MTU | MTU = 9000 | |
| vDS – Promiscuous Mode | Enabled for CMG Control Fabric Port-Group | |
| vCD OvDC – Allocation Pool Model | Reservation Pool (CPU = 100%; MEM = 100%) | |
| vCD Org – User Additional Rights | Preserve ExtraConfig Elements | Allow import ExtraConfig Params in OVF |

Table 22: VMware Host Performance Tuning

- The CMG control switch fabric and the management network require the security settings to allow Promiscuous mode.
- The CMG internal control network requires an MTU size of **9000**. To set the MTU size use the vCenter client. Under the **Networking tab**, select **vDS** and then navigate to **Edit Settings**
- The transmit (Tx) and receive (Rx) buffers of the pNIC must be increased to reduce the possibility of packet drop, especially with burst traffic. It is recommended that the Tx buffer ring is set to a higher value than the Rx buffer ring.

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### 7.5.1 Edge Services Gateway (ESG):

If VxLAN, or other network overlay technology, is used for CPC Userplane external networks, the use of an Edge Services Gateway (ESG) for VxLAN-to-VLAN mapping will become a performance bottleneck even if vNICs that support VxLAN offloading are used. If the use of VxLANs is preferred for external networks, the use of a hardware VTEP that is integrated with NSX is recommended instead of the ESG.

### 7.5.2 Distributed firewall (DFW):

The use of a DFW results in performance penalties both on the resource cluster and the edge cluster even if no rules are configured for CPC VMs, because the DFW applies and therefore any rule for all traffic applies to ingress and egress traffic of CPC VMs. For optimal performance, it is recommended to:

- Exclude CPC VMs from DFW via NSX configuration in NFV2.0 deployments.
- Disable the DFW function or exclude the logical switch from the DFW via NSX configuration in NFV3.2 deployments.

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# 8. Network Layout and Site Connectivity

## 8.1 CMG Logical Connectivity

NFV3.0 LB Mode architecture (VMXNET3)

1. Simplex Connectivity, N-vDS ensures uplink redundancy.
2. LB External Interfaces (1..7 vNICs).
3. Separate Enhanced Data Path N-vDS for high throughput vNICs.

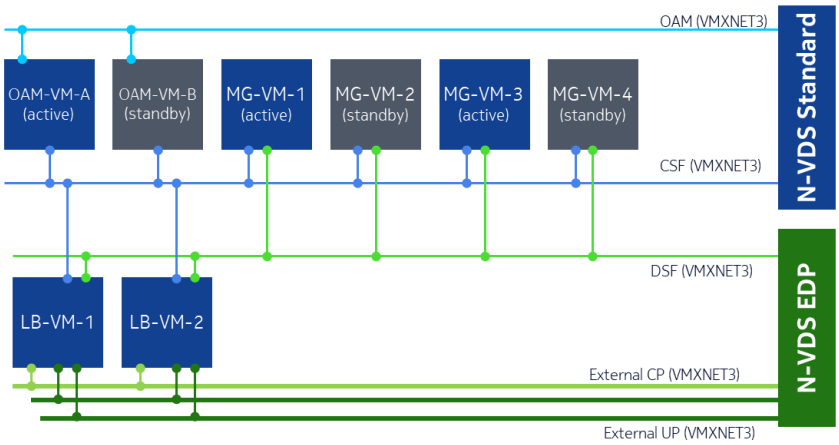



**Figure 49:** NFV3.0 LB Mode architecture (VMXNET3)

NFV3.0 LB Less architecture (VMXNET3)

1. Simplex Connectivity, N-vDS ensures uplink redundancy.
2. LB External Interfaces (1..7 vNICs).
3. MG External Interfaces (1 or 2 vNICs) .
4. Separate Enhanced Data Path N-vDS for high throughput vNICs.

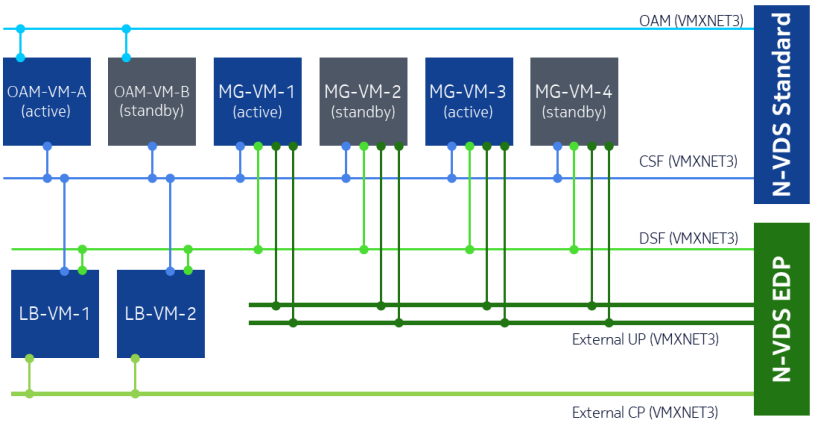



**Figure 50:** NFV3.0 LB Less Mode architecture (VMXNET3)

CMG-c VNF will be deployed in LB mode
CMG-u VNF will be deployed in LB-Less mode

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## 8.2 VRF Definition

Below are the VRFs that will be defined at the PaCo VNFs and the site route

| Site Router VRF | Traffic | Protocol / Comment |
|---|---|---|
| SFC_OAM | OAM, NTP, Bp, Ga, CMM mirror | TCP/UDP/ssh/sftp/telnet/ftp |
| SFC_SIGNAL | Gr,SGs,S6a,S13, Gx,Gy, aaa | sctp, diameter, radius .... |
| SFC_RANCP | IuC,S1-MME | sctp, RANAP, MAP ... |
| SFC_VOICE | MGW Userplane ( The relevant interface(s) will be shared by the CS Core Team ) | ... |
| SFC_ACCESS | SWu | IKEv2/ESP-Based on top of udp (VoWIFI access traffic) |
| SFC_LI | LI ( X1-1, X2 and X3 ) | N/A |
| SFC_CORECP | GNc,GNc, S10,DNS,Sv,S5c,s11c,s11u,Sx,cdbx,s2a,s2b,Gpc,s8c | GTPc/GTPv2/UDP ... |
| SFC_COREUP | 2G GnU,3G GnU, DT,GnU,S5u,S2au,S2bu,GpU,S8u | GTPu/GTPv1/UDP ... |
| SFC_RANUP | Gb, Iu-U, S1u | GTPu/ GTPv1/UDP, BSSGP ... |
| SFC_GI | Internet APN | TCP/UDP/IP ( SGi Traffic ) ... |
| SFC_IMS | ims APN | RTP ( IMS ) ... |
| SFC_CORP | corporate1 APN | TCP/UDP/IP ( SGi Traffic ) ... |
| SFC_ABIS | ABIS | |
| SFC-IUB | IUB | |

Table 23: VRF Definition

The following VRFs are non-PS Core VRFs

i. SFC_VOICE,
ii. SFC_ABIS
iii. SFC_IUB

Direct Tunnel (DT), that is, the link between RNC and CMG-GGSN will be implemented, then we will need to have a route leak between RANUP and COREUP, below are the reasons :

i. Without DT : RNC user plane is terminated on CMM (SGSN)
ii. With DT : RNC user plane is terminated on CMGu (GGSN)
iii. RNC Userplane is on RANUP, CMG Userplane is on COREUP

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# 8.3 VRF Logical Diagram

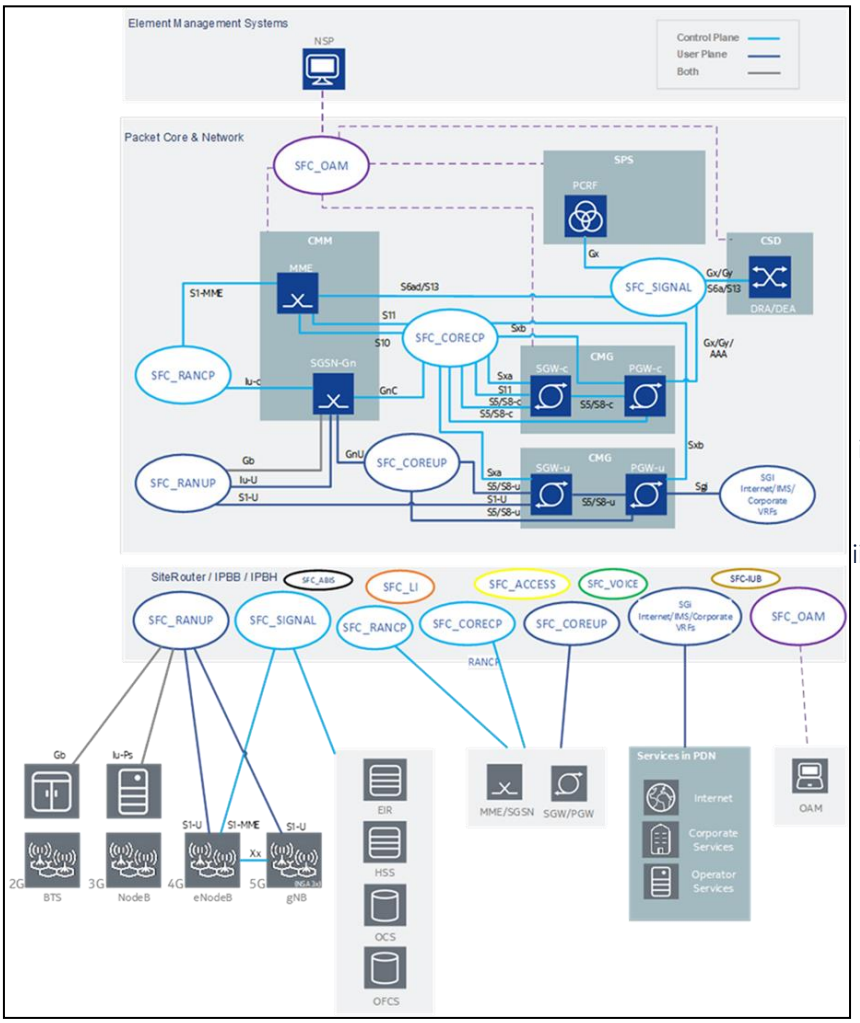



**Figure 51:** VRF Logical Diagram

Direct Tunnel (DT), that is, the link between RNC and CMG-GGSN will be implemented, then we will need to have a route leak between RANUP and COREUP, below are the reasons :

i. Without DT : RNC user plane is terminated on CMM (SGSN)
ii. With DT : RNC user plane is terminated on CMGu (GGSN)
iii. RNC Userplane is on RANUP, CMG Userplane is on COREUP

# 8.4 Routing Protocols

### 8.4.1 **CMM**

All CMM networks use default route on each vlan at the CMM. At the Arista gateway, it is a local route.

### 8.4.2 CMG

CMG OAM network use a static route at the CMG. At the Arista gateway, it is a local route.
The other CMG networks are p2p networks running eBGP.

- ✓ eBGP will be enabled on these links
- ✓ bfd will be enabled on these links
- ✓ ECMP will be enabled

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# 8.5 Overlay Design

## 8.5.1 CMM – single-homing interfaces

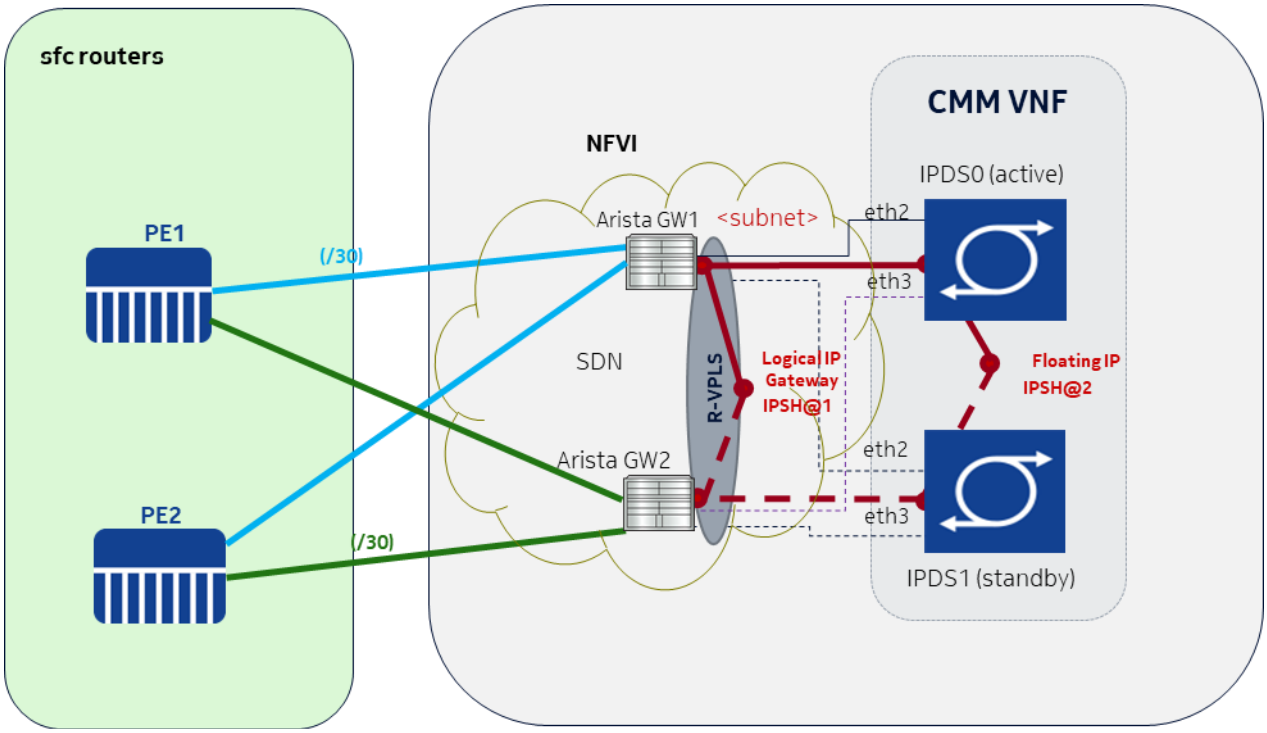



**Figure 52:** CMM Overlay Design for single-homing interfaces

Each network uses duplicated interface thus is assigned to 2 vNIC (eth). Both eth ports are attached to the same subnet (or L2 segment). Active/standby bond interface resiliency mechanism is implemented by the guest OS at the CMM, which handles the active/standby status of each interface.

i. Between the CMM VMs and the Arista GW1 there might be other L2 devices / IP nodes so the link between the IPDS and GW1 is a logical L3 link
ii. The Arista GW1/GW2 can be active/active respectively
iii. The VRRP is one way of redundancy, but it is also possible to implement other redundancy mode, the key point is that incase GW1 is down then GW2 should handle the traffic and vice-versa
iv. The number of links between the arista GW and PE will depend on Arista's team decision

Note: OAM connectivity is aligned with single-homing connectivity concept

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## 8.5.2 CMM – SCTP multi-homing interfaces

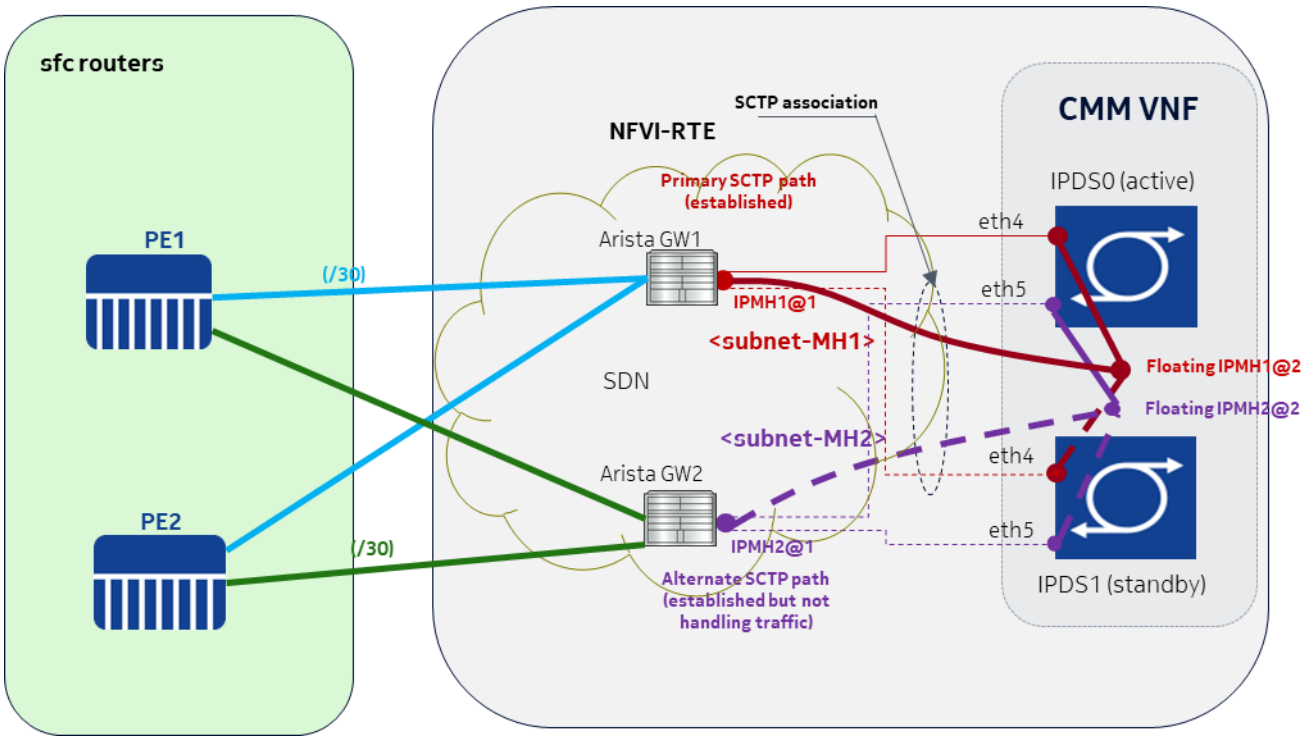



**Figure 53:** CMM Overlay Design for multi-homing interfaces

MH: 2 networks and each use simplex interface thus assigned to only 1 vNIC (eth). No L2 aggregation in SDN. The CMM guest OS is responsible for handling the interface active/standby status

i. Between the CMM VMs and the Arista GW1 there might be other L2 devices / IP nodes so the link between the IPDS and GW1 is a logical L3 link
ii. The Arista GW1/GW2 can be active/active respectively. The VRRP is one way of redundancy, but it is also possible to implement other redundancy mode, the key point is that incase GW1 is down then GW2 should handle the traffic and vice-versa
iii. The number of links between the arista GW and PE will depend on Arista's team decision

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### 8.5.3 CMG – LB VMs connectivity

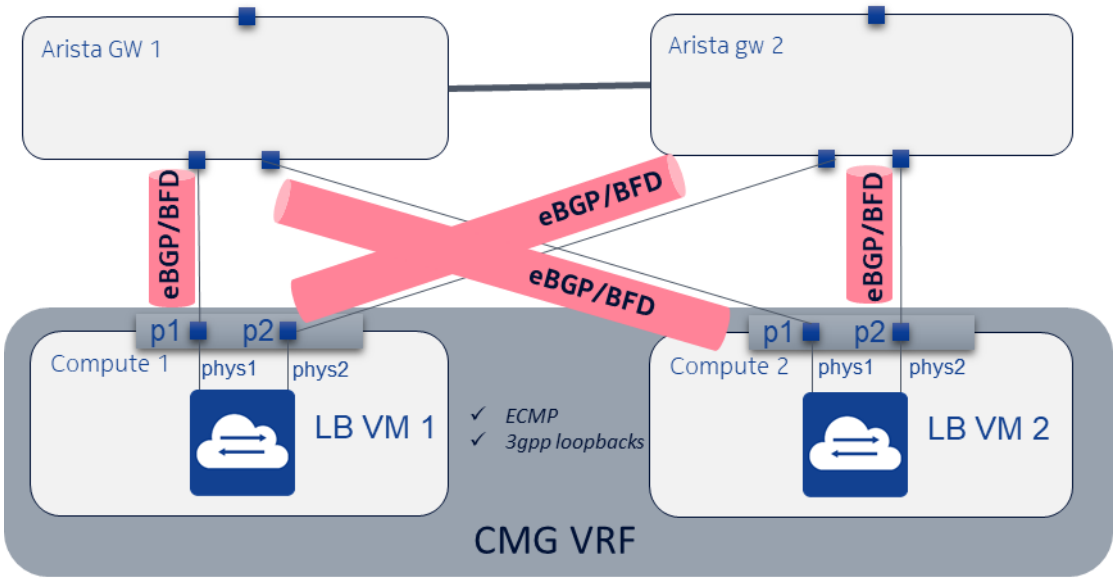



**Figure 54:** CMG Overlay Design for LB VM connectivity

### 8.5.4 CMG – MG VMs connectivity

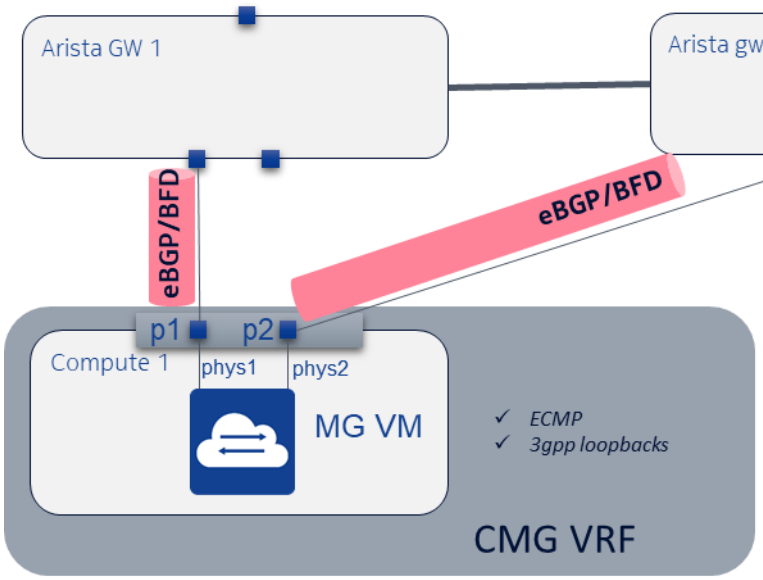



**Figure 55:** CMG Overlay Design for MG VM connectivity

## 8.6 IPVLAN Planning

The relevant information will be found on the CiQ

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# 9. Quality of Service

This section covers different architectural aspects of the end-to-end Quality of Service concept within the Cloud Packet Core architecture with varying level of detail. In general, industry specifications shall be referred to for further details; these other specifications enable the reader to acquire the full understanding of the end-to-end Quality of Service concept and architecture. As a packet core solution, the QoS assumptions provided bellow take heavily from the GPRS Quality of Service mechanisms and concepts defined within TS 23.107/23.207 and from the specification for the PCC architecture defined in TS 23.203. QoS enforcement within the Cloud Packet Core must be managed across two domains:

1. The fist domain deals with QoS the mechanisms defined by 3GPP to guarantee the reservation of resources for different types of subscriber/bearer within the Network Functions that make up the 3GPP Packet Core architecture.
2. The second domain handles Service prioritization within the VNF Infrastructure, which is comprised by the HW/Hypervisor associated to computational resources and the OVS and packet-forwarding elements related to the could networking.

To provide IP QoS within the CPC, it is necessary to manage the QoS within each domain. Given that each realm uses different QoS techniques, the VNFs delivering against MME/SGSN/SGW/PGW capabilities support translation functions to map 3GPP QoS classes into IP based DSCP marking.

## 9.1 Traffic classification within the 3GPP Packet Core Architecture

Within the 3GPP Packet Core architecture, the PDN connectivity service is provided by a bearer, called a PDP context in Pre-Release 8 (GPRS/UMTS) and an EPS bearer from Release 8 onwards. As with PDP context, an EPS bearer is the level of granularity for bearer level QoS control in the Packet Core. That is, all traffic mapped to the same bearer receives the same level packet forwarding treatment between UE and EPC (for example, scheduling policy, queue management policy, rate shaping policy). Providing different bearer level packet forwarding treatment requires separate dedicated bearers, which may be established on demand, and guarantee the reservation of resources to comply with the intended QoS Class.

Even when dedicated bearer QoS support is not in the scope of the Cloud Packet Core deployment, as in the current case, QoS class definition within the CPC must be aligned with 3GPP so that the overall cloud solution provides built-in support for the future addition of Guaranteed Bit Rate services like VoLTE (requiring no re-engineering of e2e QoS in the cloud when such services are introduced).

### 9.1.1 UMTS QoS parameters

3gpp defines four UMTS QoS classes to offer standardized measurable characteristics for implementing QoS:

- Conversational class (only 3G)
- Streaming class
- Interactive class Traffic Handling Priorities 1, 2, 3
- Background class

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These traffic classes can be divided into real-time classes (conversational and streaming) and non-real-time classes (interactive and background).

The following table lists the value ranges of the UMTS bearer service attributes. The value ranges reflect the capability of UMTS network. The values applied in each scenario are specific to a given service or telecom use-case and will not be addressed here.

| Traffic class | Conversational class | Streaming class | Interactive class | Background class |
|---|---|---|---|---|
| **Maximum bitrate (Gbps)** | <= 10 | <= 10 | <= 10 | <= 10 |
| **Delivery order** | Yes/No | Yes/No | Yes/No | Yes/No |
| **Maximum SDU size (octets)** | <=1 500 or 1 502 | <=1 500 or 1 502 | <=1 500 or 1 502 | <=1 500 or 1 502 |
| **SDU format information** | -5 | -5 | | |
| **Delivery of erroneous SDUs** | Yes/No | Yes/No | Yes/No | Yes/No |
| **Residual BER** | $5*10^{-2}$, $10^{-2}$, $5*10^{-3}$, $10^{-3}$, $10^{-4}$, $10^{-5}$, $10^{-6}$ | $5*10^{-2}$, $10^{-2}$, $5*10^{-3}$, $10^{-3}$, $10^{-4}$, $10^{-5}$, $10^{-6}$ | $4*10^{-3}$, $10^{-5}$, $6*10^{-8}$ | $4*10^{-3}$, $10^{-5}$, $6*10^{-8}$ |
| **SDU error ratio** | $10^{-2}$, $7*10^{-3}$, $10^{-3}$, $10^{-4}$, $10^{-5}$ | $10^{-1}$, $10^{-2}$, $7*10^{-3}$, $10^{-3}$, $10^{-4}$, $10^{-5}$ | $10^{-3}$, $10^{-4}$, $10^{-6}$ | $10^{-3}$, $10^{-4}$, $10^{-6}$ |
| **Transfer delay (ms)** | 100 – maximum value | 300 – maximum value | | |
| **Guaranteed bit rate (Gbps)** | <= 10 | <= 10 | | |
| **Traffic handling priority** | | | 1,2,3 | |
| **Allocation/Retention priority** | 1,2,3 | 1,2,3 | 1,2,3 | 1,2,3 |
| **Source statistic descriptor** | Speech/unknown | Speech/unknown | | |
| **Signalling Indication** | | | Yes/No | |
| **Evolved Allocation/Retention priority** | | | | |
| **- Priority Level** | 1-15 | 1-15 | 1-15 | 1-15 |
| **- Pre-emption Capability** | Yes/No | Yes/No | Yes/No | Yes/No |
| **- Pre-emption Vulnerability** | Yes/No | Yes/No | Yes/No | Yes/No |

Table 24: UMTS bearer attributes

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## 9.1.2 EPS Bearer QoS Attributes

3GPP Release 8 specification introduced a new QoS model for mobile Packet Core, for which CPC provides built-in features to support.

| | |
|---|---|
| **EPS Bearer QoS Parameters (To be defined per Bearer)** | Default Bearer/Dedicated Bearer |
| | Guaranteed Bit Rate (GBR) or Non-Guaranteed Bit Rate (N-GBR) |
| | Maximum Bit Rate (MBR) |
| | Traffic Flow Template (UL/DL-TFT |
| | **QoS Class identifier (QCI)** |
| | Allocation/Retention Priority (ARP) |
| **EPS Bearer QoS Parameters (To be defined per User)** | Aggregate Maximum Bit Rate (AMBR) |

Table 25: EPS Bearer QoS Attributes

Under the traffic classes discussion topic, the QoS Class identifier (QCI) parameter is used in Release 8 to differentiate against at least 9 classes of traffic.

QCI is a single scalar parameter which points to a set of QoS parameters. A QCI is associated with a priority, specific delay and packet loss values and whether the service has a guaranteed bit rate (GBR). These characteristics are used by the EPS nodes (eNodeB, S-GW, P-GW, MME) to guide them in deciding how a particular Service Data Flow (SDF) is to be processed. QCI replaces the four traffic classes (Conversational, Streaming, Interactive, Background) used in GPRS/UMST system.

| QCI | Resource Type | Priority | Packet Delay Budget | Packet Error Loss Rate | Example Services |
|---|---|---|---|---|---|
| 1 | GBR | 2 | 100 ms | 10-2 | Conversational Voice |
| 2 | | 4 | 150 ms | 10-3 | Conversational Video (Live Streaming) |
| 3 | | 3 | 50 ms | 10-3 | Real Time Gaming |
| 4 | | 5 | 300 ms | 10-6 | Non-Conversational Video (Buffered Streaming) |
| 5 | Non-GBR | 1 | 100 ms | 10-6 | IMS Signalling |
| 6 | | 6 | 300 ms | 10-6 | Video (Buffered Streaming), TCP-based (e.g., www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.) |
| 7 | | 7 | 100 ms | 10-3 | Voice, Video (Live ), Interactive Gaming |
| 8 | | 8 | 300 ms | 10-6 | Video (Buffered Streaming) TCP-based (e.g., www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.) |
| 9 | | 9 | | | |

Table 26: Mapping of QCI to standardized traffic characteristics

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### 9.1.3 Release 8 to pre-Release 8 QoS mapping

CPC allows for configurable QoS mapping to interwork between the LTE and 2G/3G networks, providing alignment on the QoS attributes that should be assigned to user plane traffic.This feature-set is delivered by CMM and provides flexible mapping between the LTE and R99 quality of service (QoS) provisioning to allow operators to select QoS mapping to provide the same user experience as when a subscriber moves between the LTE and 2G/3G networks.

- In inter radio access technology (inter-RAT) procedures where a UE moves from the LTE to UMTS/GERAN, the MME must map the evolved packet system (EPS) QoS parameters for each bearer into the set of QoS parameters used in the GPRS core network (generally referred to as R99 QoS or pre-Rel 8 QoS).
- The MME must also map R99 QoS parameters into EPS QoS parameters for each bearer when a UE moves from the UMTS/GERAN to LTE.

By default, CMM implements Release 8 to pre-Release 8 QoS mapping as specified in 3GPP TS 23.401 Annex E and summarized in Table 27 and Table 28

| 4G QCI | Traffic Class | SDU error ratio | Transfer Delay | Traffic Handling Priority |
|---|---|---|---|---|
| 1 | Conversational | $1x10^{e-2}$ | 100 | |
| 2 | Conversational | $1x10^{e-3}$ | 150 | |
| 3 | Conversational | $1x10^{e-3}$ | 80 | |
| 4 | Streaming | $1x10^{e-6}$ | 300 | |
| 5 | Interactive | $1x10^{e-6}$ | 100 | 1 |
| 6 | Interactive | $1x10^{e-6}$ | 300 | 1 |
| 7 | Interactive | $1x10^{e-3}$ | 100 | 2 |
| 8 | Interactive | $1x10^{e-6}$ | 300 | 3 |
| 9 | Background | $1x10^{e-6}$ | 300 | |

Table 27: CMM default LTE to 2G/3G QoS mapping

| Traffic Class | 4G QCI |
|---|---|
| Conversational_Speech | 1 |
| Conversational_Unknown_HighDelay | 2 |
| Conversational_Unknown_LowDelay | 3 |
| Streaming_Speech | 4 |
| Streaming_Unknown | 4 |
| Interactive_THP1_Signalling | 5 |
| Interactive THP1_nonSignalling | 6 |
| Interactive_THP2 | 7 |
| Interactive_THP3 | 8 |
| Background | 9 |

Table 28: 2G/3G to LTE QoS mapping

This deployment adheres to the 3GPP guidelines for Release 8 to pre-Release 8 QoS mapping, as such CMM default configuration requires no changes in this regard.

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## 9.2 Service Prioritization within the NFV Infrastructure

As a base principal, the current CPC solution should be dimensioned to avoid any congestions scenarios within the cloud infrastructure. However, QoS mechanisms shall be in place to distinguish between different types of traffic and map them to predefined Classes of services that are handled with different priority within the NFV Infrastructure in case congestion does occur.

The QoS traffic class definition within the CPC should be aligned with the end-2-end QoS policy in place in the Service Provider's network.

For the sake of simplicity, given that the VRF definition used in CPC already holds traffic priority considerations, the overlay Routing domains declared in the site router (VPRNs) can be directly associated with a specific QoS class. The traffic classes summarized in Table **29** have been defined under the scope of the current deployment.

| Overlay Routing Domain | QoS Class | Traffic type |
|---|---|---|
| (Reserved not used) | Conversational | Services that needs tight delay and jitter values. This class is reserved for VoLTE type of services. |
| (Reserved not used) | Streaming | Used mission critical services not highly sensitive to delay. Reserved for Push-to-talk type of services. |
| **Core-CP** | Interactive | Sensitive traffic which needs reserved bandwidth to guarantee service requirements. Traffic related with GPRS/EPS session management should be mapped to this QoS Class. OAM traffic should also occupy the lower ties of this traffic class, so that system admins may still operate the solutions in case of congestion. |
| **Signalling** | | |
| **OAM** | | |
| **RAN-UP** | Background | Services that use large packet sizes, not as sensitive to delay and jitter. User plane traffic relating to Mobile Broad band internet should be mapped to this QoS Class. |
| **Core-UP** | | |
| **SGi** | | |

Table 29: QoS classes for Cloud Packet Core

In this regard, QoS definition within CPC follows the same recommendations set by GSMA for IPX transit networks (IR.34 Jul 2018), where even though they share the same names, QoS Classes defined for CPC must not be confused with the UMTS QoS classes summarized in 9.1.1.

CPC NFVI uses the Diffserv model to enforce prioritization of specific applications/services across VNFs and NFVI. DiffServ networks classify packets into subset of "classes", grouped within the QoS classes defined in Table **29**, based on the DiffServ codepoint (DSCP) in a packet's IP header. This is known as behavior aggregate (BA) classification (RFC 2475). At each DiffServ router, packets are subjected to a "per-hop behavior" (PHB), which is invoked by the DSCP label.

In the Diffserv model there is no end-to-end QoS reservation. QoS is achieved by classifying and marking traffic on the IP end-points (in the VNFs and external NEs) and then having network elements in the cloud handle the corresponding IP packets based on their DSCP markings.

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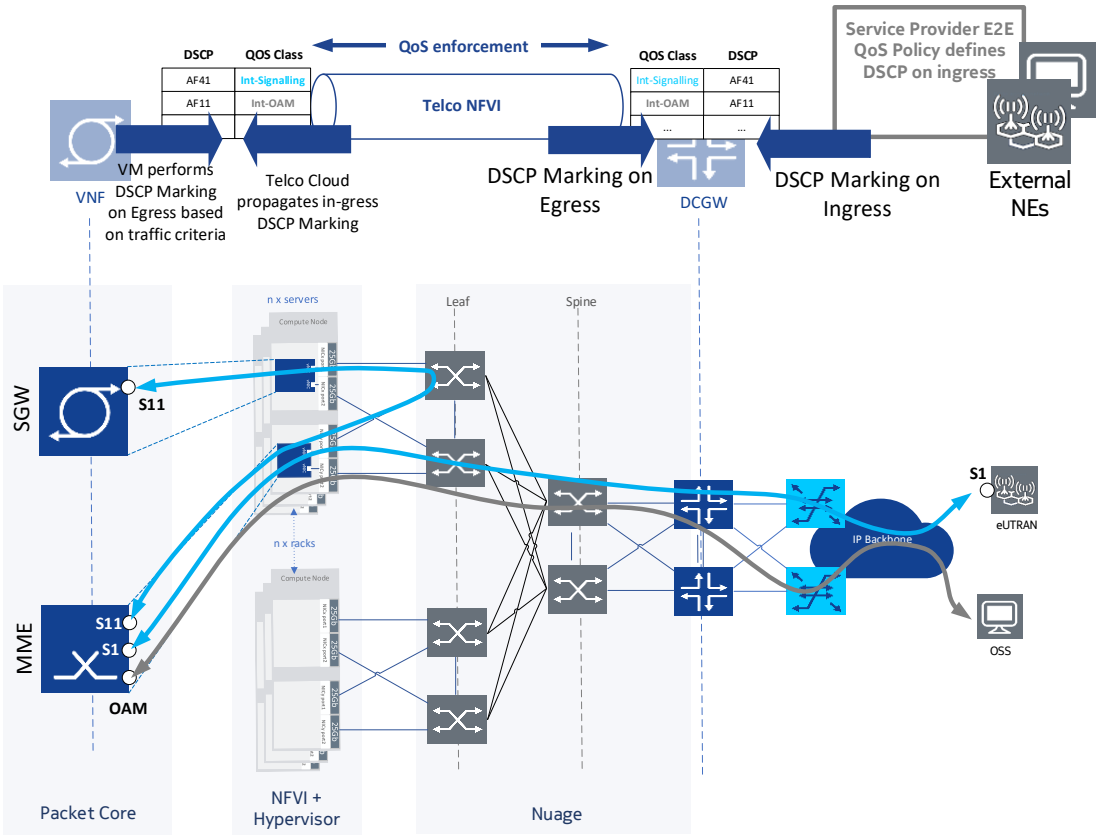



Figure 56: Diffserv model applied in the CPC architecture

To translate QoS requirements related to 3GPP applications to cloud networking elements, **CPC VNFs shall use DIffserv and mark IP packets on egress with predefined DSCP codes defined in Table 30.** Complementarily, NFVI, VIM and SDN CPC components shall adhere to the diffserv model by enforcing the following principles when forwarding packets to/from VNFs:

- The basic principle of diffserv QoS: “Mark on ingress, shape on egress”.
- When the packet is received, mark it with the appropriate QoS value on all the relevant network layers used to transport traffic.
- When queuing the packet for transmittal, apply measures (shaping, selective discard, etc.) to ensure higher priority traffic receives preferential access to scarce resources.
- Do not shape on ingress only egress.

Furthermore, cloud network fabric shall have mechanisms in place to verify if DSCP packet marking is committing to the set of DSCP ranges allowed on a specific subnet/VPRN, following the criteria specified in Table **30**. In case the DSCP marking for an incoming packet does not match any of the expected queues, cloud network infrastructure shall remark the DSCP bits in the incoming packet to the lowest DSCP priority allowed on the subnet/VPRN in question.

For network segments in the cloud using L2 transport, configurations shall be in place to coordinate a translation from DSCP CoS to PCP during packet forwarding procedures, so that QoS class may also be mapped into ethernet headers. PCP, commonly referred to as the “802.1p bits”, uses the 3bit ToS (Type of Service) bits defined in the 802.11D-1998 standard, to add QoS at the MAC level.

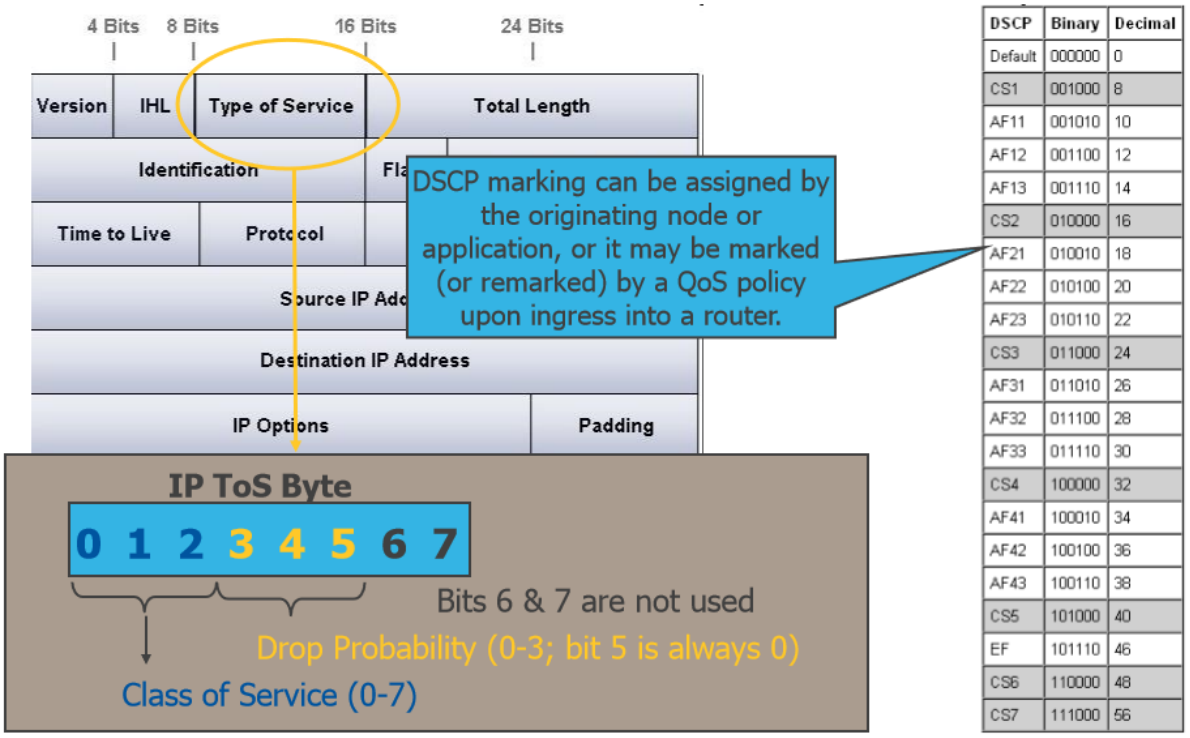
In addition to services related to VNF applications, the NFVI shall also reserve and prioritize network resources for traffic associated with Cloud management functions running on underlay networks.

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## 9.2.1 QoS Classes and DSCP marking

**The proposed Traffic Class/DSCP/COS mapping aligns itself with common telecom practices for 3GPP networks by following the recommendations of GSMA for IPX transit networks (IR.34 Jul 2018).**




| DSCP | Binary | Decimal |
|---|---|---|
| Default | 000000 | 0 |
| CS1 | 001000 | 8 |
| AF11 | 001010 | 10 |
| AF12 | 001100 | 12 |
| AF13 | 001110 | 14 |
| CS2 | 010000 | 16 |
| AF21 | 010010 | 18 |
| AF22 | 010100 | 20 |
| AF23 | 010110 | 22 |
| CS3 | 011000 | 24 |
| AF31 | 011010 | 26 |
| AF32 | 011100 | 28 |
| AF33 | 011110 | 30 |
| CS4 | 100000 | 32 |
| AF41 | 100010 | 34 |
| AF42 | 100100 | 36 |
| AF43 | 100110 | 38 |
| CS5 | 101000 | 40 |
| EF | 101110 | 46 |
| CS6 | 110000 | 48 |
| CS7 | 111000 | 56 |

Figure 57: IP Header DSCP Marking

The combination of all QoS concepts and information addressed in previous sections results in the following table which relates Routing Domains (overlay/underlay), Cloud-infrastructure/3GPP services, QoS classes and corresponding DSCP/PCP marking.

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| Routing Domain | Application/service | QoS Class | DSCP (PHB) | PCP |
|---|---|---|---|---|
| L2 domain | NFVI - Provisioning | Not defined | | |
| L2 domain | NFVI – internal APIs | Interactive-4 | AF41 | 100 |
| L2 domain | NFVI - Storage | Interactive-3 | AF31 | 011 |
| L2 domain | NFVI – Storage Management | Interactive-3 | AF31 | 011 |
| NFVI_Management | NFVI - Management | Interactive-2 | AF21 | 010 |
| Signaling | MME – S6a/d | Interactive-3 | AF31 | 011 |
| | MME – S13’ | Interactive-3 | AF31 | 011 |
| | MME-S11 | Interactive-3 | AF31 | 011 |
| | PGW-Gx | Interactive-3 | AF31 | 011 |
| | PGW-Gy | Interactive-3 | AF31 | 011 |
| | DRA-diameter | Interactive-3 | AF31 | 011 |
| | SPS-Gx | Interactive-3 | AF31 | 011 |
| RAN-CP | MME-S1 | Interactive-3 | AF31 | 011 |
| | SGSN-Iu_PS-C | Interactive-3 | AF31 | 011 |
| | SGSN-Gb_Auto | Interactive-3 | AF31 | 011 |
| CORE-CP | MME-S11 | Interactive-3 | AF31 | 011 |
| | MME-S10 | Interactive-3 | AF31 | 011 |
| | MME-S3 | Interactive-3 | AF31 | 011 |
| | SGSN-S16 | Interactive-3 | AF31 | 011 |
| | SGSN-S3 | Interactive-3 | AF31 | 011 |
| | SGSN-S4-C | Interactive-3 | AF31 | 011 |
| | SGW-S11 | Interactive-3 | AF31 | 011 |
| | SGW-Sxa | Interactive-3 | AF31 | 011 |
| | SGW-S4-C | Interactive-3 | AF31 | 011 |
| | SGW-S5/S8/Gn/Gp -C | Interactive-3 | AF31 | 011 |
| | PGW-Sxb | Interactive-3 | AF31 | 011 |
| | PGW-S5/S8/Gn/Gp -C | Interactive-3 | AF31 | 011 |
| Management | SGSN-Bp | Interactive-1 | AF11 | 001 |
| | PGW-Ga | Interactive-1 | AF11 | 001 |
| | CMM-OAM | Interactive-1 | AF11 | 001 |
| | CMG-OAM | Interactive-1 | AF11 | 001 |
| | SPS-OAM | Interactive-1 | AF11 | 001 |
| | CSD-OAM | Interactive-1 | AF11 | 001 |
| RAN-UP | SGW-S1 | Background | BE | 000 |
| | SGW-S12 | Background | BE | 000 |
| | SGSN-Iu_PS-U | Background | BE | 000 |
| | SGSN-Gb_NS | Background | BE | 000 |
| CORE-UP | SGW-S5/S8/Gn/Gp-U | Background | BE | 000 |
| | PGW-S5/S8/Gn/Gp-U | Background | BE | 000 |
| | SGSN-S4-U | Background | BE | |
| | SGW-S4-U | Background | BE | |
| SGi | Default marking (any) | Background | BE | |

Table 30: Service to DSCP to FC mapping

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When Bearer level QoS and GBR is part of the scope of the deployment the following mapping of UMTS/EPS QoS is enforced within the cloud network infrastructure.

| EPS | UMTS QoS | | | Cloud Packet Core | DSCP | COS |
|---|---|---|---|---|---|---|
| QCI | Traffic Class | THP | Sig. Ind. | QoS Class | | |
| 1 | Conversational | N/A | N/A | Conversational | EF | 5 |
| 2 | | | | | | |
| 3 | | | | | | |
| 4 | Streaming | N/A | N/A | Streaming | AF41 | 4 |
| 5 | Interactive | 1 | Yes | Interactive -4 | AF31 | 3 |
| 6 | | - | No | Interactive -3 | AF32 | 3 |
| 7 | | 2 | No | Interactive -2 | AF21 | 2 |
| 8 | | 3 | No | Interactive -1 | AF11 | 1 |
| 9 | Background | N/A | N/A | Background | BE | 0 |

Table 31: Mapping of QoS across EPS-UMTS and IP domains

Where, QCI to UMTS CoS mapping follows GSMA IR.34 recommendations. The table above is provided for reference only, given that only Bearer of QCI type 9 are to be validated under the scope of this deployment.

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## 9.3 QoS control per roaming partner

Often Service providers require to prioritize network resources for Home subscribers in detriment of visiting roamers. This may happen for a multitude of reasons, be it the nature of the roaming agreement itself or the need to guarantee resources for a specific service.

Cloud Packet core may be configured to enforce QoS Control on predefined roaming scenarios, enhancing QoS control for roaming subscribers by introducing the possibility to define a set of QoS-restrictions for each PLMN. This enhancement enables more flexible QoS management of roaming users in mobile packet core.

Roaming QoS is controlled differently, depending on the subscribers' s access type:

- For 4G and 5G NSA option 3X access MME QoS functionality is used
- For 2G and 3G S4-SGSN QoS functionality is used

QoS control for roaming subscribers is mentioned for the sake of completeness, given that it is not in scope of the current deployment.

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# 10. OAM (Operation and Maintenance)

## 10.1 Out-Of-Band vs In-Band Management Interfaces

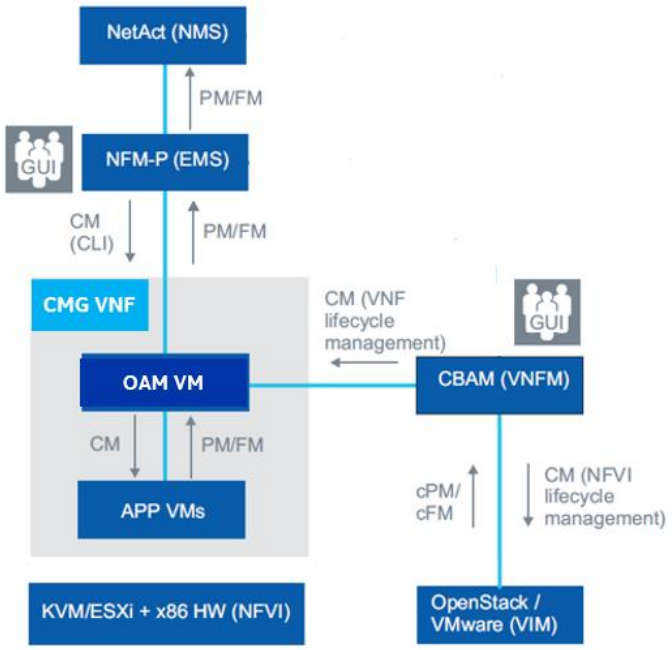
An Out-of-Band (OoB) management is management system separated from the system or solution it is used to manage. In simple words it is usually dedicated, physically separated network where all the physical network elements are plugged with their management ports (like console, iLOM, etc.). It usually requires some console routers and jump-host allowing access to NEs management ports over IP network.

On the other hand, an In-Band (IB) management involves managing devices through the common protocols such as telnet or SSH, using the network of the system as a media. It is a common way that provides identity-based access controls for better security. Its drawback is that in case of failures or overload of network or application, in-band management is being lost.

In NFV environment, all the physical NEs should be connected to OoB management to ensure access and configuration/alarms/logs inspection when facing problems with running systems and services.

SFC CMG and CMM will integrate with NFM-P via the OAM VLAN. The NFM-P and Traffic Platforms will be configured with CMG Outband Management IP.

The NFM-P implements the role of EMS and the CBAM implements the role of VNF manager for application lifecycle. The NFM-P interfaces with NetAct to forward alarms and PM data. The NFM-P GUI can also be launched from NetAct, which plays the role of the NMS. This way, a combination of a RAN (managed by NetAct) and an EPC network (managed by the NFM-P) is integrated under NetAct.




The NSP will also be able to do backup and restore of configuration function for LWS CMG.



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## 10.2 Time Synchronization

Within Telecom solutions NTP is a critical function that is intended to synchronize all systems within a few milliseconds of Coordinated Universal Time(UTC). NTP is designed to mitigate the effects of variable network latency.

Telecom Timers, Logging, and Fault Management applications require accurate time stamps. This allows for accurate correlation of events and alarms.

NTP service should be available both in out-of-band and in-band management networks – while the first is used for NTP by NFVI itself, second is required by VNFs and VMs that do not have access to out-of-band management network.

## 10.3 Fault Management

Fault Management (FM) role in NFV deployment should be considered at different layers:

- NFV Infrastructure (e.g. AirFrame servers, switches, routers)
- VNFs (e.g. CMM/CMG reported faults)

Below table presents mapping between CPC BP network elements and related EMS/NMS system in scope of Fault Management. Base Fault Management system in CPC BP is NSP platform thus recommended as first level integration whenever is supported by NEs/VNFs (required plugins or adaptors exist and are maintained as a part of release cycles). First level means NSP can be further integrated with other Network Managements Systems, like NetAct, via NSP NorthBound Interface (NBI). However, those systems and integration are out of scope of current CPC BP release.

Column “Protocol” indicate what protocol (and its version) is supported by VNF or NE for alarm notification and used in CPC BP (where applicable, in brackets other available options are listed).

| Domain/NEs | CPC BP 1st level EMS | Protocol |
|---|---|---|
| CMM | NSP | SNMPv3/NetConf |
| CMG | NSP | SNMPv3 (SNMPv1/v2) |

Table 32: Domain/NE to EMS mapping – Fault Management

## 10.4 Performance Management

Performance Measurement and Management (PMM) data helps to understand service quality, its consumption and trends, to review dimensioning and to plan capacity extensions on time. The “service” term could be understood very widely, from end user up to physical infrastructure resources, dependent on the level the collected data refer to.

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In addition, when PMM data are correlated with Fault Management information, it helps to understand failure or maintenance works impact on services and helps to make decision on service restoration steps.

The performance management is done via the collection of statistical data counters, divided in two indicators: Key Performance Indicator (KPI) and Key Capacity Indicator (KCI), which can be further retrieved by the external systems (like NSP/NFM-P or NetAct).

Following table summarizes PMM data format for each CPC BP domain and indicates systems to be used for PMM data retrieval.

| Domain/NEs | 1st level EMS | Protocol |
|---|---|---|
| CMM | NSP | NetConf Notify /XML sFTP |
| CMG | NSP | XML/sFTP |

Table 33: Domain/NE to EMS mapping – Performance Management

Cloud Packet Core Blueprint preferred PMM data collection system is NSP, which, dependent on operator needs, could expose those information to external systems (e.g. NetAct) for further post-processing, including conversion or customized KPIs creation, their presentation in user-friendly manner (like diagrams, tables), reports generation or presentation to other systems and tools (e.g. big data, trends analysis).

PMM data are usually stored locally as files by VNFs/NEs and then send to or retrieved by external systems. The way that data is stored and retrieved, as well as its naming convention, structure/format or granularity, is specific to systems it is generated by and their configuration. The guidance on recommended configuration is included in Low Level Design documentation while all available options can be found customer documentation.

### 10.4.1 KPI/KCI

The counters, Key Performance Indicators (KPIs) and Key Capacity Indicators (KCI) are used for post processing of the system's behaviour and performance with respect to Control Plane signaling, User Plane traffic patterns and system resources usage during peak and non-peak hours.

Statistical data about the system behavior helps operators to understand the network activity and to plan their network to handle the signaling, data bandwidth demands and system resources needs.

In this release, relevant KPIs and KCIs for service health monitoring and dimensioning are determined and described in Low Level Design, with recommendation on their granularity and aggregation schemas.

## 10.5 Backup & Restore

The goal of Backup & Restore (B&R) functionality is to provide capability to save and store NE/VNF/service-related configuration & state that in case of failures allows operator to restore impacted services.

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Backup procedure, by nature, is realized on-the-fly (without stopping or locking a system) and should not impact ongoing services and realized functions. In some scenarios, configuration changes could be restricted during backup execution period.

Restore procedure, in the contrary to backup, can impact ongoing services; dependent on the NE could even trigger some component, processes or whole solution restart.

Backup procedures usually comprise generating data copies of different flavours, like:

- Service-level configuration
- Networking-level configuration
- VNFCs-specific configuration, incl. databases
- logs, records and statistics information

B&R, dependent on NE/VNF, could be performed directly on NE/VNF or triggered by external systems like NSP or NetAct based on predefined rules (e.g. frequency, time, scope). Dependent on VNFs/NEs, backup can be transferred outside of VNF/NE in pull or push mode, e.g. by NSP. Nevertheless, NSP and NetAct should not be considered as backup storage solution but rather the way to schedule and retrieve it from VNF or NEs and then exposed/stored outside of NFVI.

Backup and Restore solution will be Nokia Archive.

All the details describing recommended B&R policies per NE/VNF, their scope and frequency are deeply described in respective Low Level Documents of this CPC Blueprint. LLDs indicate as well expected service and system impact during B&R procedures, its duration, related procedures (manual and scheduled) and supported external systems.

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# 11. Acronyms

| Term | Explanation |
|---|---|
| AAA | Authentication, Authorization & Accounting |
| ADSL | Asymmetric Digital Subscriber Line |
| APN | Access Point Name |
| ARP | Address Resolution Protocol |
| ARPU | Average Revenue Per User |
| BGP | Border Gateway Protocol |
| BNG | Broadband Network Gateway |
| BRG | Bridged Residential Gateway |
| CAPEX | Capital expenses |
| CDP | Connected Device Platform |
| CE | Customer Edge router |
| CPE | Customer Premises Equipment |
| CSR | Customer Service Representative |
| DC | Dual Connectivity |
| DC | Intra-E-UTRA Dual Connectivity |
| DEM | Dynamic Experience Manager |
| DHCP | Dynamic Host Configuration Protocol |
| DLNA | Digital Living Network Alliance |
| DNS | Domain Name System |
| DPI | Deep Packet Inspection |
| DRB | (user) Data Radio Bearer |
| ENB | Evolved Node B |
| EN-DC | E-UTRA-NR Dual Connectivity |
| EPC | Evolved Packet Core |
| EPS | Evolved Packet System |
| E-UTRA | Evolved Universal Terrestrial Radio Access |
| E-UTRAN | Evolved Universal Terrestrial Radio Access Network |
| FWA | Fixed Wireless Access |
| GPON | Gigabit-capable Passive Optical Network |
| GRE | Generic Routing Encapsulation |
| GUA | Global Unicast Address |
| HAG | Hybrid Access Gateway |
| IE | Information element |
| IFOM | IP Flow Mobility (3GPP) |

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| Term | Explanation |
|---|---|
| IPTV | Internet Protocol Television |
| LI | Lawful Interception |
| M2M | Machine To Machine |
| MAC | Medium Access Control |
| MCG | Master Cell Group |
| MCG | Master Cell Group |
| MN | Master Node |
| MPLS | Multi-Protocol Label Switching |
| MP-TCP | Multipath TCP |
| MR-DC | Multi-RAT Dual Connectivity |
| NAT | Network Address Translation |
| NE-DC | NR-E-UTRA Dual Connectivity |
| NFV | Network Function Virtualization |
| NGEN-DC | NG-RAN E-UTRA-NR Dual Connectivity |
| NR | New Radio |
| NSA | Non-Standalone |
| ONT | Optical Network Terminal |
| OPEX | Operational expenses |
| OS | Operating system |
| OSS | Operation support system |
| OTT | Over The Top |
| P2P | Point-to-Point |
| PCRF | Policy and Charging Rule Function |
| PDCP | Packet Data Convergence Protocol |
| PDN | Packet Data Network |
| PE | Provider Edge router |
| PPP | Point-to-Point Protocol |
| QCI | QoS Class Identifier |
| QoE | Quality of Experience |
| QoS | Quality of Service |
| QoS | Quality of Service |
| RAT | Radio Access Technology |
| RB | Radio Bearer |
| RLC | Radio Link Control |
| RRC | Radio Resource Control |
| RS/RA | Router Solicitation / Router Advertisement |

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| Term | Explanation |
|---|---|
| RTT | Round Trip Time |
| SA | Standalone |
| SCG | Secondary Cell Group |
| SCG | Secondary Cell Group |
| SFC | Service Function Chaining |
| SFF | Service Forwarding Function |
| SIB | System Information Block |
| SIP | Session Initiation Protocol |
| SMP | Service Management Platform |
| SN | Secondary Node |
| SOAP | Simple Object Access Protocol |
| SRB | Signaling Radio Bearer |
| SSC | Self Service Console |
| SSG | Service Steering Gateway |
| TDF | Traffic Detection Functions |
| UDP | User Datagram Protocol |
| UI | User Interface |
| UPnP | Universal Plug and Play |
| UTRAN | Universal Terrestrial Radio Access Network |
| VCPE | Virtual CPE |
| VDSL | Very-high-bit-rate Digital Subscriber Line |
| VLAN | Virtual Local Area Network |
| VM | Virtual Machine |
| VPN | Virtual Private Network |
| VRS | Virtual Routing and Switching |
| VSC | Virtual Services Controller |
| VSD | Virtual Services Directory |

Table 34: Abbreviations

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